

UNIVERSITY OF BUEA

**FACULTY OF ENGINEERING
AND TECHNOLOGY**

**DEPARTMENT OF
COMPUTER ENGINEERING**

**A MULTIMODAL APPROACH BASED ON ADVERSARIAL
MACHINE LEARNING FOR ANALYSIS OF SOCIAL MEDIA
MESSAGES**

By

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A Thesis submitted to the Faculty of Engineering and Technology of the
University of Buea in partial fulfilment of the requirements
for the degree of Doctor of Philosophy (PhD)
in Software Engineering.

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May, 2025

DEDICATION

To my parents Mr. Tchokote Pierre Lothin and Mme Tchokote Christine. I also dedicate this work to all those who strive to make the online world a safer and more inclusive space for everyone.

UNIVERSITY OF BUEA**FACULTY OF ENGINEERING
AND TECHNOLOGY****DEPARTMENT OF COMPUTER
ENGINEERING****CERTIFICATION**

The Thesis of **TCHUINDJANG TCHOKOTE Emmanuel Ludivin** (FE21P065) entitled “**A Multimodal Approach Based on Adversarial Machine Learning For Analysis of Social Media Messages**” submitted to the Department of Computer Engineering, Faculty of Engineering and Technology of the University of Buea in Partial Fulfilment of the Requirements for the Award of the Doctor of Philosophy (PhD) Degree in Software Engineering under the Supervision of

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ACKNOWLEDGMENTS

This thesis represents the culmination of several years of academic inquiry and intellectual development. It has also been a journey marking the transition from the early stages of academic exploration to the threshold of scientific maturity. As with any study of this magnitude, its successful completion would not have been possible without the guidance, support, and contributions of numerous persons to whom I am profoundly indebted.

I wish to express my deepest appreciation to Professor Elie FUTE TAGNE, Associate Professor, Head of Department, for his willingness to serve as my supervisor. His rigorous academic standards, invaluable advice, unwavering support, and sustained encouragement throughout the duration of this research have been essential to its development and realization.

My sincere thanks are also extended to the members of the pre-defense committee, who, despite their demanding professional commitments, devoted the necessary time and attention to thoroughly examine this thesis and offered constructive and enlightening feedback that strengthened its scientific merit.

I am also particularly thankful to the members of the final defense committee, who generously agreed to evaluate this work amidst their multiple responsibilities. Their critical observations and thoughtful suggestions will undoubtedly contribute to further enriching the scholarly impact of this research.

I would also like to extend my sincere gratitude to all the faculty members of the Department of Computer Engineering at the University of Buea, who have played a critical role in shaping my academic development since my first year. Their guidance,

encouragement, and willingness to share their expertise have significantly contributed to my intellectual growth.

I am also indebted to my parents, Mr. and Mrs. TCHOKOTE, for the great sacrifice they have made to support my academic journey. Their unwavering support, wise advice, love, and prayers have been fundamental to my progress and accomplishments. My heartfelt thanks go to all the members of my family, whose continuous encouragement and advice have been a source of strength throughout this journey.

Special acknowledgments are also due to the TCHUILO and the NKOTTO family, especially Papa TCHUILO Isaac and Mama TCHUILO Marie, whose consistent support and encouragement have been deeply appreciated and equally Col. NKOTTO, who has been like a second father to me.

I wish to thank my friends and peers (juniors, classmates, seniors, team mates, and coaches alike) for their friendship, solidarity, and moral support throughout my university studies. I am especially grateful to Dr SOP DEFFO Landry, Dr KAMDJOU Hughes, Coach OBEN Rudolf, Coach MELINGUI Patrick, ACHINEK Snonta, FORTU Sullivan, whose companionship and encouragement have accompanied me at every stage of this academic journey.

ABSTRACT

The proliferation of harmful information, such as hate speech and online harassment, has increased in recent years due to social media's explosive expansion. The prevalence of online violence against women, particularly on social media, underscores the urgent need for effective tools to mitigate misogynistic content. Existing methods, often reliant on keyword-based or rule-based approaches, struggle with the nuances of language and context, leaving them vulnerable to adversarial attacks. In this study, we proposed AdverMIRA, a Robust Multimodal Intelligence for Recognizing Abuse that uses hybrid fusion to integrate RoBERTa with additional transformer layers for text feature extraction and RESNET for visual feature extraction. We incorporated adversarial training techniques to enhance AdverMIRA's resilience against various non-technical and technical adversarial attacks. Also, to effectively address class imbalance and improve classification accuracy, we designed HateXcel, a robust model that uses the Distanced based Synthetic Minority Over-sampling Technique (D-SMOTE) and Incremental Principal Component Analysis (PCA) combined with adversarial machine learning techniques. AdverMIRA and HateXcel models significantly outperforms current techniques. Experimental results demonstrate that AdverMIRA consistently outperforms baseline models on the MAMI dataset. For non-technical attacks, it achieved a macro F1-score of 98.99%, accuracy of 99%, and precision of 97.77%. Under technical adversarial scenarios, AdverMIRA attained a macro F1-score of 97.97%, accuracy of 98%, and precision of 97.98% for FGSM; 95.9%, 96%, and 91.66% for PGD; and 73.7%, 74%, and 69.56% for AdvDrop, respectively. Compared to earlier studies on the FBHM dataset, HateXcel showed a noteworthy 18% gain in accuracy over the base model. For technical attacks, HateXcel attained a macro F1-score of 89.73%, accuracy of 89.80%, and precision of 90.9% for FGSM; 69.71%, 70.2%, and 77% for PGD; and 68.3%, 68.4%, and 70% for AdvDrop, respectively. We stored the model's prediction on the test set for each dataset used on all the attacks in JSON files for human-level judgment.

Keywords: Online Misogyny Prevention, PCA, D-SMOTE, Adversarial Robustness, Multimodal Classification, Hate Speech Detection

TABLE OF CONTENTS

DEDICATION	ii
CERTIFICATION	iii
ACKNOWLEDGMENTS	iv
ABSTRACT	vi
LIST OF FIGURES	iv
LIST OF TABLES	vi
LIST OF ABBREVIATIONS.....	vii

CHAPTER ONE

GENERAL INTRODUCTION

1.1 Context and Motivations.....	1
1.2. Problem Statement.....	5
1.3. Objectives	6
1.3.1. Main Objective	6
1.3.2. Specific Objectives	6
1.4. Contributions.....	7
1.5. Structure of Thesis	8

CHAPTER TWO

ANALYSIS OF SOCIAL MEDIA MESSAGES: STATE OF THE ART

2.1. Introduction.....	9
2.1.1. Analysis of Social Media Messages	9
2.1.2. Multimodal Deep Learning Frameworks.....	23
2.1.3. Adversarial Attacks.....	29
2.1.4. Adversarial Training	36
2.2. Research Frontiers	42
2.3. Partial Conclusion.....	43

CHAPTER THREE

ADVERMIRA: A ROBUST MULTIMODAL INTELLIGENCE FOR RECOGNIZING ABUSE

3.1. Introduction.....	45
3.2. Introducing the MIRA Approach.....	48
3.2.1. Feature Extraction Module	49
3.2.2. Classifier Module.....	56
3.2.3. Introducing the AdverMira approach	56
3.3. Experiment and Results	62
3.3.1. Dataset	62
3.3.2. Experimental Setup.....	64
3.3.3. Evaluation Metrics.....	65
3.3.4. Results	66
3.4. Conclusion	73

CHAPTER FOUR

HATEXCEL: A ROBUST HATE CONTENT DETECTION MODEL ON FBHM DATASET

4.1. Introduction.....	75
4.2. Introducing the HateXcel Approach.....	77
4.2.1. Data Preparation and Processing	77
4.2.2. Training Procedure	82
4.3. Experiment and Results	83
4.3.1. Dataset.....	83
4.3.2. Experimental setup	86
4.3.3. Classification Results.....	87
4.4. Conclusion	94

CHAPTER FIVE

GENERAL CONCLUSION

5.1. Summary	95
5.2. Critical Analysis of Results	97
5.3. Difficulties Encountered	98
5.2. Further Works	99
REFERENCES	101
LIST OF PUBLICATIONS	108
Appendices.....	109
Appendix A: Model Prediction Outputs on MAMI dataset	118
Appendix B: Model Prediction Outputs on FBHM dataset.....	120

LIST OF FIGURES

Figure 1. Common Social Media Platforms, [3].....	1
Figure 2. Applications of social media analysis, [3].....	4
Figure 3. An example of a trained model for hate post detection on social media.....	5
Figure 4. Framework of the proposed DNN-based model for cyber aggression, [11] ...	10
Figure 5. Overall methodology of proposed model, [12]	12
Figure 6. Working process of the FSSDL-CBDC model, [13].....	13
Figure 7. Consumer behavior model, [14].....	14
Figure 8. Proposed methodology for suicidality detection, [15]	15
Figure 9. HABERTOR with two sources and ensemble of two heads, [16]	17
Figure 10. The Framework of TextFirewall, [9].....	19
Figure 11. Adv4SG: Adversarial Text Attacks for Privacy Protection, [17].....	20
Figure 12. Model diagram, [18].....	21
Figure 13. Overview of the Approach proposed by Kumari et al, [20]	24
Figure 14. Multimodal Architecture for Antivaccine Message Analysis, [21]	25
Figure 15. Abstract view of multimodal offense and troll detection system, [22]	26
Figure 16. Hate Speech Detection and Target Identification Flow, [23].....	27
Figure 17. Adversarial attack on image, [25]	29
Figure 18. Adversarial training, [32]	37
Figure 19. VILLA: Vision-Language representation learning framework, [31]	38
Figure 20. The overall architecture of MANN, [33].....	39
Figure 21. A schematic showing the overall setup of experiments, [34].....	40
Figure 22. Overview of the MIRA framework	49
Figure 23. ResNet50 architecture, [50].....	50
Figure 24. ReLu activation function, [51]	52

Figure 25. Illustration of pooling operation, [52]	52
Figure 26. RoBERTa Infrastructure, [53]	54
Figure 27. Example of memes found in the dataset.....	58
Figure 28. Different forms of non-technical attacks on image considered.....	59
Figure 29. Example memes from the MAMI dataset. (a: Left, b: Right)	63
Figure 30. Top 50 words vs Frequency	64
Figure 31. Ablation study of the proposed model	70
Figure 32. Posts Detected by MIRA and AdverMIRA.....	71
Figure 33. Memes predicted with the same output across the 3 attacks	72
Figure 34. Failure cases on the MAMI test set	73
Figure 35. Principal Component Analysis, [71]	77
Figure 36. Schematic representation of D-SMOTE, [68].....	80
Figure 37. Flowchart of the HateXcel approach.....	82
Figure 38. Multimodal "mean" memes and benign cofounders, [65].....	85
Figure 39. Top 50 words of the FBHM train dataset.....	86
Figure 40. Evaluation of Model under Non-Technical Attacks.....	90
Figure 41. Model's performance on technical attacks.....	91
Figure 42. Some examples of posts detected as not-hateful	92
Figure 43. Examples of posts detected as hateful by our model.....	92
Figure 44. Similar prediction across all 3 attacks	93
Figure 45. Failure cases on the FBHM dataset	94

LIST OF TABLES

Table 1. Summary of machine learning approaches in analysis of social media.....	16
Table 2. Summary of adversarial methods in social media analysis.....	22
Table 3. Summary of multimodal techniques for analysis of social media messages	28
Table 4. Summary of typical attacks on images and their principles.....	33
Table 5. Dichotomy of attacks on text	36
Table 6. Summary of approaches with multimodal adversarial training	41
Table 7. Adversarial attacks on text.....	59
Table 8. MAMI Dataset sizes and label distribution.....	62
Table 9. Adversarial performance across training parameters.....	67
Table 10. Performance Comparison of Models on MAMI Validation Memes.....	68
Table 11. Average execution time of adversarial methods during adversarial training.....	69
Table 12. Accuracy before and after adversarial training	70
Table 13. FBHM dataset sizes and label distribution.....	84
Table 14. Results of different models on FBHM dataset.....	88
Table 15. Models' performance after applying technical adversarial attacks	89

LIST OF ABBREVIATIONS

AA: Auto Attack

AI: Artificial Intelligence

AML: Adversarial Machine Learning

BCO-FSS: Binary Coyote Optimization based Feature Subset Selection

BERT: Bidirectional Encoder Representations from Transformers

BPDA: Backward Pass Differentiable Approximation

CG-ES: Standard Evolution Strategies

CNN: Convolutional Neural Network

CW: Carlini and Wagner

DEA-RNN: Dolphin Echolocation Algorithm Recurrent Neural Networks

DCT: Discrete Cosine Transform

DMM: Deep Multimodal Models

DNN: Deep Neural Networks

D-SMOTE: Distance-based SMOTE

FBHM: Facebook Hate Meme Dataset

FGM: Fast Gradient Method

FGSM: Fast Gradient Sign Method

FSSDL-CBDC: Feature Subset Selection with Deep Learning based cyberbullying detection and categorization

GA: Genetic Algorithm

GAP++: Generative Adversarial Perturbations

GBDA: Gradient Based Distributional Attack

GGNR: Good Grammar No Redo

IDCT: Inverse Discrete Cosine Transform

JSMA: Jacobian-based Saliency Map Attack

LA: Linear Regression

L-BFGS: Large Broyden-Fletcher-Goldfarb-Shanno algorithm

LSTM: Long Short-Term Memory

MANN: Multimodal Artificial Neural Network

MAMI: Multimedia Automatic Misogyny Identification

MLP: Multi-Layer Perceptron

MMN: Multi-modal Neural Network

NLP: Natural Language Processing

OCR: Optical Character Recognition

PCA: Principal Component Analysis

PWWS: Probability Weighted Word Saliency

SMOTE: Synthetic Minority Over Sampling Technique

SOTA: State-of-the-Art

SSA-DBN: Salp Swarm Algorithm Deep Belief Network

UAP: Universal Adversarial Perturbations

VGG: Visual Geometric Group

VSNR: Vowel Search No Redo

WBNR: Word Balance No Redo

WSNR: Word Segmentation No Redo

ZOO: Zeroth Order Optimization

CHAPTER ONE

GENERAL INTRODUCTION

1.1. Context and Motivations

Social media has become a major communication channel in everyday life and has drastically changed the way people interact and communicate with each other. Among the social media platforms highlighted in Figure 1, the most popular widely used are Facebook, YouTube, WhatsApp, Instagram, TikTok, WeChat and Twitter. These platforms serve as essential channels for connecting individuals, sharing ideas, and facilitating interactions between businesses and customers through targeted advertisements, review gathering, and feedback collection [1]. With this newfound power as described in [2], people are also free to generate and share content and are also able to post messages that can reach a wide audience and even have an impact on public opinion. As a result, it is increasingly important to understand the implications of social media messages and carry out careful analysis of them.

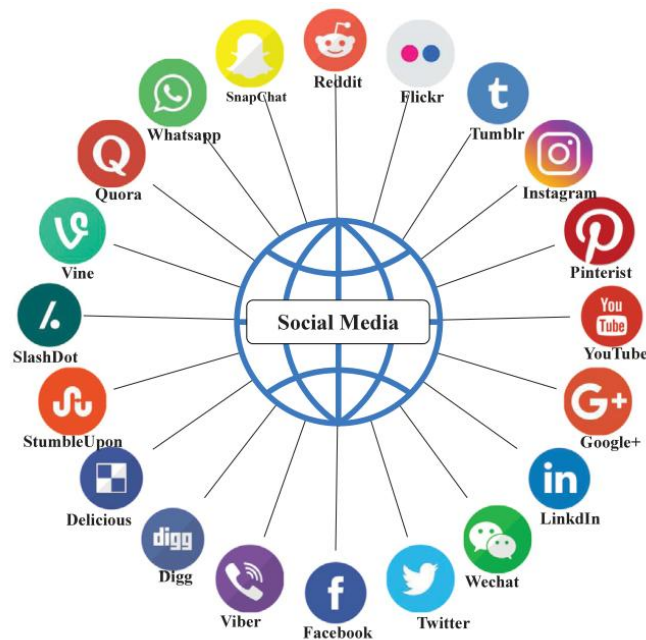


Figure 1. Common Social Media Platforms, [3]

According to Athira et al., when carrying out an analysis of a message on social media [4], it is important to consider both the content of the message as well as the context in which it was posted. Examining the message itself often involves breaking it down into its constituent parts to analyze the language and tone it conveys. It is also important to look at the message from a variety of perspectives [5], what does it mean for the sender, for the recipient, and for the larger society? It is crucial to look at the implications of the message, what it could potentially lead to, and the potential impact on public opinion. In terms of context, it is essential to consider the author, their motivations, and the audience. Who are they talking to, why are they talking to them, and what could be the implications of their message for different groups of people? It is also useful to consider the timing of the message and the overall timeline in which it was sent. What else was happening in the world when this message was posted, and how did that affect the perception of the message?

Also, it is important to consider how the message was received by its audience. It is possible to take feedback from social media platforms [5], such as likes and comments, to get an idea of how people respond to the message. People's reactions to the message can provide insight into how it was understood and interpreted by the public, and whether or not it was successful in conveying its intent. Analyzing social media messages is an important task, as it helps us understand the potential impact on public opinion and to prevent potential harm. An example of the effects of hate speech in Cameroon is the Anglophone crisis [6] that has been going on since 2016 and has been largely fueled by hate speech and posts on social media platforms. The crisis began as protests by teachers and lawyers in the Anglophone regions, demanding reforms to the educational and legal system that they felt discriminated against Anglophones.

However, instead of addressing the issues, some individuals and groups resorted to hate speech and posts, expressing sentiments of incitement and calling for secession. This escalated the crisis, leading to violence, displacement of people, loss of property, and loss of lives.

Also, hate posts have been used to stigmatize and discriminate against people from different ethnic and linguistic backgrounds in Cameroon. For instance, [7] in 2019 the politicization of ethnicity is driving polarization and hostility between ethnic groups, particularly between the Beti and Bulu communities and the Bamileke community. The rise of hate speech online resulted in communal attacks, as seen in the southern town of Sangmelima.

According to Herdiansyah et al [8], sentiment analysis is one of the most important research trends in the world of health (and/or mental health) that can identify disease characteristics through textual analysis of social media. Cultural factors are one of the serious obstacles that view people with mental disorders as people who cannot function fully, need to be avoided, have problems and are given a negative social stigma. Detecting depressed users online using sentiment analysis as a tool can improve identification and early intervention so that treatment can be started earlier. Also, the stigma associated with mental illness in our community can be minimized.

The reach to a large audience and freedom of generating and sharing content poses several challenges (rumors, hate speech, misinformation, opinion mining). The diagram in Figure 2 shows some applications of social media analysis.

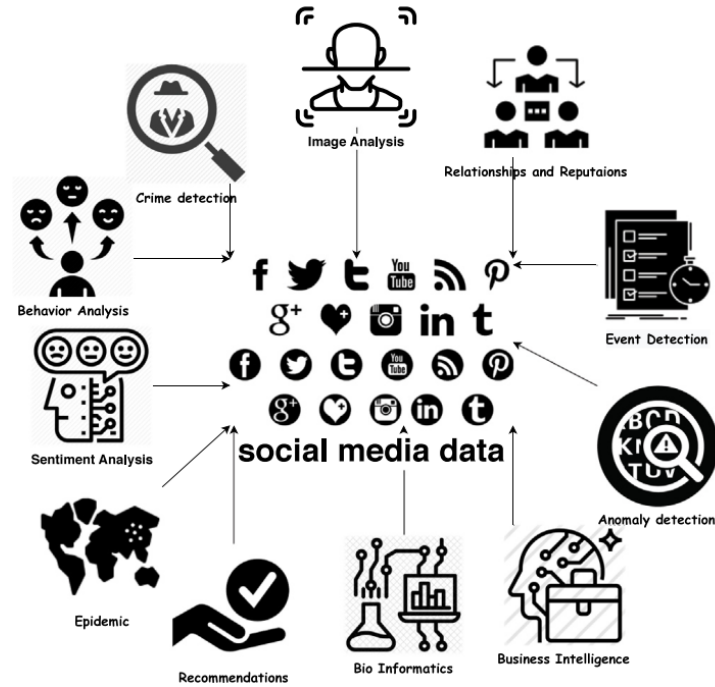


Figure 2. Applications of social media analysis, [3]

In recent years, deep neural networks (DNNs) are continuously developed to solve these challenges. However, research has shown that DNN-based models are vulnerable to adversarial examples, which are often imperceptible to human eyes and can cause erroneous outputs [9]. As reported by Wang et al. [9], such adversarial examples have been identified in various data modalities, including images, text, and audio, all of which represent common forms of content shared across social media platforms. This vulnerability brings serious security concerns to users who deploy DNN-based systems on social media platforms to detect and address issues such as misinformation, abusive content, and automated spam.

Within social media websites, hate post detection systems are deployed to detect hate speech in posts but these systems are vulnerable to adversarial attacks and these attacks lead these systems to make erroneous outputs, i.e., misclassifying a post as cyberbullying.

1.2. Problem Statement

Considering the increasing concerns over global violence, several efforts have been put to identify the potential sources and reduce the spread of hate post over social networks. Given the fact that communication over these social medias is mostly common through images and text, interesting hate post detection techniques on text and images have been proposed but these detection methods are also subject to multimodal adversarial attacks (image and text attacks). The goal of such attacks is to disturb the classification capabilities of the classifiers resulting in misclassification of abusive and toxic content. These attacks limit the accuracy of the classifiers.

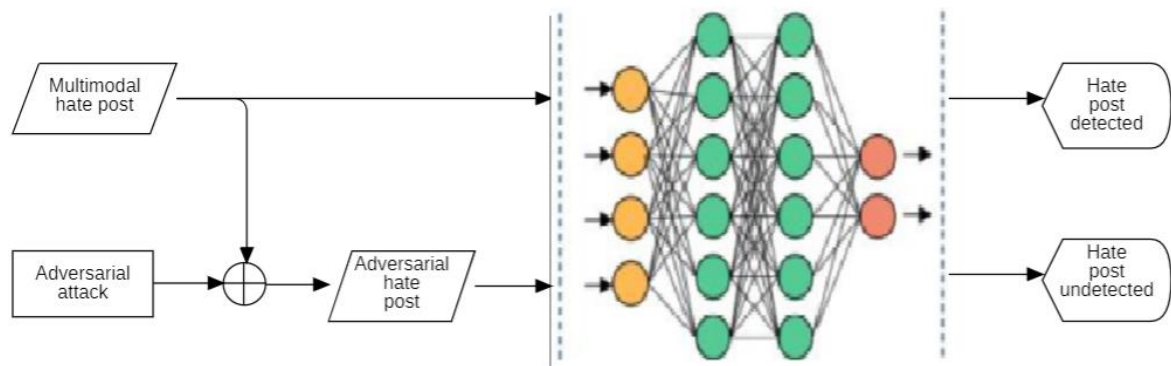


Figure 3. An example of a trained model for hate post detection on social media.

In Figure 3, it can be seen how a hate post can go undetected and spread throughout social media if the post is disguised. This disguise can be done either on the image or text to fool the detector and it may be known or not known to the detector model. A hate post not effectively detected and blocked from others users like teenagers on social media can seriously affect the mental health and may in extreme cases lead to suicide. For instance, [10] Hana Kimura, 22 years, killed herself in May 2020 after being subjected to abuse and harassment on social media following her appearance on a popular TV show. Also, a user posts a comment on a news article about a political

leader, expressing their disagreement to their policies, the comment contains some sarcasm and irony but no abusive language, but the system fails to understand the tone and the context of the comment and flags it as a hate speech. The user receives warning and their comment is removed. The user will feel censored and silenced by the platform. These scenarios show challenging it is to design an AI system that can accurately detect hate speech in different cultures and contexts.

1.3. Objectives

1.3.1. Main Objective

The main objective of this work is to build a robust multimodal defense system capable of countering known adversarial attacks during the analysis of social media messages. This multimodal system will be able to respond rapidly to potential threat while adapting to emerging threats on the fly. It will also provide visibility and transparency by providing actionable insights and reporting to make informed decisions regarding the cyber security posture of social media platforms.

1.3.2. Specific Objectives

To achieve our main goal, we are going to do a segmentation of the work and specifically we have the following specific objectives:

- Conduct a systematic literature review on analyzing social media messages for hate post detection and identify the main gaps and challenges in existing research.
- Design a novel algorithm for performing adversarial training on text and image.
- Evaluate model performance and efficiency compared to existing methods.

1.4. Contributions

In this thesis, the following contributions are made to the field of social media analysis and online content moderation.

- **Contribution 1: MIRA, A multimodal framework for misogyny detection**

Existing models often treat text and visual data separately, limiting detection accuracy in complex online content such as misogyny, racism, hate etc. We propose MIRA, a novel multimodal framework that integrates advanced text (RoBERTa with additional transformer layers) and visual (ResNet50) feature extraction techniques. MIRA significantly outperforms current state-of-the-art models in detecting misogynistic content on social media platform demonstrating the potency of multimodal learning.

- **Contribution 2: HateXcel, Overcoming class imbalance with D-SMOTE and Incremental PCA**

Hate speech datasets like the Facebook Hate Meme (FBHM) dataset, typically suffer from severe class imbalance which undermines the classification performance. To address the challenge of class imbalance in hate speech datasets we employ a combination of Distance-based Synthetic Minority Over Sampling Technique (D-SMOTE) and Incremental Principal Component Analysis (PCA), to augment and balance the training data effectively. The results obtained showed a notable improvement in classification performance of HateXcel, contributing to a fairer and a more accurate hate speech detection

- **Contribution 3: Enhancing models robustness via adversarial training**

Deep learning models are vulnerable to adversarial examples that can compromise reliability in real-world applications. To improve the resilience of the models (MIRA and HateXcel), we incorporated adversarial training to the two frameworks. These frameworks demonstrated an enhanced resilience against both technical and non-

technical adversarial attacks. The experimental results we carried out on validated benchmarks datasets such as MAMI and FBHM, shows that the proposed approaches achieve state-of-the-art performance. This also highlights the importance of combining multimodal data and adversarial machine learning to create more secure and reliable online content moderation systems that will ultimately contribute to a safer digital environment.

1.5. Structure of Thesis

The remainder of this thesis is organized as follows: in Chapter 2, we provide a comprehensive literature review that covers existing state-of-the-art approaches in content moderation on social media, multimodal hate speech detection, and adversarial learning. In Chapter 3, we focus on enhancing the multimodal classification robustness of our framework AdverMIRA to detect online misogyny through adversarial training. In Chapter 4, we describe the approach (HateXcel) used in addressing class imbalance and resilience in multimodal hate post detection through the combination of incremental PCA, SMOTE and adversarial multimodal learning. In Chapter 5, we provide a summary of the findings and the significance of our different approaches in advancing robust multimodal analysis of social media messages. We also discuss the implications of the findings, the limitations of the current study, and the potential directions for future research.

CHAPTER TWO

ANALYSIS OF SOCIAL MEDIA MESSAGES: STATE OF THE ART

2.1. Introduction

In this section, we discuss some related works on the analysis of social media messages, multimodal learning, adversarial attacks, and adversarial training. We also present a table summarizing the existing approaches for analyzing social media messages.

2.1.1. Analysis of Social Media Messages

Social media analysis is the procedure of studying social compositions through the use of networks and graph theory [3] . According to Quinlan et al. (as cited in reference [5] in Balaji et al [3]), social media analysis can be categorized into several domains based on its applicability to machine learning techniques. Some of the applications of social media analysis considered by Balaji et al [3] are anomaly detection, behavioral analysis, bioinformatics, business intelligence, crime detection, event detection, opinion and emotion analysis. The following subsections will describe some existing machine learning and adversarial machine learning approaches that apply to the said application domains.

a) Machine Learning Approaches

Behavior analysis is a natural science that seeks to understand the behavior of individuals. One way of analyzing the behavior of a user is by identifying the behavior of that user on social using machine learning on social media analysis for opinion mining, text forensics, etc. According to Singh et al. (as cited in reference [82] in Balaji et al [3]), they developed a method to identify the behavior of normal users from spammers in social media. They worked mostly for identifying pornographic spammer

from the twitter data using NodeXL. They applied few machine learning algorithms to classify the data and discovering the behavior of illegitimate users.

While prior studies such as those by [3] have outlined broad application domains of social media analysis, more recent efforts have shifted focus towards the detection of harmful online behaviors. In this context, Khan et al. [11], proposed a deep learning model that extracts eight novel emotional features from the textual tweets and uses a newly designed deep neural network with only three number of layers to identify aggressive statements on social media. The proposed DNN model was tested on the Cyber-Troll dataset. The combination of the word embedding and eight different emotional features were fed into the DNN for significant improvement in recognition while keeping the DNN design simple and computationally less demanding. The model Khan et al [11] proposed achieves an F1 score of 97% outperforming several machine learning and deep learning models. The framework of the DNN model for cyber aggression detection is shown in Figure 4.

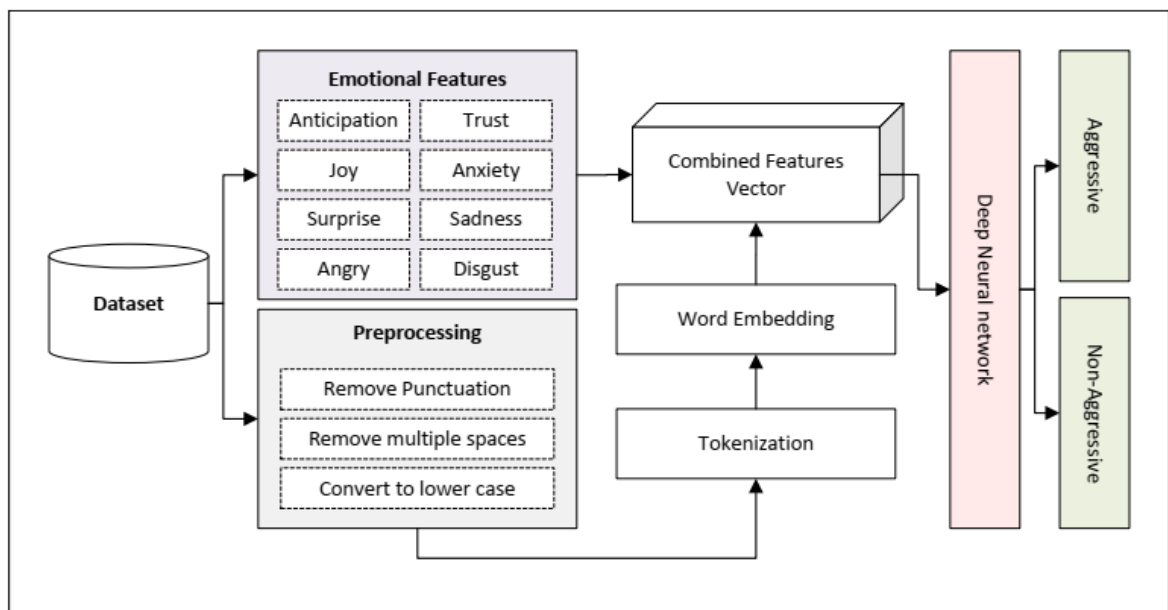


Figure 4. Framework of the proposed DNN-based model for cyber aggression, [11]

Subsequent research has explored hybrid modelling techniques to further enhance performance and optimize training efficiency. For instance, Murshed et al [12], proposed a hybrid model (DEA-RNN) that combines Elman type Recurrent Neural Networks (RNN) with optimized Dolphin Echolocation Algorithm (DEA) for fine-tuning the Elman RNN's parameters and reducing training time in the context of cyberbullying detection. They evaluated the hybrid model thoroughly using a dataset of 10000 tweets and compared its performance to those of the state-of-the-art algorithms. The results demonstrated that the DEA-RNN model achieved a high average accuracy of 90.45% and an F1-score of 89.25%, indicating a strong balance between precision and recall. These outcomes underscore the effectiveness of integrating DEA with RNN, particularly in enhancing parameter optimization and reducing the overall training time.

Although the proposed model obtained higher performance rates than the other considered models, specifically, the feature compatibility of DEA-RNN reduces when the volume of input data exceeded the initial dataset size, suggesting potential challenges in adapting the model to larger-scale or more diverse datasets. Moreover, the study was limited only to a single data source, the Twitter dataset, which may limit the generalizability of the findings to other social media platforms or communication environments. Furthermore, they performed analysis solely on the textual content of the tweets, without incorporating user-level behavioral features such as posting patterns, interaction history, or metadata, which could provide deeper contextual insights into cyberbullying behavior. The overall methodology of the proposed hybrid model is shown in Figure 5. The model includes the following phases: data collection, data annotation, preprocessing and data cleansing, feature extraction and feature selection, and classification.

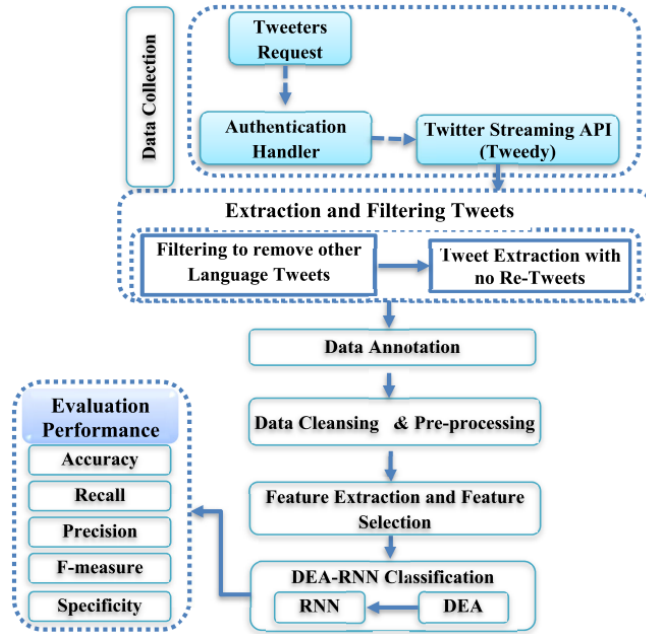


Figure 5. Overall methodology of proposed model, [12]

In the regard to emphasize the importance of feature selection in improving detection accuracy and reducing model complexity, Neelakandan et al [13], proposed a Feature subset selection with deep learning-based cyberbullying detection and categorization (FSSDL-CBDC) approach for social networks that combines deep learning with feature subset selection. The suggested FSSDL-CBDC technique consists of a number of phases, including preprocessing, feature selection, and classification among others. Additionally, as shown in Figure 6, a binary coyote optimization (BCO)-based feature subset selection (BCO-FSS) technique is employed to select a subset of features that will increase classification performance by using the BCO algorithm. The salp swarm algorithm (SSA) is used in conjunction with a deep belief network (DBN), which is known to as the SSA-DBN model, to detect and characterize cyberbullying in social media networks and other online environments. The SSA-DBN model receives and classifies the feature-reduced subset in the same time frame as the other models. When compared to the classic DBN model, the usage of the SSA to fine-tune the DBN model's hyperparameter resulted improved outcomes.

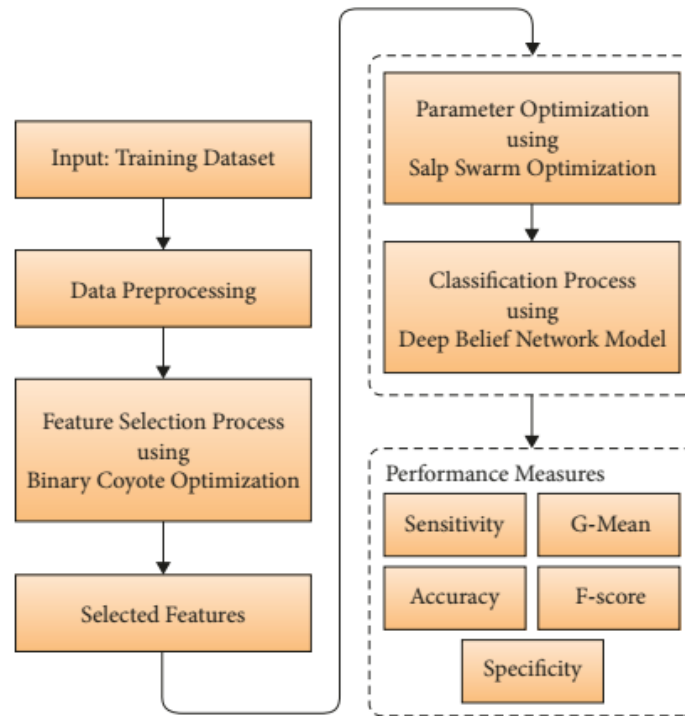


Figure 6. Working process of the FSSDL-CBDC model, [13]

The SSA-DBN model has exhibited superior accuracy to other algorithms, with a 99.983% accuracy rate.

While previous studies primarily focused on detecting harmful or abusive behavior on social media through deep learning and feature optimization techniques, other researchers have explored the broader potential of social media data for behavioral analytics. For example, Chaudhary et al [14], used the concept of big data technology to process data and analyze it to predict consumer behavior on social media platforms based on some parameters and criteria. They analyzed consumer perception, attitude towards the social media platform. They predicted consumer behavior from the social media data like Facebook, Youtube, Linkedin, Twitter, Instagram and Pinterest. Their model is helpful for businesses to predict consumer behavior about the product using social media data. they also found out that the decision tree is the best model for

consumer behavior prediction on social media achieving a maximum accuracy of 98%.

The diagram in Figure 7 below describes the consumer behavior model used.

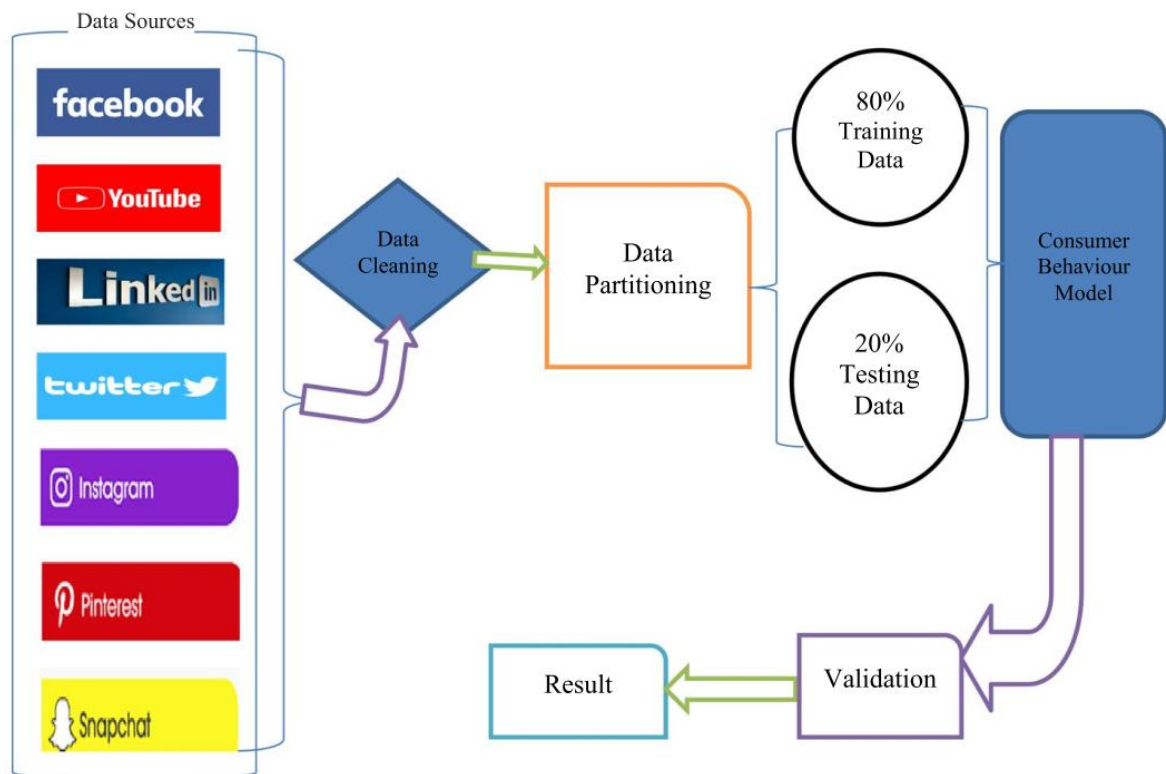


Figure 7. Consumer behavior model, [14]

In this model, they predicted the behavior of consumers by doing big data analytics. The data are cleaned by removing noises, errors, duplication and outliers to make the quality data and to achieve good results. The machine learning algorithms was used to process the data for prediction of consumer behavior on social media. The supervisor learning algorithm was used for prediction. The limitation of this model is that it can't be used on a daily basis consumer data as it has shown very poor results.

Beyond commercial and behavioral applications, recent research has also turned to the use of social media analytics for addressing critical mental health issues. In this context, Rabani et al [15], developed a multi-class machine learning classifier for identifying suicidal risk levels in social media posts. They proposed an enhanced feature

engineering approach for suicidal risk identification that was used to extract features from a novel dataset collected from Twitter and Reddit platforms. They used three machine learning algorithms for classification i.e., Support Vector Machine, Random Forest and Extreme Gradient Boosting. The diagram in Figure 8 describes the four major steps.

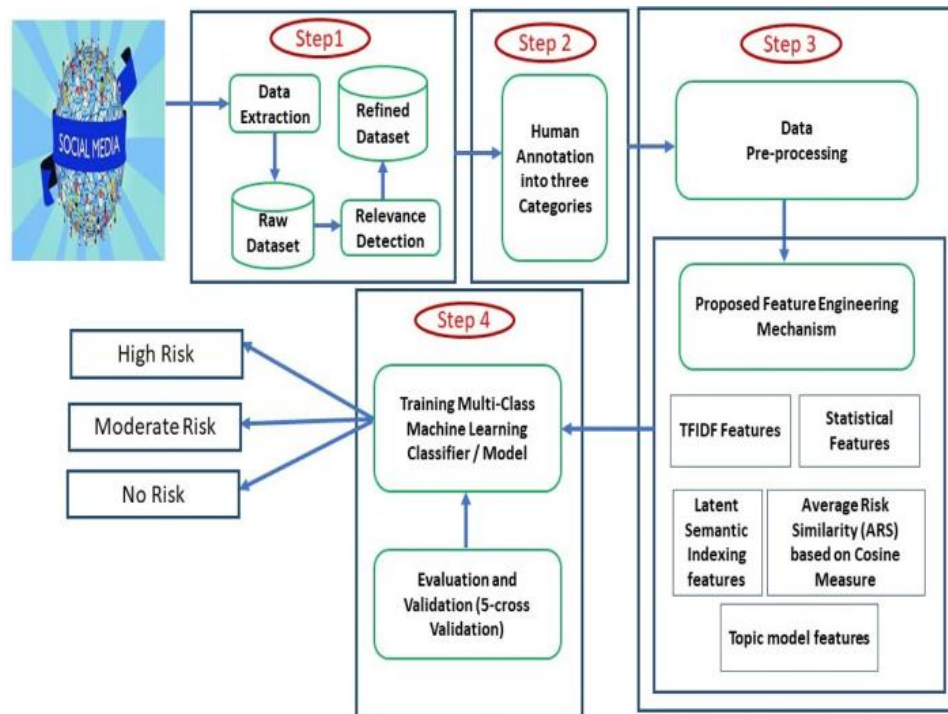


Figure 8. Proposed methodology for suicidality detection, [15]

In the first step, relevant data is extracted from social networking sites. The second step is about annotating the posts into three levels of risk based upon the annotation scheme that is devised in consultation with mental health experts. The third step involves preprocessing the posts to remove irrelevant and redundant information and proposing the feature extraction mechanism to extract the relevant features for training the multi-class machine learning model. The fourth step is about classifying the posts and evaluating the model using different metrics. The study demonstrates significant improvements compared to previous research that used classical feature extraction mechanisms. The best performing algorithm, Extreme Gradient Boosting, achieved an

overall accuracy of 96.33%. The different machine learning approaches are summarized in Table 1.

Table 1. Summary of machine learning approaches in analysis of social media

Study	ML Approach	Accuracy (%)	Limitations
Khan et al, 2022, [11]	DNN	97	- Model needs to be trained on a larger dataset for aggression detection. - Dataset wasn't expanded to other languages.
Murshed et al, 2022, [12]	DEA-RNN	90.45	- Analysis was performed only on the content of the tweet and not considering the users behavior. - Cyberbullying is detected only in the textual contents of the tweets, whereas image, video and sound are other types of media. - The study was limited only to the Twitter dataset exclusively.
Neelakandan et al, 2022, [13]	FSSDL-CBDC	99.983	The performance of the FSSDL-CBDC technique can be enhanced by including outlier identification and feature reduction techniques in the algorithm.
Chaudhary et al, 2021, [14]	Decision Tree	98	The model can't be used on a daily basis consumer data
Rabani et al, 2023, [15]	SVM, RF, XGB	96.33	The work is not extended to photos and videos related to suicidal ideation

b) Adversarial Machine Learning Approaches

Adversarial machine learning (AML) is a powerful technique for analyzing social media data. It applies a range of methods to detect, classify and analyze behavioral patterns from text and images online, allowing for improved understanding and targeting of user behaviors. To guard against attacks on hate speech detection models, several interesting defense strategies have been introduced in literature. The majority of the proposed methods rely on adversarial training to cope with the perturbations. For instance, Tran et al [16] used adversarial training to guard against Fast Gradient Method (FGM) attack by extending FGM with a learnable and fine-grained noise magnitude where noise (perturbation) is added to misleading samples. The model they created for hate speech detection understands the language of the hate speech datasets better, is 4-5 times faster than the traditional BERT model, uses less than 1/3 of the memory and has a better classification performance.

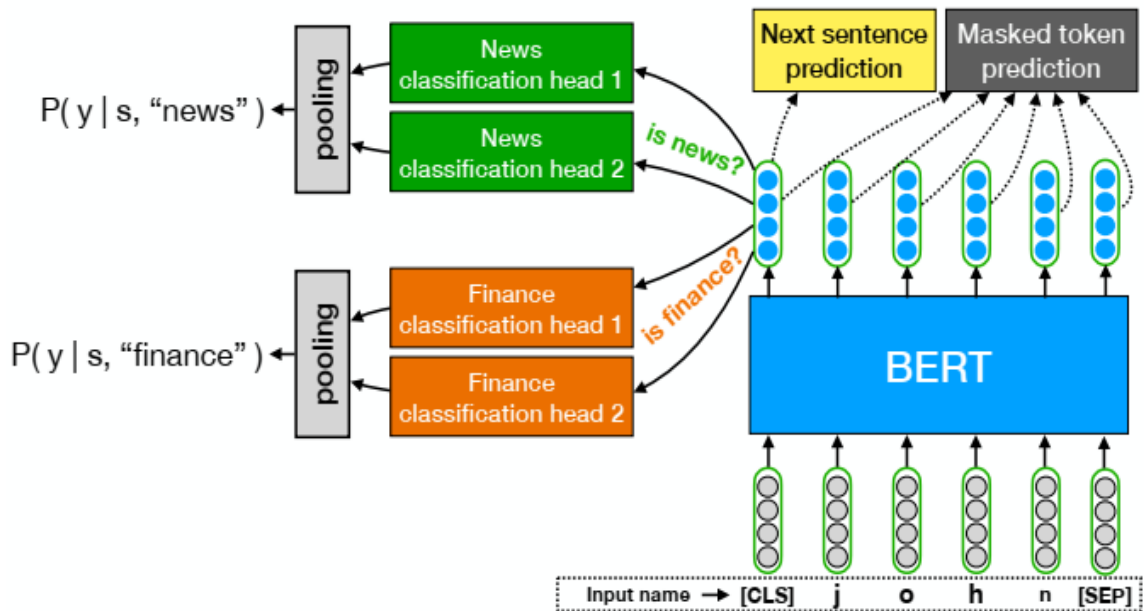


Figure 9. HABERTOR with two sources and ensemble of two heads, [16]

As shown in Figure 9, the HABERTOR has separated classification heads for different input sequences of different sources but with a shared language understanding

knowledge. Intuitively, instead of measuring the same probabilities $P(y|s)$ for all input sequences, it injects additional prior source knowledge of the input sequences to measure $P(y|s, \text{"news"})$ or $P(y|s, \text{"finance"})$.

Beyond approaches like that of Tran et al. [16], which enhance model robustness through improved adversarial training mechanisms, other researchers have explored alternative defense strategies grounded in model behavior and interpretability. For instance, Wang et al, [9] introduce a novel adversarial defense tool known as *TextFirewall*, specifically designed to enhance the robustness of sentiment analysis algorithms against adversarial text inputs. Unlike traditional methods that rely heavily on retraining models or modifying their internal structures, *Textfirewall* operates as an external tool that can be deployed without altering the architecture of the target model. Its primary mechanism is based on detecting inconsistencies between the sentiment predicted by the model and a computed impact value, which quantifies the contribution of individual words to the overall sentiment polarity.

TextFirewall effectively identifies adversarial texts generated by the three state-of-the-art attacks and outperforms previous defense techniques. Specifically, *TextFirewall* achieves an average accuracy of 90.7% on IMDB dataset and 96.9% on Yelp dataset in defending the three SOTA attacks. It could be deployed as a third-party tool without modifying the target model and agnostic to the specific type of adversarial texts. The diagram in Figure 10, shows the framework of TextFirewall. The sizes of yellow and gray areas in a consistent match respectively indicate the proportion of positive and negative polarity in an input. The yellow area is larger than the other, so the sentence is

recognized as positive by the model [9], which is different from the prediction of the target model.

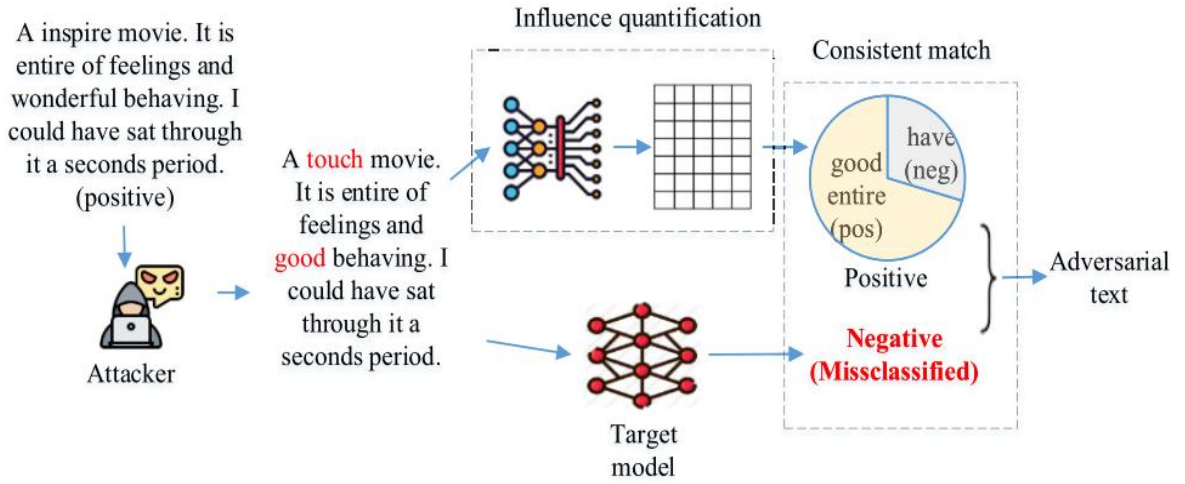


Figure 10. The Framework of TextFirewall, [9]

While many adversarial techniques aim to defend models against malicious perturbations, some researches have explored the use of adversarial strategies for socially beneficial applications. Li et al [17], provide a compelling example of how adversarial machine learning techniques can be harnessed not only for defense against malicious attacks but also for advancing socially beneficial goals. They developed a novel framework Adversarial Attacks for Social Good (Adv4SG) which reimagines adversarial attacks as a tool for protecting user privacy in natural language processing applications.

Specifically, they address the challenge of safeguarding personal attribute privacy in social media contexts by formulating the problem as an adversarial attack on NLP-based attribute inference systems. These systems typically analyze users' text data to infer sensitive personal characteristics such as gender, age, or political orientation. To counter this, Adv4SG generates subtle, semantically-preserving word-level perturbations in the original input text. These modifications are carefully constrained to retain the overall

meaning and grammatical integrity of the text, while reducing the effectiveness of attribute inference models. As a result, the perturbed inputs mislead the inference model, preventing accurate identification of targeted personal attributes. In doing so, Li et al. present an innovative perspective on adversarial methods—employing them not as threats, but as protective tools to enhance privacy in publicly shared text data. The diagram in Figure 11 illustrates the overview of Adv4SG.

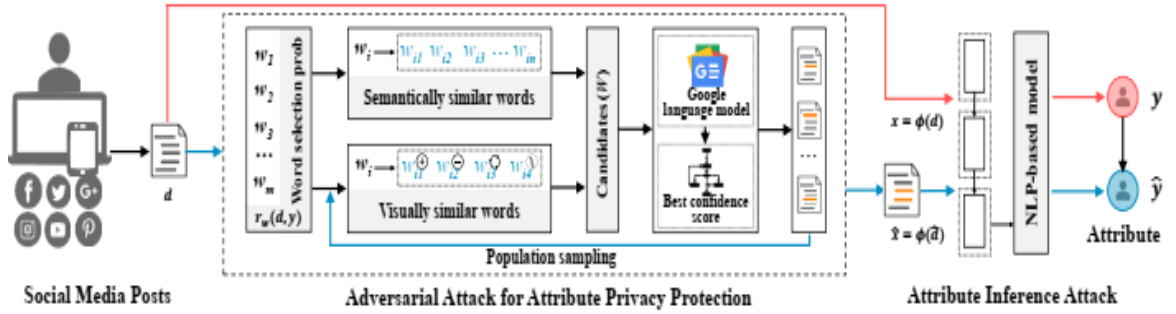


Figure 11. Adv4SG: Adversarial Text Attacks for Privacy Protection, [17]

Adv4SG directly perturb the tokens in the text with guidance towards the misclassification of the target attribute through a self-trained NLP model, where the end-to-end learnability constraint and the black-box setting are naturally satisfied. They focused in perturbing the texts at word level because the implicit information of the texts can be better encoded from the latent representations using word embedding than characters and also because the search space of possible changes over words is much smaller than characters, such that word-level perturbation is significantly more computationally tractable than character-level perturbation. They achieved a success rate of 97.40% on Twitter-location, 74.03% on Twitter-gender, 82.28% on Blog-gender, 88.61% on Blog-age inference task.

Complementing such privacy-centered adversarial methods, Amin et al [18], proposed to increase NLP model's resilience against adversarial attacks by developing an

adversarial attack strategy with a larger attack duration and better performance. They went on to first train a CNN using the IMDB dataset, which is made up of labeled movie reviews that represent positive and negative sentiments on movie reviews. Following the CNN model training, they generated adversarial samples from the IMDB dataset using the FGSM attack and obtained 13,200 samples out of the 15,000 chosen samples. They trained an LSTM model on both the original IMDB data and the adversarial examples generated using FGSM such that the model can learn to distinguish between the original data and the adversarial examples, thereby accurately identifying adversarial examples attacks while avoiding false positives. The diagram in Figure 12 describes the model diagram for their approach.

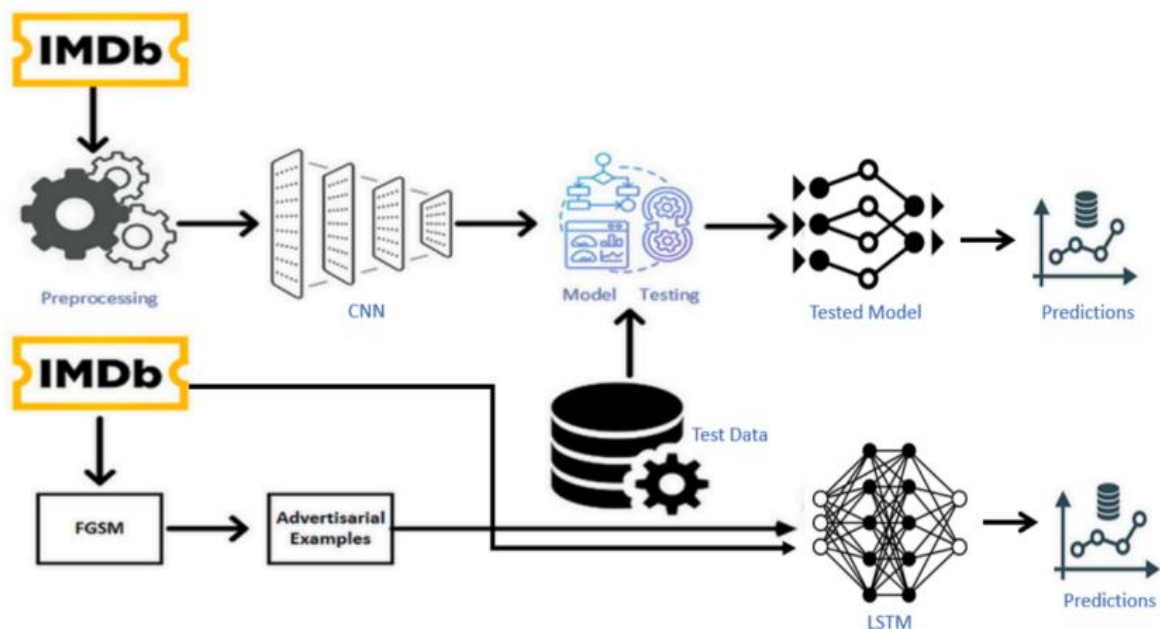


Figure 12. Model diagram, [18]

They obtained an accuracy of 97% for the CNN in classifying reviews, whereas the LSTM model demonstrated an accuracy of 96% in identifying hostile assaults. From the results obtained, they implied that the model's resilience and ability to identify

adversarial assaults can be enhanced by including adversarial samples in the original dataset. The studies reviewed in this section is summarized in Table 2.

Table 2. Summary of adversarial methods in social media analysis

Study	Method	Attack	Defense	Accuracy (%)
Tran et al, 2020, [16]	BERT	Fast Gradient Method (FGM) word and word boundary	Adversarial training	94.77 on Yahoo dataset 94.12 on Yahoo News dataset 95.72 on Yahoo Finance dataset
Wang et al, 2021, [9]	DNNs	Deepwordbug, GA, PWWS	Considers inconsistency between the model's output and the impact value	90.7 on IMDB dataset 96.9 on Yelp dataset
Li et al, 2023, [17]	NLP	Inference attacks	Adversarial for social good (Adv4SG)	97.40 on Twitter-location 74.03 on Twitter-gender 82.28 on Blog-gender 88.61 on Blog-age
Amin et al, 2024, [18]	NLP, LSTM	Fast Gradient Sign Method (FGSM)	Adversarial training	96 on IMDB dataset

2.1.2. Multimodal Deep Learning Frameworks

Multimodal learning [19] is a framework for building models that make predictions based on different types of modalities. Important challenges in multimodal learning are the inference of shared representations from arbitrary modalities and cross-modal generation via these representations. However, this requires taking into account the heterogenous nature of multimodal data. Most of the Deep Multimodal Models (DMMs) that have achieved impressive performances can be categorized as traditional fusion-based models and transformer based fusion models.

Given the growing interest in applying DMMs to real-world tasks, researchers have sought innovative strategies to address the fusion of heterogeneous data sources in socially impactful applications. One such effort is by Kumari et al [20], who developed a multimodal neural network tailored for hate speech detection on social media. Their approach exemplifies how traditional fusion-based techniques can be applied and extended to address nuanced societal challenges. A fine-tuned VGG16 model was employed to extract visual features from images, and a 1-D convolutional neural network to extract features from the text contents. These features were then optimized by a genetic algorithm to select the better performing subset of features for further model processing. Despite these strengths, the textual context information was not considered to extract during the model training. The diagram in Figure 13 shows an overview of the approach. They achieved an F1-score of 78% which 9% improvement over earlier reported results.

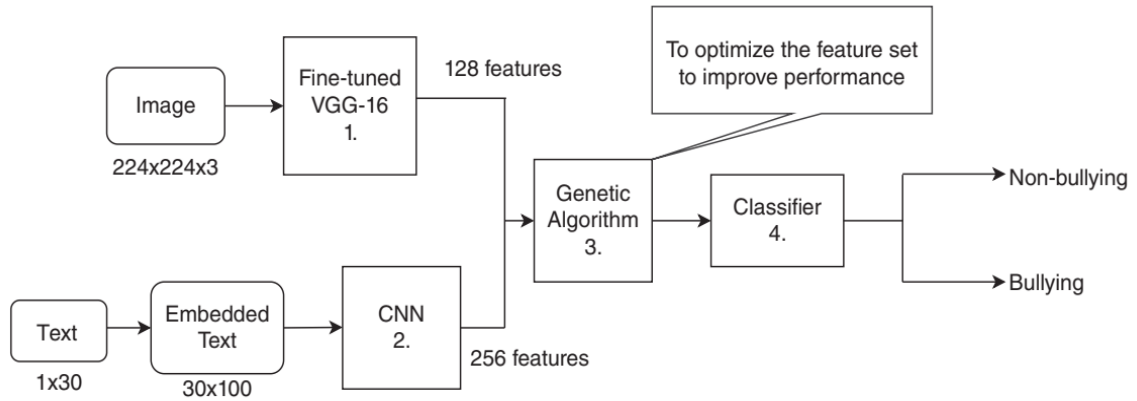


Figure 13. Overview of the Approach proposed by Kumari et al, [20]

Subsequent studies have explored more refined attention mechanisms to further improve the interpretability and accuracy of multimodal models in related domains. In this regard, Zuhui et al [21], proposed a deep learning network that leverages both visual and textual information. They built a new semantic and task-level attention mechanism to help their model focus on the essential contents of a post that signals antivaccine messages. The proposed model consists of three branches as shown in Figure 14, and can generate comprehensive fused features for predictions. To further enhance the classification performance of their model, they incorporated an ensemble learning strategy that integrates outputs from multiple classifiers to improve robustness and generalization.

They evaluated the model's performance on a real-world social media dataset that consists of more than 30,000 samples collected from Instagram between January 2016 to October 2019. Their final network achieves above 97% testing accuracy and outperforms other relevant models, demonstrating that it can detect a large number of antivaccine messages posted daily. The model's strong accuracy shows its effectiveness in recognizing subtle, evolving multimodal signals in anti-vaccine content.

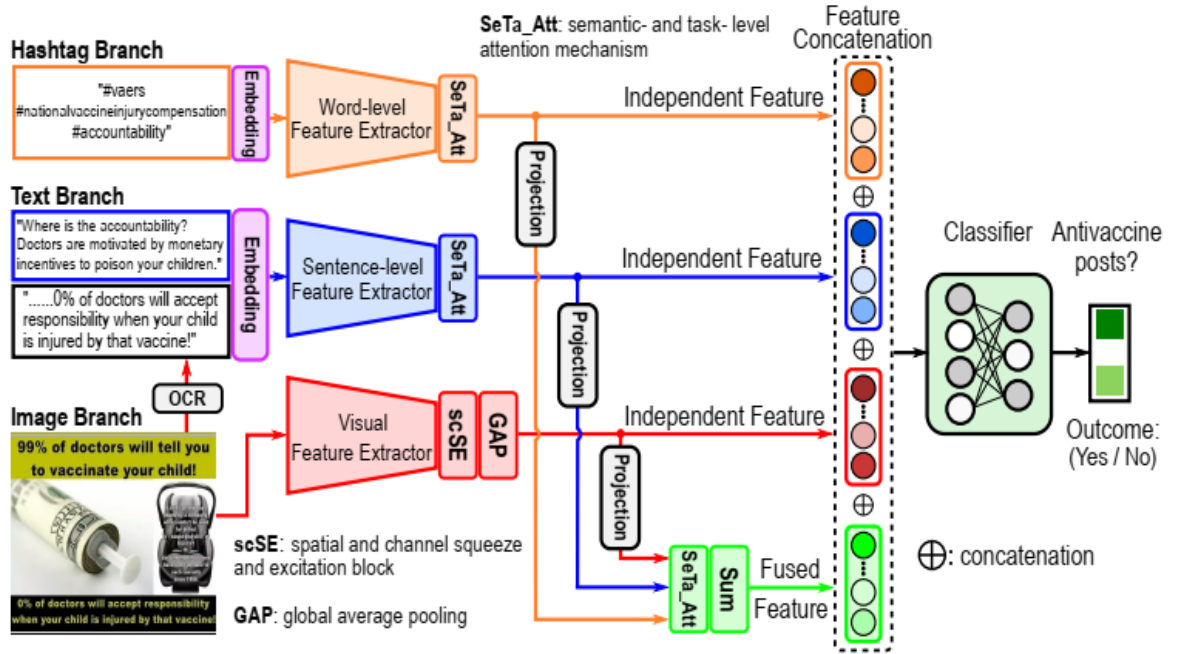


Figure 14. Multimodal Architecture for Antivaccine Message Analysis, [21]

Expanding on the use of attention mechanisms and multimodal fusion in content analysis, recent work has also explored ensemble-based strategies to further enhance predictive performance. Hossain et al [22], proposed a comprehensive framework that utilizes the weighted ensemble technique to assign weights to the participating visual, textual and multimodal models for analyzing memes. They employed the state of the art visual (i.e., VGG19, VGG16, ResNet50) and textual (i.e., BERT, XLM-R, Distill-BERT) models to make the constituent modules of the framework. Moreover, they used early and late fusion to combine the visual and textual features to develop the multimodal models. The diagram in Figure 15 shows an abstract view of the multimodal model proposed. After analyzing all models' performance on the two datasets, this work proposed a weighted ensemble technique for classifying memes. The proposed technique can readdress the softmax probabilities of the participating models based on their previous outcomes on the datasets. The experimented results revealed that the proposed technique outdoes the unimodal (i.e., image, text), multimodal, and average

ensemble models by obtaining the highest weighted f1 score of 66.73% (MultiOFF dataset) and 58.59% (TamilMemes dataset), respectively. Moreover, the comparative analysis indicated that the proposed technique outcomes are approximately 13% (in ‘MultiOFF’) and 1.69% (in ‘TamilMemes’) ahead compared to the current state of the art systems. Thus, results ensured the effectiveness of the proposed technique in detecting offensive and troll memes based on multimodal information.

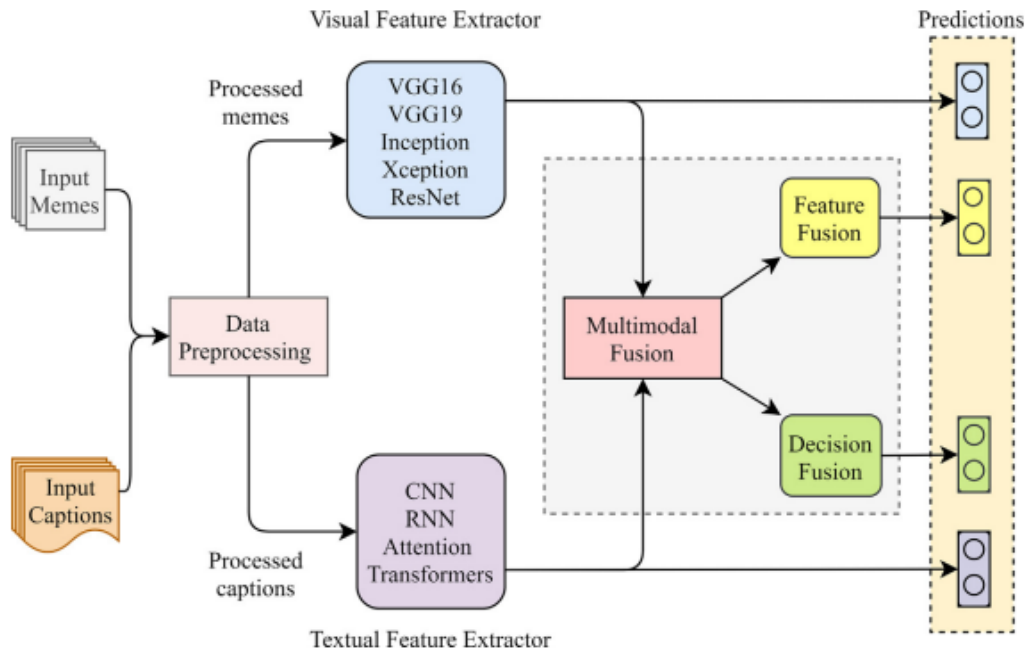


Figure 15. Abstract view of multimodal offense and troll detection system, [22]

Extending beyond the fusion-based framework proposed by authors in [22] for meme analysis, Singh et al [23], went on to investigate the detection of hate speech in multimodal data that comprised of text-embedded images using advanced deep learning models. They conducted experiments using four state-of-the-art classifiers: XLM-Roberta-base, BiLSTM, XLNet base cased and AL-BERT, on the CrisisHateMM dataset. The details of the process flow of their hate speech detection and target identification processes are shown in Figure 16. A preprocessing step is performed before model input to prepare the textual data. They fine-tuned the four models on the

training set to enhance hate speech identification, then they demonstrated the effectiveness of the proposed approach in hate speech detection and target identification by generating testing predictions using the fine-tuned models.

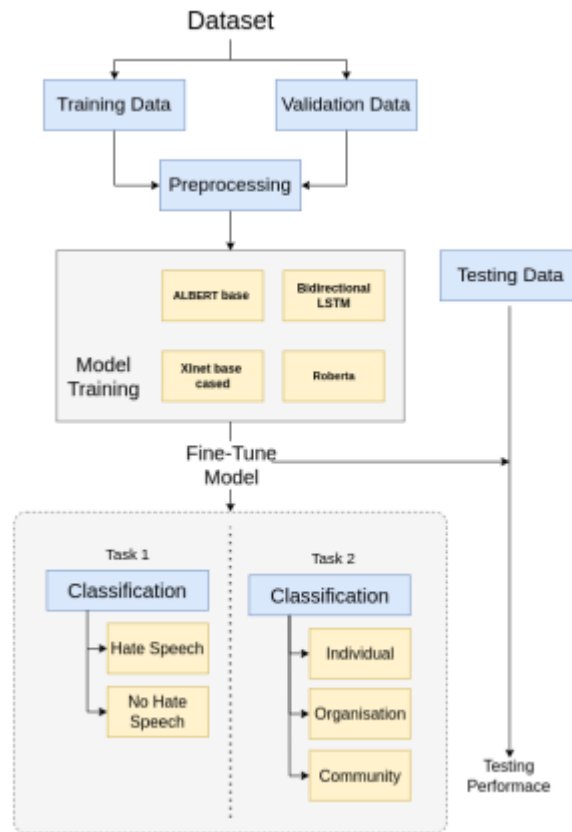


Figure 16. Hate Speech Detection and Target Identification Flow, [23]

From the results obtained, they found out that XLM-Roberta-base exhibits superior performance, outperforming the other classifiers across all evaluation metrics, including an impressive F1 score of 84.62%. Their study emphasizes the importance of integrating advanced deep learning models to improve detection accuracy of the system, particularly when analyzing diverse content types such as text, images, and videos. The techniques discussed in this section are summarized in Table 3.

Table 3. Summary of multimodal techniques for analysis of social media messages

Study	Technique	Accuracy (%)	Limitations
Kumari et al, 2020, [20]	VGG16-CNN	78	• Components of the posts such as user's information, network information were not considered. • Multi-lingual comments could be used with images to extract better features • Explore other optimization algorithms for better feature engineering as this can improve overall accuracy of cyberbullying post detection
Zuhui et al, 2021, [21]	DNN	97	• Proposed model is inaccurate when applied to antivaccine post without sufficient context information. • More memory cells can be added to force the model to remember essential features during the model training phase.
Hossain et al, 2022, [22]	VGG19-DistillBERT	66.3	• Explore visual attention and transformers to capture strong visual and textual features.
Singh et al, 2023, [23]	XLNet-Roberta-base	84.63	• Proposed method doesn't address potential biases, promoting fairness and safeguarding user rights. • There's a need to develop more robust and generalized hate speech detection solutions

2.1.3. Adversarial Attacks

Adversarial attacks are designed to deceive AI systems and simulate the malicious users attack action by adding adversarial pixels to images or replacing words and characters in the text [24] [25]. Recent research [26] found that the DNNs have a fatal flaw: a small amount of specially designed perturbations will fool the model, while these perturbations are typically invisible or visible but inconspicuous. These perturbed inputs called adversarial examples/samples, enable attackers to disrupt the expected behavior of models, leading to undesirable consequences and security risks for these systems deployed in the real world, especially in areas that are safety-oriented. In Figure 10, we can have an example of adversarial attacks on text by replacing words in a text, and it demonstrates how this replacement affects the prediction of the classifier. In Figure 17, how adversarial attacks are performed on an image is described.

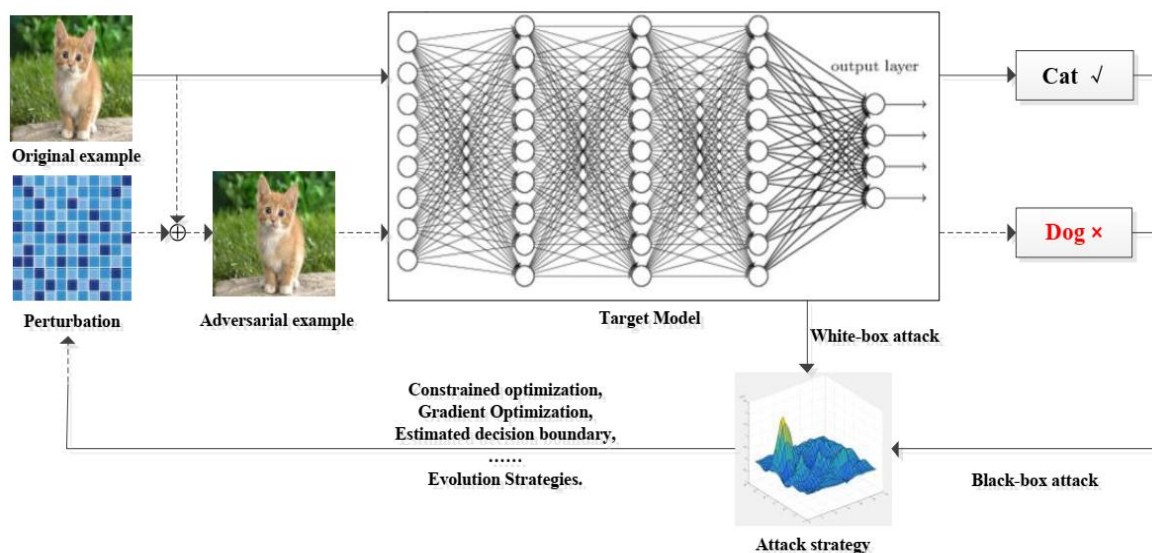


Figure 17. Adversarial attack on image, [25]

As shown in Figure 17, after adding an adversarial perturbation to the original example to form an adversarial example, the target model incorrectly identifies “Cat” as “Dog”, and the human eye can’t detect the difference between the original example and the adversarial example. The white-box attack can obtain the structural information of the

model, while the black-box attack can only obtain the output of the target model by inputting the original data to the target model. The attack strategy will be introduced together with the actual attack method.

a) Adversarial Attacks on Images

Adversarial attacks originated in the field of computer vision. The large BFGS (L-BFGS) method proposed by Szegedy et al (as cited in reference [23] in Chen et al [24]) solved the optimization problem of misleading the model for the adversarial examples of the image classification task. L-BFGS was effective but the computational task was high, this inspired Goodfellow et al (as cited in reference [8] in Chen et al [24]) to propose a simpler solution namely fast gradient sign method (FGSM). This method settled the perturbation as the product of the gradient sign and the step size, which increased the loss of the model. The Jacobian-based saliency map attack (JSMA) proposed by Papernot et al (as cited in reference [24] in Chen et al [24]) used the Jacobian matrix of the neural model to evaluate the output sensitivity of the neural model to each input component and gave greater control to the adversarial examples under the given perturbation.

While early adversarial attack methods in computer vision focused on adding perturbations to clean inputs to mislead neural networks, more recent approaches have explored alternative paradigms for generating adversarial examples. In this context, Duan et al [27] introduced AdvDrop, a novel technique that deviates from the traditional additive strategy by instead crafting adversarial examples through the selective removal of existing image information. They demonstrated the effectiveness of AdvDrop by extensive experiments and showed that this type of adversarial example is more difficult to defend by current defense systems.

According to the goal of the attack, we can distinguish between white-box attacks and black-box attacks. White-box attacks assume that the attacker has full access and knowledge of the model, such as its architecture, parameters and gradients. Black-box attacks assume that the attacker has limited or no access to the model and can only query its input-output behavior. According to the method of the attack, we can distinguish between gradient-based attacks and score-based attacks. Gradient-based attacks use the gradient information of the model to generate adversarial perturbations that maximize the loss or minimize the confidence of the model. Score-based attacks use the output scores of the model to generate adversarial perturbations that change the predicted class or rank of the input.

Distance Metrics are used to measure the distance between two points. To compute the distance between two vectors, the norm of the difference between those two vectors needs to be evaluated using a norm function. A norm $\mathbf{L}_p$ is a function that measures the magnitude of a vector. Formally, the norm function $\mathbf{L}_p$ of x can be defined as:

$$\|x\|_p = (\sum \|x_i\|^p)^{1/p}, \exists: p \in \mathbb{R}, p \geq 1 \quad (2.1)$$

The distance metrics are used within the process of generating the adversarial attacks by quantifying their similarities. L_0 , L_1 , L_2 , and L_∞ distance metrics have been widely adopted by state-of-the-art adversarial attack algorithms. Perturbation norms are measures of the magnitude or size of the changes made to the input data. we have the following norms [28]:

- **$\mathbf{l}_0$ -norm attacks:** L_0 -norm attacks are a type of adversarial attack that aim to manipulate input data by changing the smallest possible number of features, typically pixels in image data or words in text, while still causing the model to

make incorrect predictions. The L_0 distance is used to calculate the vector size by measuring the total number of the non-zero elements of a given vector. These attacks minimize the number of pixels that are modified in the input data. L_0 can be defined as:

$$\|x\|_0 = (i|x_i \neq 0) \quad (2.2)$$

- **l_2 -norm attacks:** In the context of adversarial attacks, l_2 -norm attacks quantify the overall magnitude of change between the original input and its adversarially perturbed version. The L_2 distance, also known as the Euclidean distance is widely used in machine learning. L_2 measures the distance between the class labels of the original input data and the perturbed data. These attacks minimize the Euclidean distance between the original and the perturbed input data, thereby ensuring that the adversarial example remains visually or semantically similar to the original input. This makes the perturbations more difficult to detect, both for human observers and automated defense systems. L_2 can be defined as:

$$\|x\|_2 = \left(\sum_{i=1}^n \|x_i\|^2 \right)^{1/2} \quad (2.3)$$

- **l_∞ -norm attacks:** The L_∞ distance, also known as the “max norm”, returns the maximum magnitude of the difference between each element in a vector. These attacks minimize the maximum absolute difference between the original and the perturbed input data. It is used as a sufficient constraint over the size of perturbations that could be added to generate the perturbed image. L_∞ can be defined as:

$$\|(x - x')_\infty\| = \max(|x_1 - x'_1|, \dots, |x_n - x'_n|) \quad (2.4)$$

Table 4 summarizes common image-based adversarial attacks by goal and method.

Table 4. Summary of typical attacks on images and their principles

Adversarial Attack	Attack Type	Perturbation Norm	Attack Strategy
Fast Gradient Sign Method (FGSM) Goodfellow et al, (2015)	White-box	l_∞	Gradient-based
Carlini and Wagner (C&W) Carlini et al, (2017)	White-box	$l_0l_2l_\infty$	Constrained optimization
Projected Gradient Descent (PGD) Madry et al, (2017)	White-box	l_2l_∞	Iterative Gradient-based
One-Pixel Su et al, (2019)	Black-box	l_0	Estimated decision boundary
Zeroth Order Optimization (ZOO) Chen et al, (2017)	Black-box	l_2	Migration mechanism
Universal Adversarial Perturbations (UAP) Moosavi et al, (2016)	White-box	l_2l_∞	Gradient optimization
Backward Pass Differentiable Approximation (BPDA) Athalye et al, (2018)	Black-box	l_2l_∞	Gradient approximation
Generative Adversarial Perturbations (GAP++) Mao et al, (2020)	White-box	$l_0l_2l_\infty$	Generative model
Standard Evolution Strategies (CG-ES) Feng et al, (2020)	Black-box	$l_0l_2l_\infty$	Evolution strategies
Adversarial Drop (AdvDrop) Duan et al, (2021)	White-box	l_2l_∞	Constrained optimization

b) Adversarial Attacks on Text

Due to the inherent differences between visual and textual data, countermeasures for images can't be directly applied to text data. Ebrahimi et al (as cited in reference [26] in Chen et al [24]) proposed a character-level attack method HotFlip, which used the directional derivative represented by one-hot input to estimate which character to replace, and combined beam search to find the right combination of character changes. Jia and Liang (as cited in reference [27] in Chen et al [24]) generated adversarial examples by adding some meaningless sentences at the end of the paragraph. Gao et al. (as cited in reference [28] in Chen et al [24]) proposed DeepWordBug generate adversarial examples against recurrent neural network (RNN) models, which used a scoring function to calculate the importance of words in a sequence under a black-box scenario and character-level modifications to make spelling mistakes. Since spelling errors were easy to detect and correct, Jin et al. (as cited in reference [29] in Chen et al [24]) proposed a black-box attack method TextFooler, which performed synonym substitution for important words and checked the semantic similarity of sentences to fool the system.

Building on earlier character-based and synonym-based attacks that target RNNs or simpler NLP models, more recent work has extended these techniques to target transformer-based architectures, which underpin many state-of-the-art language models. In this context, Guo et al [29], introduced the first general purpose gradient-based attack against transformer models. Instead of searching for a single adversarial example, they searched for a distribution of adversarial examples parameterized by a continuous valued matrix, enabling gradient-based optimization. They empirically showed that their white-box attack attains state-of-the-art attack performance on a variety of natural

language tasks, outperforming prior work in terms of adversarial success rate matching imperceptibility as per automated and human evaluation. Their framework, gradient-based distributional attack (GBDA) consists of two key components that circumvent the difficulties of gradient descent for discrete data under perceptibility constraints. First, instead of constructing an adversarial example, they searched for an adversarial distribution. Second, they enforced perceptibility and fluency using BERTScore and language model perplexity, respectively, both of which are differentiable and can be added to the objective function as soft constraints.

While Guo et al.’s gradient-based approach achieves strong performance in a white-box setting by leveraging full access to the model gradients, there remains a growing interest in developing effective black-box attack strategies that don’t require internal model information. In this direction, Chang et al [30], proposed a new adaptive black-box text adversarial example generation scheme, TextGuise. They designed a keyword selection method in which word scores are calculated by combining context semantics to select the appropriate keywords to modify. To maintain semantics, new keyword substitution rules are designed in combination with the characteristics of text and popular text expressions. The best modification strategy is adaptively selected through a query model to reduce the magnitudes of disturbances. TextGuise can automatically select replacement strategies that efficiently generate adversarial examples with good readability for various text classification tasks. They showed that attack experiments with TextGuise yield high attack success rates that can surpass 80% when the perturbation ratio does not exceed 0.2. A summary of adversarial attacks on text is shown in Table 5.

Table 5. Dichotomy of attacks on text

Adversarial attack	Attack type	Perturbation norm	Algorithm
HotFlip Ebrahimi et al, (2018)	White-box	Letters	Greedy
Jia&Liang Jia et Liang, (2017)	Black-box	Letters	Greedy
Gradient-Based Distributional Attack Guo et al, (2021)	White-box	Tokens	Gradient-based
TextGuise (Chang et al, 2023)	Black-box	words	Adaptive example generation scheme

2.1.4. Adversarial Training

Adversarial machine learning is an active research area. Algorithms are developed to either attack existing models by constructing adversarial examples, or train robust models to defend against adversarial attacks. Among existing defense approaches, adversarial training is a general strategy to empower models with state-of-the-art robustness in different settings. Existing research [31] mostly focuses on adversarial training for image classification, and the general notion is that robustness is often at odds with accuracy. Due to the growing dominance of large-scale pre-training, very few recent works started to explore adversarial training in the pre-training stage.

Adversarial training is a data optimization approach to defend deep neural networks against adversarial attacks. At present [25], adversarial training is one of the main defense methods for adversarial examples, lack research results on large-scale datasets, and various defense measures on ImageNet will be broken by specific white-box attacks. Xie et al (as cited in reference [64] in [31]) shows that model accuracy on clean images can be improved if a separate auxiliary batch norm is used for adversarial

examples. Parallely, there exist some studies on applying adversarial training to language modelling (as cited in reference [61] in [31]) and natural language understanding (as cited in reference [17],[8],[34] in [31]). Figure 18, shows a diagram of a conventional adversarial training process [32], it uses hand-crafted attack strategy to generate adversarial examples (AE).

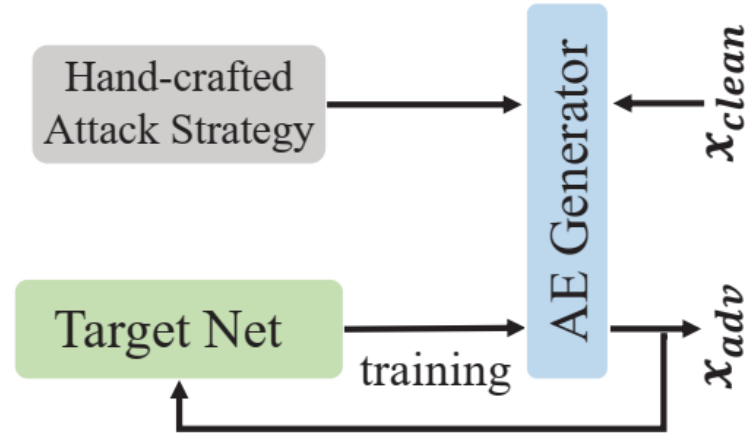


Figure 18. Adversarial training, [32]

a) Multimodal Adversarial Training

As multimodal learning systems become increasingly prominent, particularly those integrating vision and language, ensuring their robustness against adversarial attacks is very important. While most adversarial training methods target unimodal data, Gan et al [31], did the first known effort on large-scale adversarial training for vision-and-language (V+L) representation learning. Instead of adding adversarial perturbations on image pixels and textual tokens, they proposed to perform adversarial training in the embedding space of each modality. To enable a large-scale training, they adopted the ‘free’ adversarial training strategy and combined it with KL-divergence based regularization to promote higher invariance in the embedding space. Figure 19 shows an overview of their proposed framework.

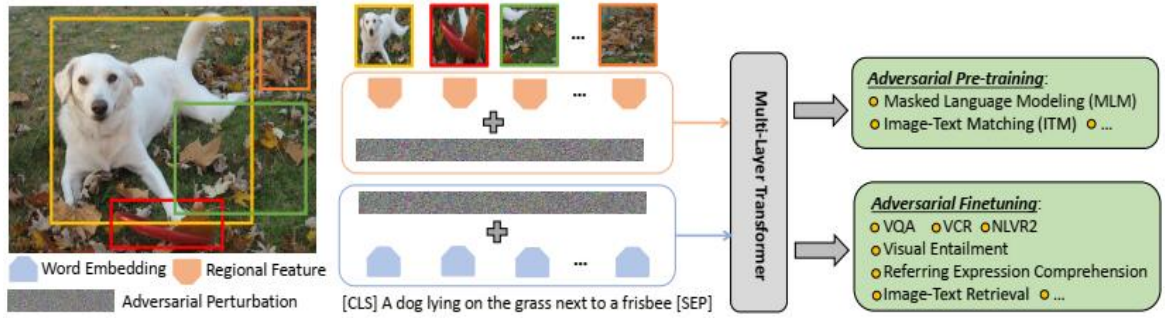


Figure 19. VILLA: Vision-Language representation learning framework, [31]

They used Auto Attack (AA) to evaluate the robustness of their model, since it is a reliable and strong attack method that consists of three white-box attacks and a black-box attack. Their model achieved 57% test robustness on the CIFAR-10 database. Also, the model achieved new state-of the art on a wide range of tasks, including visual question answering, visual commonsense reasoning, image-text retrieval, referring expression comprehension and visual entailment.

Beyond improving general robustness in multimodal models, adversarial training has also been applied to domain-specific tasks, such as social media analysis for crisis response. Gao et al [33], proposed a multimodal adversarial neural network (MANN) to detect disaster related tweets on social media. MANN consists of three modules as shown in figure 20. They defined adversarial training as a min-max game between the feature extraction module and the disaster discrimination module. On one hand, the feature extraction module cooperates with the tweet detection module to minimize loss and improve performance of detecting disaster-related tweets. On the other hand, the feature extraction module tries to fool the disaster discrimination module to maximize loss and learn disaster-invariant feature representations. They used a labeled dataset CrisisMMD as positive example, which contains tweets with both textual messages and

associated images. As for the negative examples, they collected English tweets containing pictures which were labeled by human annotators as irrelevant to disaster events. The proposed MANN achieves of accuracy of about 89%.

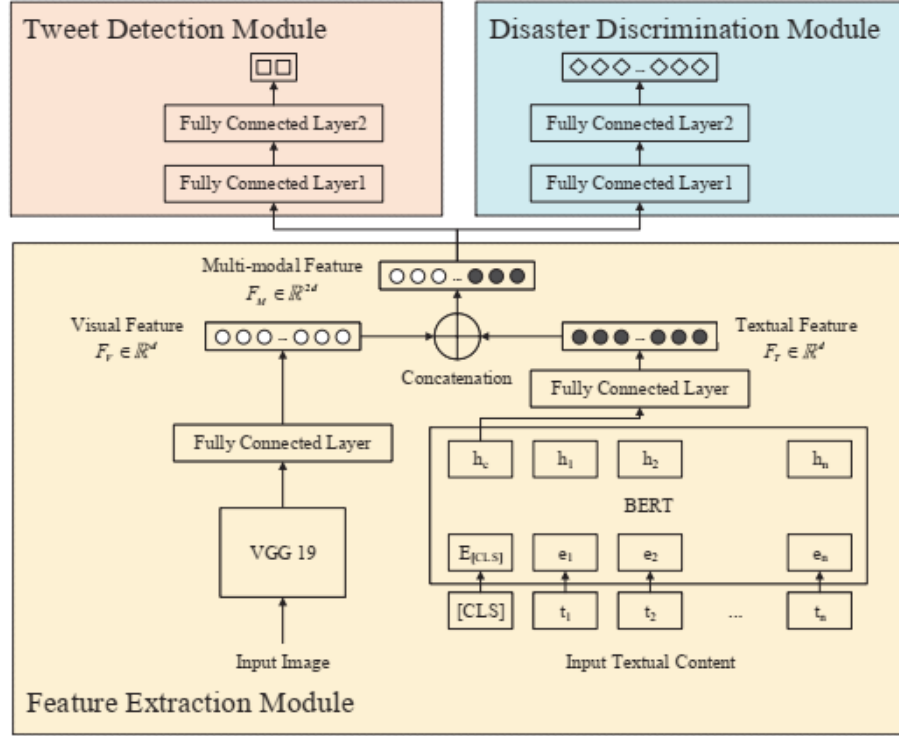


Figure 20. The overall architecture of MANN, [33]

From Figure 19, the feature extraction module consists of visual features and textual features. To extract visual features the VGG-19 network was used and to learn textual features from posts, the BERT network was employed. The tweet detection module is used to predict whether tweets are related to disasters and the disaster discrimination module is used to classify tweets as one of N disaster events.

In addition to developing robust multimodal models, understanding their vulnerabilities to adversarial attacks is crucial, especially in sensitive applications. Aggarwal et al [34], conducted a use case study to analyze the vulnerabilities of existing hateful meme detection systems against external adversarial attacks. They proposed nine different attacks to investigate the vulnerabilities of the existing hate meme detection systems to

human-induced adversarial attacks and found out that all of these models are highly vulnerable with a drop of as high 10% in macro-F1 performance in certain cases. They developed two different countermeasures to tackle such adversarial inputs namely an approach based on contrastive learning and the other approach based on adversarial training. In the former case, a function learns additional signals from simple augmentations of each data point in the training set and that is blended with hate meme detection model's objectives through a suitable loss function thus increasing the overall generalization capabilities of the model. In the latter case, in addition to available data points, a model is trained on adversarial examples that are generated by perturbing one modality of the input data while keeping another one unchanged at a time. For the image modality, they consider image-region features rather than perturbing pixels.

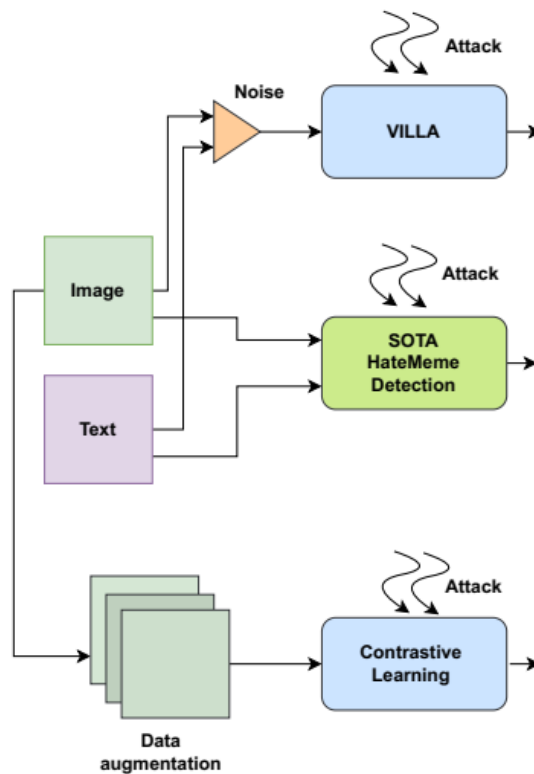


Figure 21. A schematic showing the overall setup of experiments, [34]

They specifically focused on model-independent attacks, these types of attacks are particularly concerning because they can be crafted and executed by individuals with little technical expertise. The study aimed to simulate realistic scenarios in which malicious actors could exploit plausible weaknesses in hate speech detection systems. Through their analysis, the researchers identified several loopholes in the current detection pipelines that could be manipulated to evade automated moderation. This includes subtle changes to text, images, or their combinations that can bypass content filters while retaining harmful intent. The overall experimental setup, is visually summarized in Figure 21. Table 6 shows a summary of approaches with multimodal adversarial training.

Table 6. Summary of approaches with multimodal adversarial training

STUDY	TECHNIQUE	ROBUSTNESS ACCURACY	LIMITATIONS
Gan et al, 2020 [31]	They incorporated adversarial training in both the pretraining and fine-tuning stages.	57%	Computationally expensive due to large-scale training.
Gao et al, 2020 [33]	Adversarial training is a minimax game between the feature extraction module and the disaster discrimination module	89%	They don't apply it to videos of a post
Aggarwal et al, 2023 [34]	Attempt to boost model's robustness using contrastive learning as well as an adversarial training-based method	63.10%	- Small set of attack schemes - Identify countermeasure approaches for datasets where the original images are themselves of very low quality

2.2. Research Frontiers

In this section, we discuss the current challenges and limitations of defending multi-modal (text-image) systems against adversarial attacks in social media applications. Compared with the wide range of existing adversarial attack methods and their various effects [35], there is relatively little existing work on defense strategies. For defenders, the task of completely addressing the vulnerabilities of deep neural networks is as difficult as redesigning the entire network architecture itself.

- Evolving adversarial tactics and lack of new defense methods:** Robust encoding [35] is also a defensive approach that strengthens the input model. For text, although the existing spell checkers can preprocess the input, its ability to map the typos back to the original input is limited. The current immature development of robust encoding is an opportunity, considering the trend of NLP systems to generalize the capabilities of models as much as possible, rather than limiting them to a single task. Also, attacks like data poisoning and model inversion are becoming more sophisticated and adversaries can use generative models [36] (e.g., GANs) to create subtle perturbations that will evade detection while causing misclassification. Defenses like adversarial training to keep the pace, as retraining the models on new attack patterns is resource-intensive.
- Ineffectiveness to unknown attacks:** To defend against adversarial examples [25], adversarial training is a practical and feasible method. However, a key drawback of this approach is that they assume that the threat model is known, which is often difficult to implement in practice. This leads to this defense method being less effective in the face of unknown attacks.
- Low self-iteration capability:** The applicability and adaptability of adversarial example defense technology needs to be improved. Research [25] has proven that

even the defense technology with the best defense effect will be broken by an endless stream of adversarial attack technologies. How to improve the self-iteration capability of defense technology is an urgent problem to be solved.

- **Lack of robust benchmarks and datasets:** Given the fact that a variety of datasets is used in training, testing and performance evaluation [1], there is a need for standardizing benchmark datasets. Although several large-scale datasets have been collected for common NLP tasks in recent years, there remains a need for new datasets for more challenging social networks' NLP applications.
- **Challenges in interpretability and transparency:** Modern machine learning models, particularly deep models are highly complex and non-linear. While they achieve state-of-the-art performance, their complex nature [37] creates a black-box effect, making it difficult to understand how specific inputs lead to specific outputs. Also, unlike accuracy or precision, interpretability lacks a universal metric [37]. Different stakeholders interpret “interpretability” differently. Developers may want feature importance, Users may need natural-language explanations, and regulators may require transparent decision logs. This creates confusion and inconsistencies in designing and evaluating interpretable systems.

2.3. Partial Conclusion

A hate post on social media that is not effectively detected by online detectors can have a huge impact on the mental health of its users. In some severe cases can lead to suicide. On the other hand, a post containing viable information for the general public, that is flagged as a hate post by a detector can have a real impact on the amount of knowledge received by the general public and in severe cases lead to slow economic development.

This chapter reviews the existing literature on machine learning and adversarial machine learning techniques to analyze posts on social media. It provides an overview of the theoretical framework and previous research findings related to our topic. Several interesting techniques to detect hate posts were discussed and we will use the architecture proposed by Aggarwal et al., 2023 to build our robust multimodal hate post-detection system. This work has been chosen for implementation because of the accuracy achieved in a low-resource setting. We will train a classifier to detect if a post is related to hate in normal and adversarial settings. Adversarial examples will be crafted by applying some of the most pertinent state-of-the-art attacks we covered in this work for both text and image. The adversarial examples will be fed during training. The chapter concludes with a summary of the research gaps and limitations.

CHAPTER THREE

AdverMIRA: A Robust Multimodal Intelligence for Recognizing Abuse

3.1. Introduction

Misogyny is a form of hate against women that has been spreading exponentially through the web, especially on social media platforms. The cultural belief that considers women as inferior beings creates a severe problem. In general, [38] it has been associated with attitudes biased against women, such as male privilege, gender discrimination, sexual objectification, and verbal aggressiveness. These manifestations can affect women's self-esteem and can progressively damage their mental health and physical health, leading to suicide in severe cases. Hence the automatic detection of misogynistic content on these online platforms is of increasing concern.

Misogynistic content on social media is not only conveyed through text but also using visual and audio sources or even a combination of these modalities, highlighting the necessity to tackle the multimodal problem. According to Rizzi et al [39], One of the main types of multimodal content targeting women is represented by memes, which consists of images that feature visual elements along with an overlaying text introducing a posteriori. To date, diverse methods have tackled the task of detecting automatically misogynistic language in social media. A recent review on this topic distinguishes two principal approaches [40]; models that use classical machine learning and those based on Neural Networks. The latest trends use transfer learning, through pre-trained models such as BERT (Bidirectional Encoder Representations from Transformers).

Building on technological advancements, recent research has expanded the scope of the hate content detection beyond algorithmic classification. Rodriguez et al., [41] indicate that efforts to combat hate content, especially misogyny, span various areas including online and offline detection, safety measures, and educational initiatives. In the sphere of online detection, many contributions leverage Artificial Intelligence (AI), particularly through machine learning algorithms, to identify hateful content. Additionally, multiple strategies have been developed that consider the diverse forms in which misogyny can appear, such as in text and images. Most approaches are based on pre-trained models [42], such as BERT [43] and USE [44], while a small number rely on lexical resources [45] [46] or on a combination of both methods [38]. To overcome the scarcity of labeled data, some approaches employ transfer learning strategies.

In line with the growing reliance on deep learning, and to further enhance misogyny detection, recent research has increasingly leveraged embedding techniques to capture the semantic and contextual subtleties of hateful language. Calderon-Suarez et al., [38], recently modified methods that utilize cross-domain and embedding techniques to transfer knowledge from phrases in songs to social media texts that may express misogynistic sentiments. In a study performed by Pamungkas et al. [46], they introduced a joint-learning model that integrates LSTM and BERT to address challenges associated with low-resource languages. Their primary contribution involved utilizing various datasets focused on sexism, hate speech, and offensive language to identify misogynistic content using cross-domain classification techniques. They indicated that lexical characteristics, such as derogatory terms and language related to women, serve as some of the most effective signs for identifying misogyny.

From a multimodal point of view, misogyny can be manifested by a combination of textual and visual content. Today, the most widespread forms of multimodal communication on social media are given by memes, which, despite their ironic intent, can also be used to communicate hateful content to various targets. Recent investigations from Hee et al [47] indicate that the best multimodal approach is based on the combination of textual and visual features given by transformer-based architectures (i.e., ViLBERT and Visual-BERT). Those approaches are compared to other unimodal and multimodal strategies and are proposed as a baseline for the challenge itself, confirming the need for a multimodal analysis to detect hate content. Despite the good performance achieved by those approaches (~75% accuracy), those models still perform poorly concerning human ability, underlying the necessity of new machine learning models.

Although transformer-based multimodal architecture like ViLBERT have demonstrated promising performance, they still fall short of human-level understanding, this highlights the need for more robust and generalizable models. In this context, Aggarwal et al., [34] further investigated previous studies that raise a concern about model biasing and domain-specific responses. They realized that among images, tampering has been used for malicious purposes for quite some time now. GAN-based systems are commonly used for automatic image alteration. Further, several perturbation-based algorithms have been proposed to attack the image modality. Among text-based attacks, Aggarwal et al., [34] investigated the study by Grondal et al., which added LOVE to every text input and found that the nature of the data and the labeling standards are more important than the model architecture. The research demonstrated that training with character-level features enhanced the model's robustness. Attacks have been also

developed while considering both modalities. For example, Aggarwal et al., [34] investigated the study of Vishwamitra et al., who observed that decoupling of image and text components directly defeats the sole purpose of the visual-linguistic models.

In this chapter, the primary goal is to develop a robust multimodal model to detect misogynistic posts on social media, and we intend to explore the countermeasures developed by Aggarwal et al [34], by using the Gaussian noise-based perturbations along with other adversarial attacks to generate adversarial examples to perform the training. Considering multimodality and robustness to achieve highly accurate predictions is essential for detecting misogynistic content because both false positives and false negatives can have detrimental effects. Specifically, frequent false positives will lead to backlash from social media users who feel that their posts are unfairly demoted and censored; false negatives will fail to suppress the propagation of misogynistic content.

3.2. Introducing the MIRA Approach

In this section, we first present the architecture for the multimodal neural network and we explain its ResNet50 and RoBERTa components that extract information from image and text modalities into a multimodal representation for misogyny post detection. Then, we explain the adversarial training process, which retains the multimodal neural network with crafted adversarial examples for images and text for more robust misogynistic post detection. In Figure 22, we present the overall architecture of MIRA.

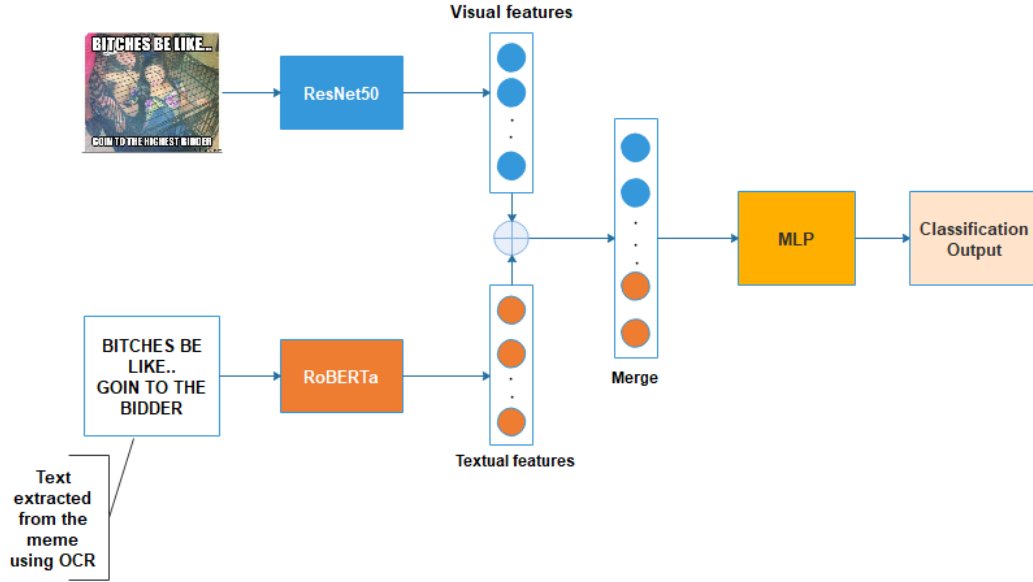


Figure 22. Overview of the MIRA framework

3.2.1. Feature Extraction Module

As illustrated in Figure 22, we combined the pretrained RoBERTa model for text classification with the pretrained ResNet50 model for image classification, integrating their outputs through a simple feedforward neural network. Alec et al., [48] demonstrated that deep convolutional neural networks (CNNs) such as ResNet50, Xception, VGG-16, and VGG-19 represent state-of-the-art approaches for image classification tasks, leveraging pre-trained weights to effectively extract visual features. Based on this evidence, we selected the pretrained ResNet50 model to extract the visual features of memes.

A) ResNet50

The overall architecture of ResNet50, as illustrated in Figure 23, is a deep residual network consisting of 50 layers. It was introduced by He et al. [49] as a solution to the vanishing gradient problem that often hinders the training of very deep neural networks. This problem arises because gradients tend to diminish exponentially as they are propagated backward through many layers, limiting the network's ability to learn

effectively. To address this, ResNet50 incorporates residual learning through shortcut connections, which allow the network to learn identity mappings and facilitate gradient flow during training.

The architecture is systematically organized into four principal components: Convolutional layers, which perform initial feature extraction; Convolutional blocks, which apply successive convolutions and non-linearities; Residual blocks, which is the core innovation enabling skip connections to bypass one or more layers; and Fully connected layers, responsible for aggregating extracted features into the final output classification. This design enables ResNet50 to achieve remarkable depth and performance while maintaining efficient training dynamics.

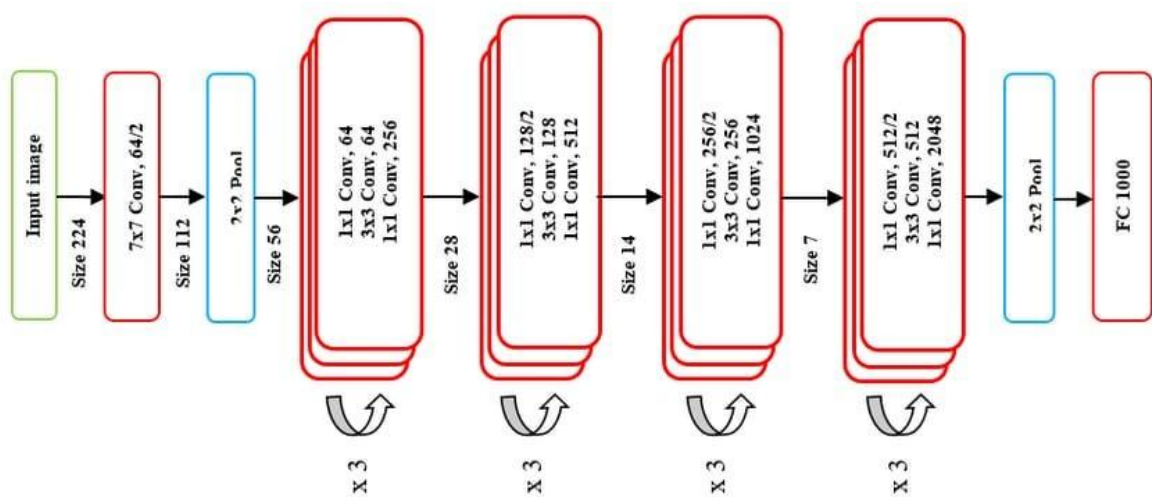


Figure 23. ResNet50 architecture, [50]

Before the image is passed into the ResNet50 model, it is resized to a fixed size, typically 224x224 pixels, which is the standard input size for ResNet models. After which the pixel values of the input image are normalized, by scaling them to a range of [0, 1] or by subtracting the mean and dividing the standard deviation of the dataset. This helps to stabilize the training process and improve the convergence.

The first few layers of the ResNet50 consist of convolutional layers that apply filters to the input image. The convolutional operation performed here helps the model extract various local features such as patterns, edges, and textures within the image. Let X be the input feature map to a convolutional layer, and W be the kernel (filter) of the layer. Let b be the bias. We can express the output feature map Y from the convolution layer as:

$$Y[i, j] = \sum_{m=0}^{k_h-1} \sum_{n=0}^{k_w-1} W[m, n] \cdot X[i + m, j + n] + b \quad (3.5)$$

Where:

- $Y[i, j]$ is the value at position (i, j) in the output feature map Y .
- $W[m, n]$ is the weight at position (m, n) in the kernel W .
- $X[i + m, j + n]$ is the input value at position $(i+m, j+n)$ in the input feature map X .
- k_h and k_w are the height and width of the kernel, respectively
- b is the bias term (a scalar). It is added to each element of the convolved output.

After each convolution, a non-linear activation function (typically ReLU) is applied to the feature map to introduce non-linearity into the model. The ReLU function is displayed in Figure 24. Any negative value in the feature map is replaced with 0. The Max pooling layers follow these convolutional layers and down-samples the spatial dimensions of the feature maps, retaining important information and reducing computational load. In Figure 25, the pooling operation is described.

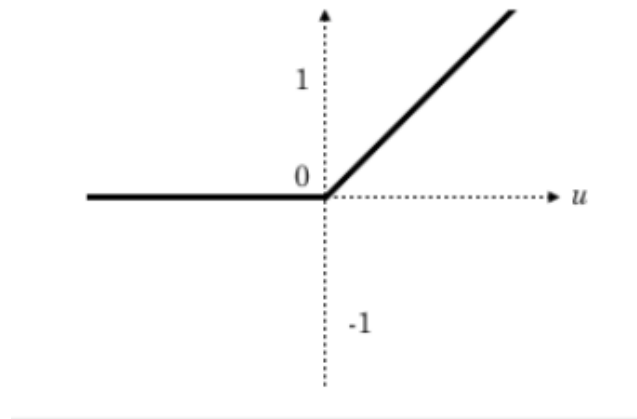


Figure 24. ReLu activation function, [51]

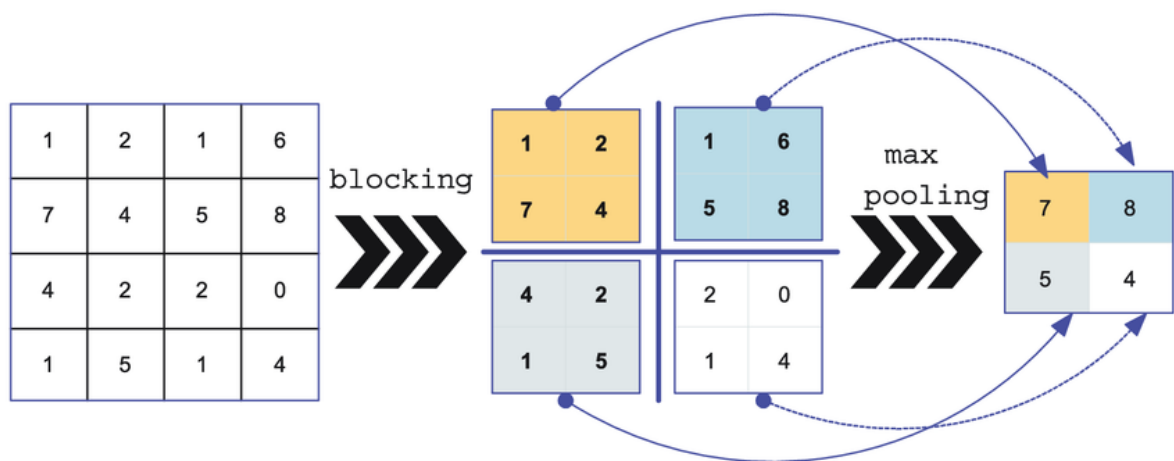


Figure 25. Illustration of pooling operation, [52]

The core of ResNet50 consists of residual blocks, which help to allow gradients to flow more easily through the network. Each residual block contains a skip connection that bypasses one or more layers. This means that the input to the block is added to the output of the block, this allows the network to learn the residual mapping instead of the original mapping. This helps to mitigate the vanishing gradient problem in training deeper networks. After each convolution and before the activation function, batch normalization is applied to stabilize the learning process thereby improving convergence.

As the image passes through the layers of ResNet50, it is transformed into a series of feature maps. Early layers capture low level features like edges, textures while deeper layers capture high-level features like shapes, objects. The spatial dimensions of the feature maps are reduced through pooling and strided convolutions, while the depth increases, allowing the model to learn more complex features. We replaced the final fully connected layers of the ResNet50 model with a new linear layer to project the output to the same dimensions as the text embedding. The output obtained at the end is an image tensor with shape (16, 2048).

B) RoBERTa

The RoBERTa model [53] is an optimized and robustly trained variant of BERT, designed to improve performance on a variety of natural language understanding tasks through more extensive pretraining and architectural refinements. It employs a multi-layer stack of transformer encoders to enhance its capacity for contextual representation learning. RoBERTa is pretrained on a significantly larger corpus than BERT, using dynamic masking and the removal of the Next Sentence Prediction (NSP) objective, which contributes to a more effective language modeling. As a result, it generates deep bidirectional representations that integrates a rich contextual knowledge across entire input sequences. The RoBERTa infrastructure, is illustrated in Figure 26. The feature extraction process involves several key stages: initial tokenization using a byte-pair encoding (BPE) tokenizer, construction of input embeddings through the integration of token, segment, and positional embeddings, and the application of a transformer architecture composed of multi-head self-attention mechanisms and feed-forward neural networks. These components work in synergy to capture complex linguistic patterns and semantic dependencies across varying contexts.

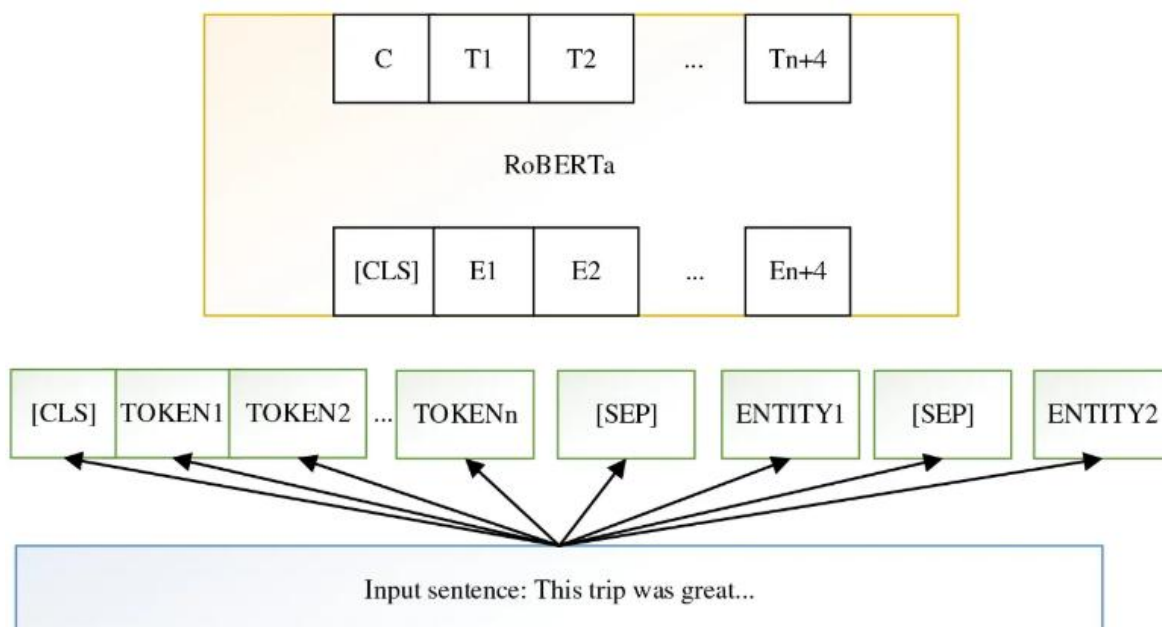


Figure 26. RoBERTa Infrastructure, [53]

Before the text is passed through RoBERTa, the input text is tokenized. RoBERTa uses a WordPiece tokenizer, which breaks down words into subword units. After the tokenization process, each token is converted into an input representation (numerical representation) using an embedding layer. The input representation for each token consists of three components namely: token embeddings, segment embeddings (used to distinguish between different sentences), and position embeddings (used to encode the position of the token in the sequence).

RoBERTa consists of multiple transformer layers (12 in our case), with each transformer layer consisting of a multi-head self-attention mechanism and a feed-forward neural network. The multi-head self-attention mechanism allows the model to focus on different parts of the input sequence, thereby weighing the importance of different words in the input sequence when representing a specific word. For each

attention head (h), the input representations (X) are linearly transformed into three matrices: Query (Q), Key (K), and Value (V).

$$Q = XW_Q, K = XW_K, V = XW_V \quad (3.2)$$

where W_Q, W_K , and W_V are learned weight matrices for the (h)-th head. The outputs from all the heads are concatenated and linearly transformed. The attention scores are computed to obtain the attention weights.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3.3)$$

where d_k is the dimensionality of the keys, and the softmax function ensures the sum of the weights is 1.

After the multi-head self-attention mechanism, the input representation for each token is passed through a feed-forward neural network. This network consists of two linear transformations with a non-linear activation function in between:

$$FFN(x) = W_2(\text{ReLU}(W_1x + b_1)) + b_2 \quad (3.4)$$

where W_1, W_2 are learned weight matrices, b_1, b_2 are bias vectors, and ReLU is the Rectified Linear Unit activation function.

After each sub-layer, layer normalization and residual connections are applied. This helps stabilize training and allows for deeper networks. After passing through all the transformer layers, the contextualized embeddings for each token in the input sequence are generated. These embeddings capture the semantic meaning of each word and their relationships with other words in the sentence. The final output of each token with shape (16, 768) is used as the feature representation.

3.2.2. Classifier Module

The representation of the multi-modal features $F_M \in \mathbb{R}^{2d}$ is formed by concatenating the visual feature representation F_V and textual feature representation F_T :

$$F_M = \text{concat}(F_V, F_T) \quad (3.5)$$

F_M is the output of the feature extraction module that can be denoted as $f_F(X; \theta_F)$. X represents the input multimedia post, and θ_F is the parameters of the module. F_M is then passed through a feedforward neural network defined by the merge block as shown in Figure 22. This merge block consists of two fully connected layers with ReLU activation functions, followed by a final fully connected layer with no activation function.

Finally, the output tensor is passed through another feedforward neural network defined by the MLP block as shown in Figure 22. This block consists of four fully connected layers with ReLU activation functions, followed by a final connected layer with no activation function.

3.2.3. Introducing the AdverMira approach

Adversarial training techniques have been widely investigated in recent years to improve the robustness of deep neural networks against adversarial examples. By adding small perturbations η to the original input x , the adversarial example x' aims to fool the classifier f such that $f(x')$ does $\neq y$. A robust deep neural network model should not change its output for these small perturbations in its input which are imperceptible to human eyes. Therefore, the adversarial training process utilizes such adversarial examples as an augmented dataset to increase robustness. In the domain of adversarial

security research [54], adversarial training has been used by almost all adversarial attack studies.

In this work, we focused on implementing non-technical and technical attacks. Likewise, in Aggarwal et al [34], we specifically focused on the model independent attacks which humans can introduce with little technical expertise. Attacks for instance like adding noise to the image and texts in the memes can be easily implemented without prior knowledge of the classification models. For technical attacks like we selected notable attacks like FGSM, PGD, and AdvDrop which are popular methods used to calculate perturbations and easily generate adversarial examples. The trade-off between strength and computational efficiency is a crucial consideration when choosing and attack strategy.

A) Non-technical Attacks

Adversarial attacks targeting both image and text are applied during the data loading phase for both training and testing. For non-technical attacks, we used the python-based library Augly, which offers a diverse set of semantic-preserving augmentation techniques. For each batch of data loaded for training, a random number is generated between 1 and 10, if the number is greater than 1 then adversarial example for both modalities are generated using one of the various augmentation operations described in Table 7 and passed to the model for training.

The probabilistic application of adversarial attacks introduces a controlled variability that helps to enhance the model’s generalization to real world multimodal noise. During training each transformation is applied in a cross-modal manner to promote training stability and semantic robustness.

Algorithm 1 Multi-modal Adversarial Training Algorithm for Non-technical Attacks

Input: labelled multi-modal input (x_i, x_t, y) , Augly operations

Output: Adversarial examples (x'_i, x'_t, y) , robust adverMIRA

begin

1: Initialize MIRA classifier f

2: **repeat**

3: For each batch (x_i, x_t, y) ,

4: Generate a random integer r in $[1, 10]$,

5: If $r > 1$ then:

6: Obtain perturbation (η_i, η_t) for (x_i, x_t) from Augly

7: Data augmentation with $(x_i^{aug}, x_t^{aug}, y^{aug}) = data_aug((x_i, x_t, y), (\eta_i, \eta_t, y))$

8: Train MIRA classifier f with augmented batch $(x_i^{aug}, x_t^{aug}, y^{aug})$

9: Else:

10: Train MIRA classifier f with (x_i, x_t, y)

11: **until** training converged

end

Let's consider the text extracted from the meme shown in Figure 27, "BITCHES BE LIKE.. GOIN TO THE HIGHEST BIDDER" We considered the different attack forms for text as shown in Table 7.



Figure 27. Example of memes found in the dataset

For the scenarios where we cannot predetermine the internal behavior of the threat model, the attacker cannot appropriately generate attacks using gradient information.

However, the attackers can introduce random modifications to the image while preserving its message. We considered the following attacks as shown in Figure 28.

Table 7. Adversarial attacks on text

Attacks on Text	Output
Replace Bidirectional	REDDIB TSEHGIH EHT OT NIOG ...EKIL EB SEHCTIB
Replace Similar Chars	B TC/-/3S B3 LIKE... GOIN +O THE /-/IGHEZ+ 13IDDEI2
Simulate Typos	BITCHES BI LIKE.... GOING TO THE HIGHEST BIDDER
Change Case	BITCHES BE like... goin TO the HIGHEST BIDDER
Insert Punctuation Chars	B.I.T.C.H.E.S. .B.E. .L.I.K.E..... .G.O.I.N. .T.O. .T.H.E. .H.I.G.H.E.S.T. .B.I.D.D.E.R
Insert White Space Chars	B I T C H E S B E L I K E G O I N T O T H E H I G H E S T B I D D E R



Figure 28. Different forms of non-technical attacks on image considered

B) Technical Attacks

For technical attacks we considered the FGSM, PGD and AdvDrop because they are easy to implement in our multimodal setting (low-resource setting). For each technical attack, we implemented a balanced approach for generating adversarial examples in the training process, and we included an equal mix of original and adversarial samples. The FGSM [55] algorithm is simple and efficient in fooling ML models, but it can be easily detected by more sophisticated countermeasures. FGSM generates adversarial examples [56] by applying small perturbations in the gradient direction of the model's loss function. The loss function L is calculated as the difference between the model's prediction and the actual label. For an input x and its label y , the loss function $L(x, y)$ is calculated, and then the gradient of this loss function with respect to the input x is obtained, denoted as $\nabla_x L(x, y)$. The perturbation is calculated using the gradient sign function and multiplied by a factor ϵ , representing the perturbation's magnitude. The adversarial image is then obtained as follows:

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x L(x, y)) \quad (3.6)$$

Where x' is adversarial image, x is original input image, y is ground truth label of the input image, ϵ is the small value to ensure the perturbations are small enough that the human eye cannot detect them but large enough that they fool the neural network, $\nabla_x L(x, y)$ is the gradient of the loss function for the input x .

The PGD method [57] is a variant of FGSM that seeks to generate more robust and difficult-to-detect adversarial examples. PGD performs multiple iterations to update the perturbed image and find the optimal perturbation. The algorithm follows these steps:

1. Define specific hyperparameters, such as the perturbation size (ϵ).

2. Calculate a perturbation δ from the gradients, add it to the image, and in the second step, calculate the gradients again, but on the already modified image from the previous step. This should give a slightly different perturbation per step, so that we don't add the same noise to the picture repeatedly.
3. During each iteration, ensure that $\|\theta\|_\infty \leq \epsilon$ is not violated, by projecting δ into the norm ball.

At first, this might seem similar to the FGSM minimal attack method, but the difference is that the algorithm doesn't stop if $\epsilon_{step} > \epsilon$, because we always calculate a new perturbation and the noise can still change even if $\epsilon_{step} > \epsilon$.

The AdvDrop attack [27] is an adversarial attack method that generates adversarial images by dropping existing details of clean images. This approach contrasts with other adversarial attacks that add extra disturbing information to clean images. AdvDrop [27] transforms images from the spatial domain to the frequency domain and then reduces certain frequency components. The attack optimizes a quantization table q by minimizing the adversarial loss $L_{adv}(\cdot)$. The objective can be formally denoted as:

$$\min_q L_{adv}(x', y), \text{ where } x' = DI(Q_{diff}(D(x), q)) \text{ st } \|q - q_{init}\|_\infty < \epsilon \quad (3.7)$$

Where x' is adversarial image, x is input image, y is ground truth label, $D(\cdot)$ is Discrete Cosine Transform (DCT), $DI(\cdot)$ Inverse Discrete Cosine Transform (IDCT), Q_{diff} is differential quantization function, q is quantization table, q_{init} is initial value of quantization table, ϵ is a bound to tune the value of the quantization table, L_{adv} is an adversarial loss

The adversarial loss L_{edu} is defined using the cross-entropy loss:

$$L_{\text{adu}} = \begin{cases} -\log(p_y(x')), & \text{for untargeted attack} \\ -\log(p_{y_{\text{adv}}}(x')), & \text{for targeted attack} \end{cases} \quad (3.8)$$

Where $p(\cdot)$ is the probability output (softmax on logits) of the target model f with respect to class y_{adv} or y .

3.3. Experiment and Results

In this section, we present the dataset used, the steps carried out, and the results of the experiments carried out to provide a model that classifies memes into misogynous or non-misogynous classes under different experimental conditions.

3.3.1. Dataset

We used the Multimedia Automatic Misogyny Identification (MAMI) dataset provided by Fersini et al. [58] during SemEval 2022. The dataset is provided upon completion of an online form available at <https://forms.gle/AGWmiGicBHiQx4q98>. The dataset consists of 12,000 memes and all memes are balanced. Table 8 describes the dataset sizes and label distribution.

Table 8. MAMI Dataset sizes and label distribution

Dataset	Size	Misogynous Memes	Non-Misogynous memes
Train	10000	5000	5000
Test	1000	500	500
Validation	1000	500	500

A) Examples of memes

Most of the time, a meme requires both textual and visual information to be correctly understood. Sometimes the text and image often contradict each other for humoristic

purposes. From Figure 29 (a), the text alone indicates a positive feeling towards buying a dishwasher and the image without the caption shows a woman in the supermarket cart. When the two are simultaneously taken into consideration, it becomes a misogynistic meme implying that women exist to function as dishwashers.



Figure 29. Example memes from the MAMI dataset. (a: Left, b: Right)

Figure 29 (b) is an example of a meme where both modalities are unnecessary to make a correct prediction. It can be seen that any of the two modalities would suffice to make the correct prediction, while some other memes may need only one of the modalities to make the correct prediction.

B) Word Frequency

We visually analyzed the memes present in the training set of the MAMI dataset to identify the top 50 words in the training set. We did that by examining the token frequency and we could see those words like hooker, prostitute, fuck highlight sexually violent tendencies towards women. Figure 30, shows the top 50 words and their respective frequencies.

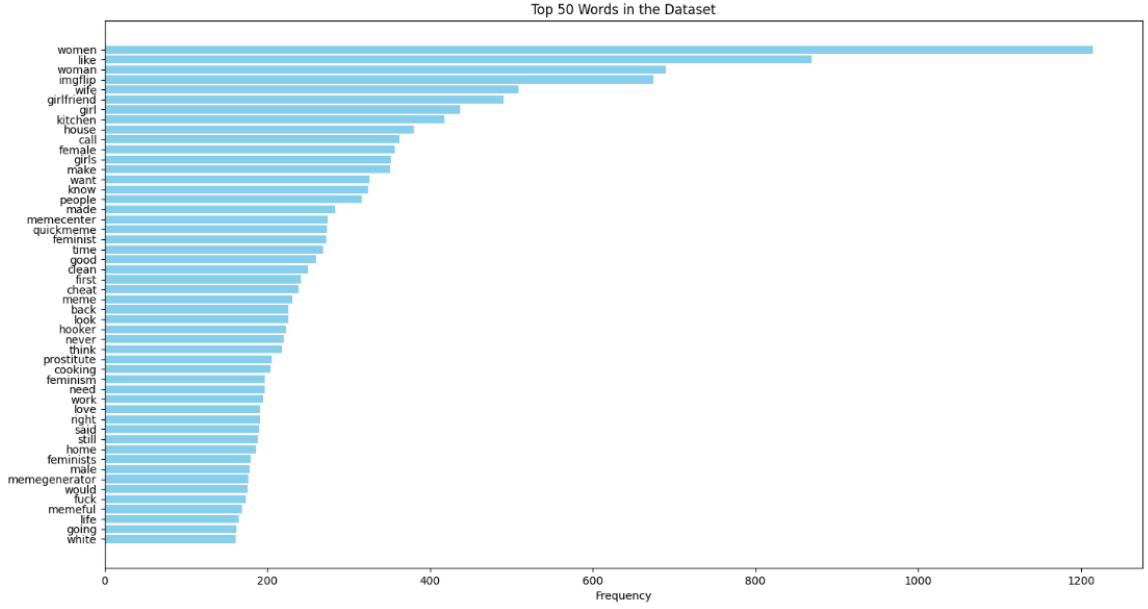


Figure 30. Top 50 words vs Frequency

3.3.2. Experimental Setup

Our method adopts a hybrid fusion approach (early and late fusion) to better capture the individual contributions of the two modalities, thereby reducing the impact of noise or errors in individual modalities, and also handling missing data more effectively, as the absence of data in one modality doesn't affect the performance of the other modality. Hybrid fusion allows for a more comprehensive, flexible, and robust analysis of social media messages by combining the strength of early, intermediate, and late fusion.

For this purpose, we suggested separate extraction pipelines for the image and text features. ResNet50 model is used to extract the image features and the popular RoBERTa model is used to obtain the text representations. Finally, we concatenated the features from both modalities and pass them through a feed-forward neural network having 128 hidden layers with ReLu activation, cross-entropy as loss function, and dropout = 0.2 to generate predictions. We train the model for 20 epochs on Adam optimizer with learning rate = $1e-5$ and weight decay = 0.1. All of the texts were

tokenized using the RoBERTa base model with a max number of tokens set to 100. We used the Adam optimizer (AdamW) with early stopping was used to avoid overfitting. These experiments were run using PyTorch 2.0, python 3.11, and Windows 10 on a laptop with Nvidia A200 GPU, and 32GB RAM. We selected the best epoch and the best hyperparameters as measured by F1 score and accuracy.

For non-technical attacks, we randomly attack both modalities during training and validation. For technical attacks we attack both modalities as well, since most of the technical attacks are more directed toward images, we keep attacking the text modality using the non-technical means while attacking the image modality as well during training and validation.

3.3.3. Evaluation Metrics

Evaluation metrics play a critical role in assessing the performance of machine learning models particularly in tasks involving imbalanced or multiclass data such as social media content analysis. The performance metric must capture both overall accuracy and class-wise effectiveness. To this end, the main evaluation metric used during training are Macro-F1, Precision and Accuracy, each offering complementary perspectives.

The F1 score, originally developed in the context of information retrieval [59], is the harmonic mean of precision and recall, balancing false positives and false negatives (Equation 3.12). While Accuracy is a widely used classification metric that measures the overall proportion of correct predictions (Equation 3.9), it may be misleading in scenarios with class imbalance [60]. To address this limitation, Macro-F1 is employed as it computes the unweighted mean of F1 scores across all classes, thereby treating each class equally regardless of its frequency [61]. Precision measures the proportion of

correctly predicted positive cases out of all positive predictions, indicating the model's accuracy in identifying positives (Equation 3.10). Recall measures the proportion of actual positives correctly identified by the model (Equation 3.11). True Positives (TP) is the number of positive predictions that are actually positive. True Negatives (TN) is the same applied to the negative class. False Positives (FP) is the number of positive predictions that are actually negative, and False Negatives (FN) is the number of negative predictions that are actually positive.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (3.9)$$

$$Precision = \frac{TP}{TP+FP} \quad (3.10)$$

$$Recall = \frac{TP}{TP+FN} \quad (3.11)$$

$$F1 = \frac{2*Precision*Recall}{Precision+Recall} \quad (3.12)$$

3.3.4. Results

In this section, we perform a quantitative and a qualitative analysis of the results obtained from the experiments we carried out. In Table 9, we report the performance of our model at different hyperparameters and Table 10 displays the performance of our model compared to several state-of-the-art models.

A) Quantitative Performance Evaluation

We reproduced the experiment carried out by Aggarwal et al on the MAMI dataset, from Table 9, it can be noted lowering the learning rate significantly improves the model's performance, with the best result achieved at 1e-5. Additionally, reducing the batch size from 16 to 6 further boosts performance, with the highest Macro-F1 Score (98.99%), Accuracy (99%), and Precision (97.77%) observed at the smallest batch size.

These results underscore the critical role of hyper-parameter tuning. The performance of our model in Table 9 is the performance reported under no adversarial training (1-5) and non-technical attacks (6-7). All the experiments were carried out for 10 epochs.

Table 9. Adversarial performance across training parameters

Experiments	Learning Rate	Batch Size	Macro- F1 Scores (%)	Accuracy (%)	Precision (%)
1	0.1	16	30.55	44	28.37
2	1e-2	16	57.78	58	56.3
3	1e-3	12	61.44	62	60.41
4	1e-5	12	82.98	83	81.82
5	1e-5	10	89.85	90	88.63
6	1e-5	7	95.95	96	93.61
7	1e-5	6	98.987	99	97.77

Comparing our model with other works in the field one noticeable observation is that our model outperforms the other state-of-the-art models. We present in Table 10 the performance of various multimodal models for hate content detection as reported by different authors. The models are compared primarily on Macro-F1 scores, Accuracy and Precision included where available. The MIRA model in this work surpasses all prior approaches, demonstrating substantial improvements. The AdverMIRA model, which incorporates adversarial training specifically designed to counter non-technical attacks, sets a new benchmark in the field of multimodal hate content detection. To comprehensively evaluate the effectiveness and robustness of our approach, we provided the Macro-F1, Accuracy, and Precision. This multi-metric evaluation provides a more holistic view of our model’s performance, which is very valuable in imbalanced or high-stakes classification tasks.

Table 10. Performance Comparison of Models on MAMI Validation Memes

Author	Model	Macro-F1 Scores (%)	Accuracy (%)	Precision (%)
Rizzi et al, (2023) [39]	MTT	N/A	81.00	N/A
Rizzi et al, (2023) [39]	MTC	N/A	80.00	N/A
Rizzi et al, (2023) [39]	Visual-BERT	N/A	84.00	N/A
Aggarwal et al, (2023) [34]	RoBERTa-ResNet (Baseline)	85.50	N/A	N/A
Aggarwal et al, (2023) [34]	Visual-BERT	85.00	N/A	N/A
Aggarwal et al, (2023) [34]	UNITER	84.99	N/A	N/A
Zhang et al, (2022) [62]	UNITER-CLIP	83.40	N/A	N/A
MIRA	RoBERTa-ResNet50	89.85	90	88.63
AdverMIRA	RoBERTa-ResNet50	98.987	99.0	97.77

Among these models, it should be noted that we considered adding adversarial examples for non-technical attacks in the training and validation set and we used a hybrid fusion approach to gain the advantages of the early, and late fusion scheme. Aggarwal et al., [34] added adversarial examples for both modalities on the test set to investigate the robustness of hate memes detection against adversarial attacks. They carried out the experiment on RoBERTa-ResNet, Visual-BERT, and UNITER. Rizzi et al., [39] proposed three models to detect hate memes but did not consider adversarial training. They proposed a first model called Multimodal Text and Tags (MTT) that combines the information given by the considered sources of information by a concatenation of the corresponding latent representations. They also proposed a second model called Multimodal Text and Caption (MTC) that analogously concatenates the embedding of the text within the meme, with the embedding of the meme caption. Zhang et al., [62] proposed an ensemble system of pretraining models, boosting methods, and rule-based adjustment to detect online misogynistic memes. They used the late sequential fusion scheme to fuse the text information into their system.

B) Ablation Study

In order to evaluate the effectiveness of adversarial training in making our model more robust, we carried out many other experiments to examine their corresponding performances. During adversarial training, we chose our perturbation factor (ϵ) to be 0.09 and our probability score to be 0.8. The choice of these values was based on the computational resource available and also because it provides a good balance between robustness and accuracy.

In Table 11 reports the execution time of the different technical adversarial attacks, for non-technical attacks, the execution time for one epoch was the same as the execution time for one epoch under no-adversarial attack.

Table 11. Average execution time of adversarial methods during adversarial training

Method	Execution Time (hrs)
FGSM	7
PGD	10
AdvDrop	2

In terms of execution time, AdvDrop method is the fastest, on the other hand PGD and FGSM require more time. PGD attack has more strength than FGSM attack which explains it requiring more time. Thou AdvDrop takes the least time to execute, it is the most effective method of the three attacks since it greatly affects the model accuracy. Table 12 reports the accuracy of the model after each of the attacks has been applied to it. The accuracy before adversarial training is obtained by applying adversarial attacks on our model with accuracy 99%.

Table 12. Accuracy before and after adversarial training

Attacks	Accuracy after adversarial attack on base model (%)	Accuracy after adversarial attack on our model (%)	Accuracy after adversarial training (%)
FGSM	83.6	97	98
PGD	58.7	75	96
AdvDrop	46.3	67	74

In Figure 31, we show the results of the experiment conducted to examine the corresponding performance of our model under various settings.

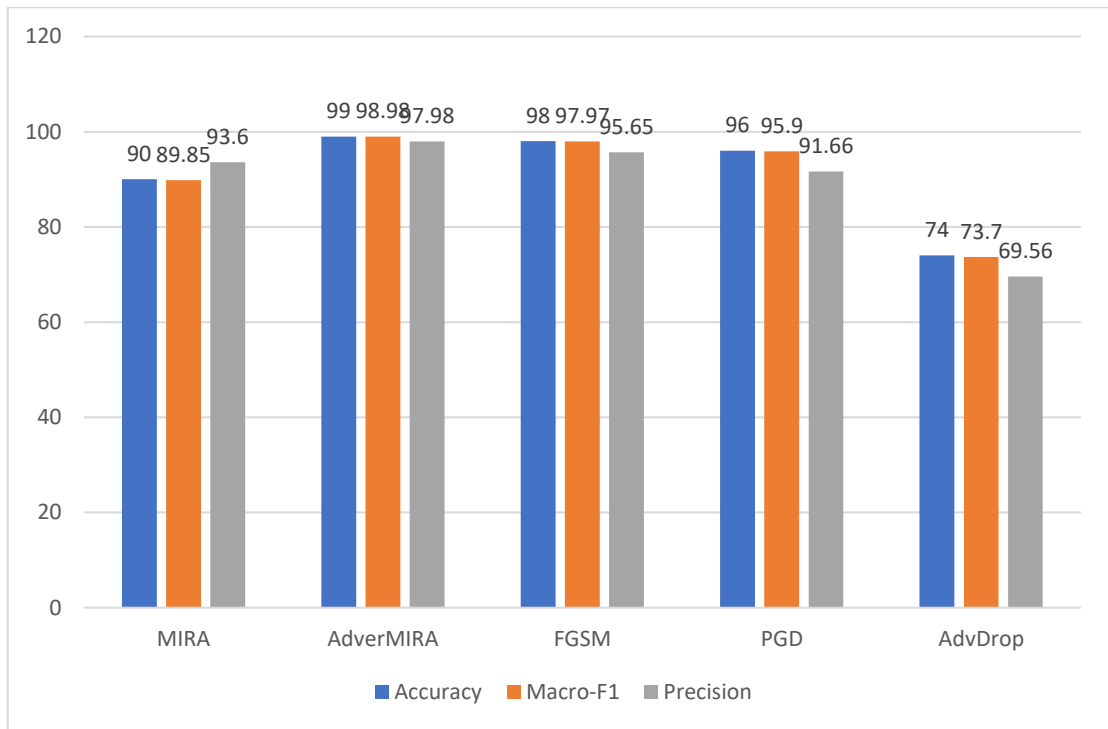


Figure 31. Ablation study of the proposed model

- **MIRA:** This was designed as the proposed multimodal network without adding adversarial examples
- **AdverMIRA:** This was designed as the proposed multimodal network that we adversarial examples from non-technical attacks during training.

- **FGSM:** This is the model designed to undergo adversarial training to be robust against FGSM attacks
- **PGD:** This is the model designed to undergo adversarial training to be robust against PGD attacks
- **AdvDrop:** This is the model designed to undergo adversarial training to be robust against AdvDrop attacks

C) Qualitative examples

In Figure 32, we present examples of post detected by our multimodal network, the left image is an example of a post detected as non-misogynous and the image on the right is an example of a post detected as misogynous. The non-misogynous post reveals the supportive nature of women whereas the misogynous post promotes violence against women. More examples of posts detected as misogynous and non-misogynous by our model are shown in Appendix A: Figure A1 for misogynous posts and Figure A2 for non-misogynous posts can be. We can observe that the model can obtain reliable cues to make accurate predictions based on the contents of the image and text provided.



Figure 32. Posts Detected by MIRA and AdverMIRA

For technical attacks, in Figure 33, the left image is an example of a post detected as non-misogynous and the right image is an example of a post detected as misogynous

across all the 3 attacks. It's assessment of whether the predictions were right or wrong is for human judgment level. More examples of posts detected across all three attacks can be found in Appendix A: Figure A3 for non-misogynous post across all 3 attacks and Figure A4 for misogynous posts.



Figure 33. Memes predicted with the same output across the 3 attacks

From the Figure 33, we can observe that our model could detect memes where the pictorial content of the meme didn't express misogyny and the lexical text did and vice-versa. Also, for posts referenced in Appendix A: Figure A4, we can observe that our model was able to get the lexical components of the text and the pictorial content related to misogyny. In Figure 34, we show some examples of failure cases, highlighting scenarios in which our model struggled to produce accurate predictions. Looking closer at these instances reveals that for cases where the textual and visual elements needed nuanced understanding was very challenging for our model to make accurate predictions. In such cases, the semantic meaning is often subtle, context-dependent, or relies on implicit cues that are not easily captured through conventional feature representations. These challenges underscore the limitations of our model in handling complex multimodal content where high-level reasoning, sarcasm, or cultural references may play a crucial role in determining the correct label

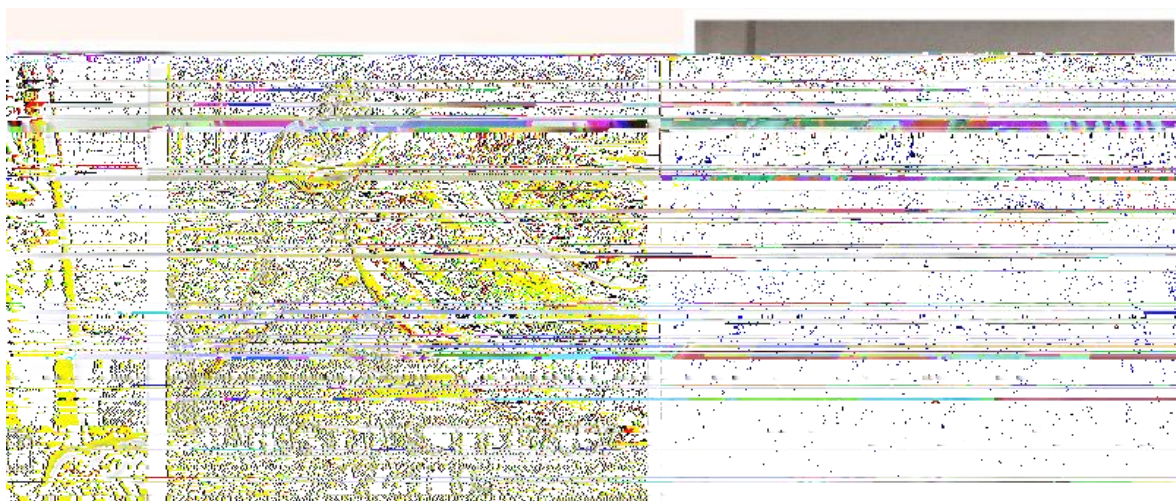


Figure 34. Failure cases on the MAMI test set

In Figure 34, the left post was detected as non-misogynous and the right post as misogynous. This reveals that our model misclassifies sarcasm (the case of the post on the right side of Figure 34) and irony due to its lack of pragmatic reasoning. The post on the left through its pictorial content shows violence towards a woman but the textual content doesn't contain any hateful feeling was not recognized by our model as misogynous because of its inability to recognize visual stereotypes.

3.4. Conclusion

In this chapter, we highlighted the need to develop robust multi-modal systems to detect hate posts online as more and more social media users are sharing text and images to spread hate while modifying the post to escape existing online detectors while preserving the message and also highlighted the significance of parameter tuning for better performance. We have developed AdverMIRA, a robust multi-modal model that uses two pre-trained deep learning models, ResNet50 and RoBERTa, for image and text feature extraction, respectively with a hybrid fusion approach. The model was also trained against some non-technical and notable technical attacks on images and text and showed good results. By incorporating adversarial examples during training, AdverMIRA could identify nuanced aspects of misogynistic content that may have been

overlooked by conventional training methods, since they are compelled to focus on critical features that are robust against perturbations.

In the next chapter, we will focus on applying the same model architecture on another dataset that doesn't contain a balanced amount of data for each class and apply techniques alongside with adversarial training to improve its accuracy and robustness to be able to detect other forms of hateful content apart from misogyny.

CHAPTER FOUR

HateXcel: A Robust Hate Content Detection Model on FBHM Dataset

4.1. Introduction

Online hate [63] encompasses a variety of prejudiced expressions aimed at individuals or groups based on characteristics such as race, religion, gender, and sexual orientation. This type of content can manifest in derogatory comments, threats, and dehumanizing language, often leading to serious emotional and psychological consequences for victims. The impact of online hate [64] is exacerbated by its diverse manifestations across various media formats, which complicates efforts to address it comprehensively. As the digital environment continues to evolve, understanding the dynamics and consequences of online hate is essential for fostering a safer online community for all users.

The Facebook Hateful Memes dataset [65] is a pioneering resource developed by Facebook AI to aid researchers in the detection of multimodal hate speech, which combines both text and images. The dataset suffers from class imbalance, thus presenting a shared challenge in machine learning, where examples are not evenly distributed among various classes. Imbalanced data presents significant challenges for machine learning algorithms, often resulting in biased models that underperform on the minority class. Numerous strategies [66] have been suggested to tackle class imbalance, yet a considerable research gap remains in effectively managing abnormal instances within the minority class while applying re-balancing techniques. Many current approaches find it difficult to sustain prediction accuracy when creating synthetic data in the context of these abnormal instances.

While various strategies have been proposed to address class imbalance, effectively handling abnormal instances within the minority class remains challenging, particularly when synthetic data is involved. To address such issues, Mohammed et al., [67] used a novel dataset called ZADA, which includes medical information from over 7000 patients in Zakho, Iraq, to build a system to assess and detect patients with disproportionate diabetes. They diagnosed diabetes using six machine learning techniques and they corrected class imbalance by using a classification analysis that included the SMOTE resampling approach with three normalization strategies. Their study showed that the performance of the classification models based on minority class distribution was much enhanced by both the resampling and normalizing techniques. Sowjanya et al, [68]discovered that SMOTE gives only moderate accuracy on imbalanced datasets. They proposed to modify SMOTE to reduce the class imbalance in terms of Distance-based SMOTE (D-SMOTE) and Bi-phasic SMOTE (BP-SMOTE) also known as instance-based sampling. Both D-SMOTE and BP-SMOTE yielded slightly higher accuracy compared to the original SMOTE.

Building on the limitations of standard SMOTE, subsequent studies explored hybrid techniques to further enhance performance on imbalanced datasets. Mulla et al., [69] performed a method that combined SMOTE oversampling with PCA to solve imbalance problems in high dimensional data. They used Logistic Regression, K-Nearest Neighbor, and Decision Tree methods on two separate datasets to evaluate their method efficiency and to determine their classifiers' performance. They used WEKA with the algorithms above to analyze raw datasets and converted datasets by PCA, SMOTE, and SMOTE+PCA. The results they obtained showed that almost all classification algorithms improved their classification accuracy using PCA, SMOTE, and

SMOTE+PCA methods, with the SMOTE method giving the most efficient result for data rebalancing and K-Nearest Neighbor having the highest classification performance among the other algorithms. The primary goal of this chapter is to develop a robust multimodal model for detecting hate online content, specifically utilizing the Facebook Hate Meme Dataset while addressing challenges related to class imbalance and adversarial robustness.

4.2. Introducing the HateXcel Approach

4.2.1. Data Preparation and Processing

In this section, we applied techniques like PCA and SMOTE to ensure the data is ready for accurate analysis and model training by addressing issues like inconsistencies, missing values, and noise.

A) Principal Component Analysis (PCA)

It is a statistical procedure [70] concerned with elucidating the covariance structure of a set of variables. In particular, it permits us to identify the principal directions in which the data varies. The main goal of PCA is to reduce the dimensionality of a dataset while maintaining the most important patterns between the variables without any prior knowledge of the target variables. Before applying PCA, the features are standardized to have a mean of zero and a variance of one.

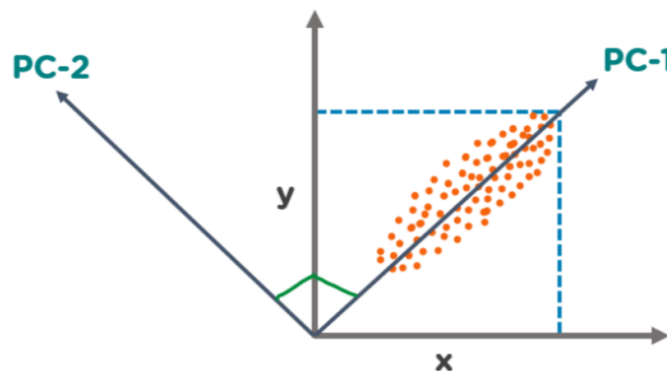


Figure 35. Principal Component Analysis, [71]

From Figure 35, we have several points plotted on a 2-D plane. There are two principal components. PC1 is the primary principal component that explains the maximum variance in the data. It captures the most information about the dataset. PC2 is another principal component that is orthogonal to PC1. PC2 captures the second most variance in the data. Each subsequent principal component continues this pattern, capturing less variance than its predecessor.

$$X' = \frac{X - \mu}{\sigma} \quad (4.1)$$

Where:

- X is the feature vector.
- X' is the standardized value.
- μ is the mean of the feature values.
- σ is the standard deviation.

Incremental PCA in our case allows for the processing of data in smaller batches, this enables us to update the principal components rather than loading the entire dataset into memory as shown in Equation (4.3). This approach is very effective when dealing with large datasets or streaming data. Before applying Incremental PCA to the FBHM dataset, the features we extracted from both modalities (image and text) are standardized as shown in Equation (4.1). This helps us to ensure that each feature contributes equally to the distance calculations during PCA. Given that our dataset $\mathbf{X}$ is represented as a matrix where each row is a sample and each column is a feature, we can find the lower-dimensional representation of $\mathbf{Z}$ such that:

$$\mathbf{Z} = \mathbf{XW} \quad (4.2)$$

Where W is the matrix of principal components obtained from PCA, and the principal components are computed by maximizing the variance of Z .

The update rule for Incremental PCA can be expressed as:

$$W_{new} = W_{old} + \Delta w \quad (4.3)$$

Where Δw represents the change in principal components based on the new batch of data.

In order to efficiently control memory use, we processed the dataset in batches. For each batch of features, we updated the scaler with the new batch of features. After scaling, we updated the principal components based on the scaled features. After processing every batch, we looped through the dataset once more to transform all features, thereby reducing the dimensionality of the features while retaining the most essential information. We created a single feature set that captures information from both modalities by concatenating the reduced picture and text features after dimensionality reduction. After that, this merged feature set was ready for additional processing, including D-SMOTE to address the class imbalance.

B) Distance-based SMOTE Implementation

SMOTE is a sampling technique used to generate synthetic data points to balance minority classes in datasets. It randomly selects neighbors and doesn't check if they are from the minority class or not. According to Sowjanya et al, [68] D-SMOTE has a greater advantage over SMOTE. They report that D-SMOTE leads to a better model performance, the synthetic sample quality are generally of higher quality than that of

SMOTE, and there is lower noise due to a structured sampling. Figure 36 shows a schematic representation of D-SMOTE

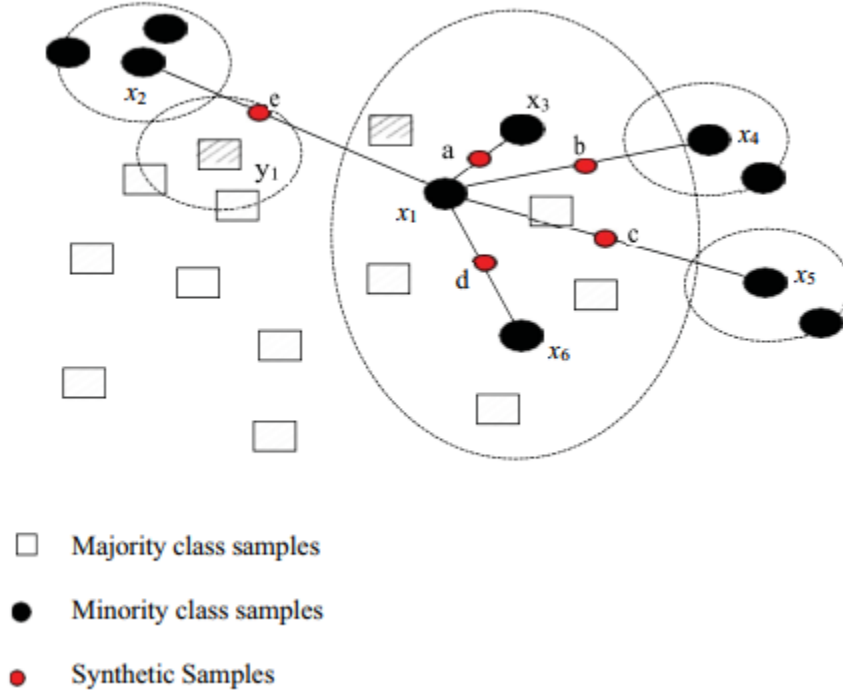


Figure 36. Schematic representation of D-SMOTE, [68]

From Figure 36, the newly generated samples are generated in the vicinity of the minority class and these samples operate within the feature space. Among the assumptions made [68] by SMOTE to identify features that are similar across minority classes, it is assumed that: each minority sample is measured in terms of its centroid (c), and the distance (d_i) between each minority sample and the centroid is determined separately for each minority sample and the centroid.

$$c = \frac{1}{n} \sum_{i=1}^n y_i \quad (4.4)$$

$$d_i = (y_i - c) \quad (4.5)$$

This is followed by the computation of the average (avg) of the distance matrix. Specifically, it is represented as a distance from the class center (c) that is greater than the average distance and the sample distance.

$$avg = \frac{1}{n} \sum_{i=1}^n d_i \quad (4.6)$$

$$Ss = \{y_i \mid d_i > avg\} \quad (4.7)$$

Using a random number generator between (0,1), a synthetic sample is generated by multiplying the difference in centroid (c) and distance (d_i) for each of the N centroids by a number between (0,1), where ' σ ' is a number between (0,1). The value of this seed is added to the value of the original seed to determine its total value.

$$nss = Ss_i + (Ss_i - c) \times \sigma \quad (4.8)$$

The algorithm to apply D-SMOTE on our dataset is discussed below.

Algorithm 2 D-SMOTE Algorithm

Input: Transformed features of original samples

Output: Combined features and labels of original and synthetic samples

begin:

- 1: Calculate class_counts from labels
- 2: Identify majority_class and minority_class
- 3: Calculate num_synthetic_samples = majority_count – minority_count
- 4: Fit NearestNeighbors on features
- 5: Initialize synthetic_samples and synthetic_labels as empty lists
- 6: Identify indices of minority class samples
- 7: For each feature in minority class samples
- 8: Get the indices of the k nearest neighbors
- 9: Generate synthetic samples
- 10: until enough synthetic samples have been generated
- 11: Combine feature and labels of original and synthetic samples

end

4.2.2. Training Procedure

In this study, we propose a new method based on combining adversarial learning with D-SMOTE (for class imbalance) and PCA (for dimensionality reduction) to enhance the robustness and improve the accuracy of our classification model. The ensemble model used here is the same as the one described in Chapter 3. Figure 37 describes the training procedure. For adversarial training, the same non-technical and technical attacks applied in Chapter 3 on the MAMI dataset were equally applied to the FBHM dataset.

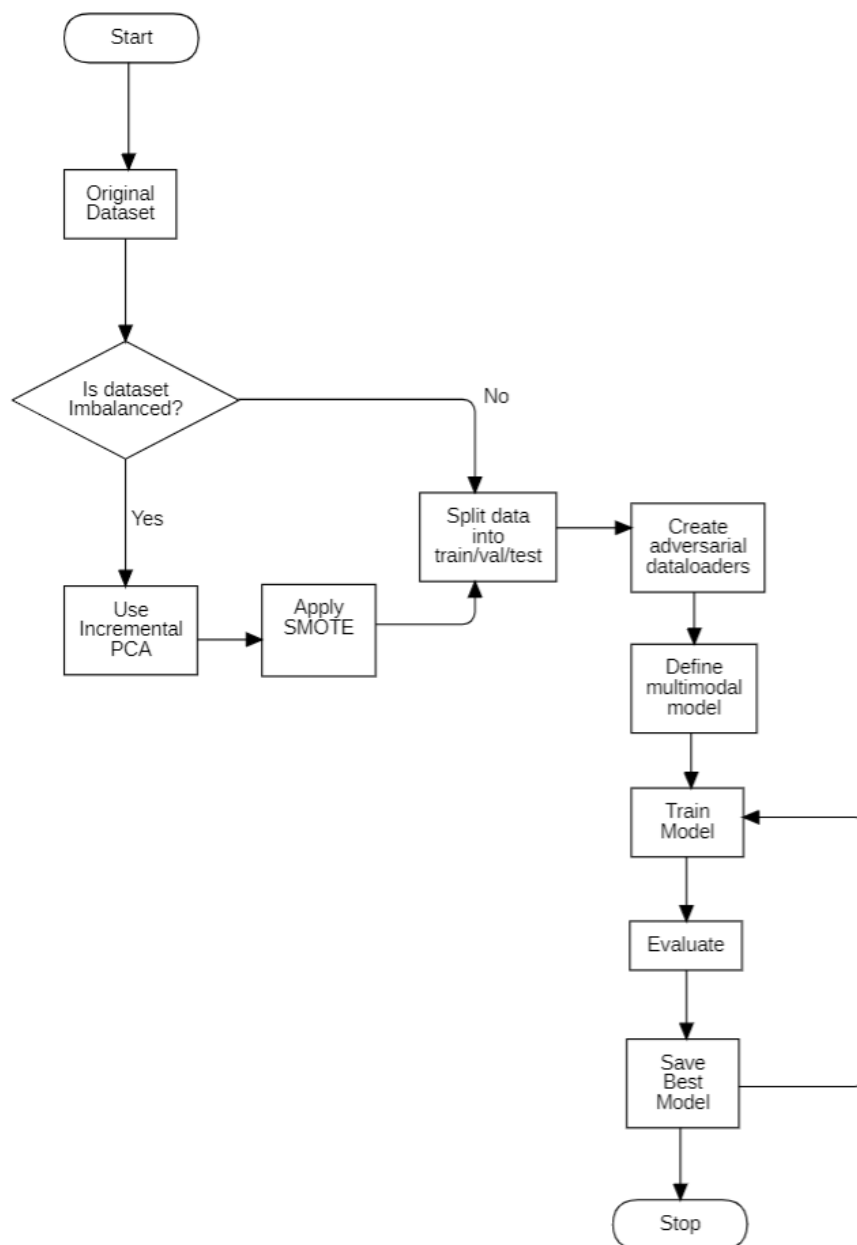


Figure 37. Flowchart of the HateXcel approach

As shown in Figure 37, the method starts with using the original dataset, and then we started by checking if it is imbalanced. If the dataset is imbalanced, then we balanced it by applying incremental PCA (to first reduce the dimensionality of the data) and D-SMOTE method. If otherwise, then we proceeded to splitting the data into training, validation and testing. We created adversarial data loaders to be able to add adversarial examples in the training set to be used during the training process. For the adversarial examples used during training, we mainly focused on the attacks described in Chapter 3 using the same parameters. For each epoch, we computed the loss, validated the model on the validation set, and stored the model with the highest validation accuracy.

4.3. Experiment and Results

In this section, we present the dataset used, the steps carried out, and the results of the experiments carried out to provide a model that classifies memes as hateful or non-hateful under various adversarial conditions and class imbalance.

4.3.1. Dataset

In this chapter, we used the Facebook Hate Meme dataset (FBHM) provided by Facebook AI [65]. The dataset contains 10,000 memes collected from Getty images and has been annotated in a semi-artificially manner by benign cofounders. The dataset is made up of five classes of memes which are multimodal hate, unimodal hate, benign image, benign text, and random not-hateful examples. We labeled the first four classes as hateful and the rest as non-hateful. We split the dataset into 85% training, 5% development and 10% test sets.

The FBHM dataset aims to help researchers develop AI systems [72] that can identify multimodal hate speech by understanding how text and images together convey meaning. The content types of the dataset include racist, and sexist memes and also

contents that incites violence or targets protected categories like religion, gender, and sexual orientation. The dataset size and label distribution is described in Table 13.

Table 13. FBHM dataset sizes and label distribution

Dataset	Size	Hateful Memes	Non-Hateful memes
Train	8500	3049	5451
Test	1000	456	544
Validation	500	251	249

In Table 13, the nature of the dataset is demonstrated. It can be seen that the class is highly imbalanced and as reported by Aggarwal et al [34], the percentage of hate is 37.56%. Training a model on such dataset will definitely make the model be more biased to the non-hateful class.

A) Example of Memes

In Figure 38, examples of memes are shown, let’s consider the sentences “Love the way you smell today” or “Look how many people love you”. These sentences are harmless, but when combined with an equally harmless image of a skunk or a tumbleweed, suddenly they become mean. This is to say that, memes are often subtle and while their true underlying meaning can be easy for humans to detect, they can be very challenging for AI systems. To illustrate the role of benign cofounders, in countering the possibility of models exploiting unimodal priors, Kiela et al, [65] introduced an alternative image or captions that make the label flip from hate to not-hateful. Using the examples of the sentences above, they flipped the label by keeping the original images but changing the text to “Look how many people hate you” and “Skunks have a very particular smell”.

Also, they replaced the skunk and tumbleweed images with pictures of roses and people respectively to flip from hateful to not-hateful.

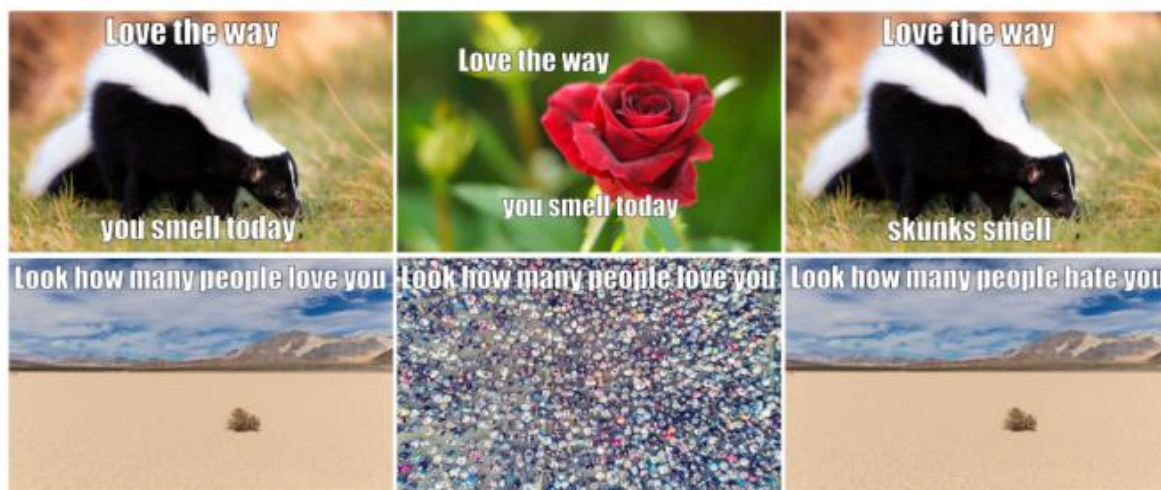


Figure 38. Multimodal "mean" memes and benign cofounders, [65]

Mean memes are found on the left side of the image, benign image cofounders are found in the middle, and benign text cofounders are found on the right side of the image.

B) Word Frequency

We conducted a visual analysis of the memes contained in the training set of the FBHM dataset to identify the most frequently occurring words. By examining the token frequency distribution across all textual content in the training data, we extracted the top 50 words that appeared most often. We noticed that several words such as black, white, and racist indicate racially charged hostility, while terms like Muslims and Jews suggest religious intolerance. Additionally, words including women and tranny reveal gender-based aggression and derogatory language. Figure 39, illustrates the top 50 words along with their respective frequency counts, providing a clear overview of the prevalent hateful terms in the dataset.

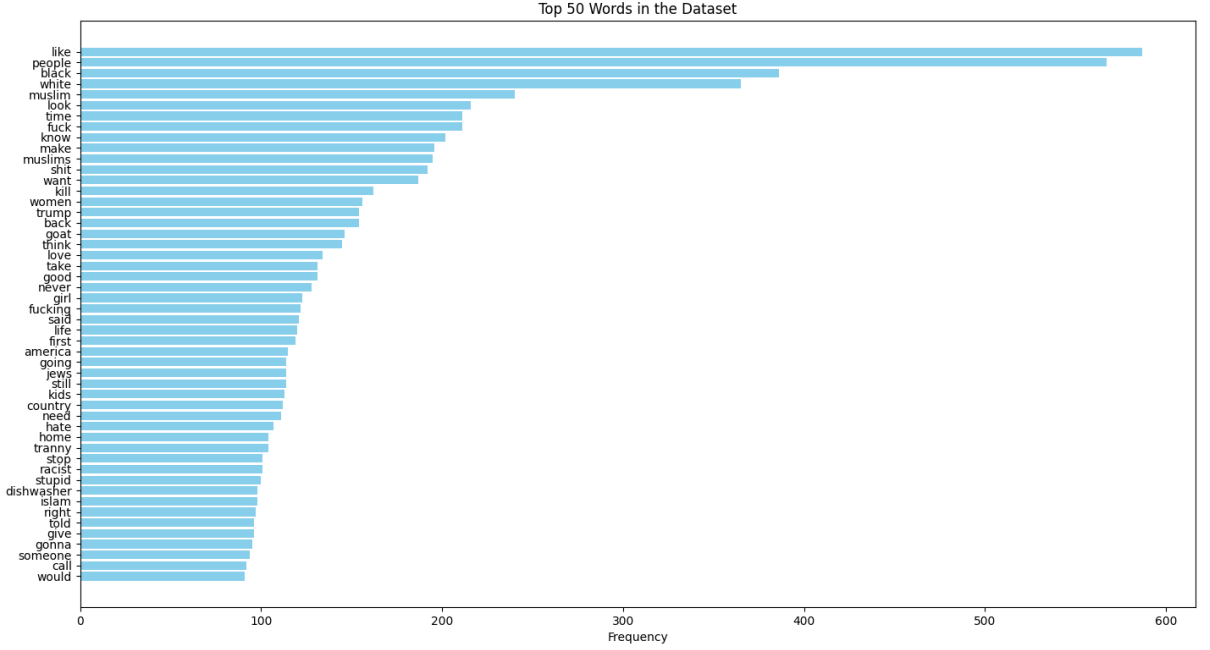


Figure 39. Top 50 words of the FBHM train dataset

4.3.2. Experimental setup

In this chapter, we used a pre-trained RoBERTa model to encode the textual input and a pre-trained ResNet50 to extract the image features. We used a late fusion approach for the multimodal features. The architecture of the model (MIRA) used is the same as the one described in Chapter 3. We set the number of components to 250 for each modality to be used during the reduction of the dimensions for the image and text features. We used a standard scaler to standardize the image and text features before applying PCA incrementally. We adopted the incremental style because we are in a low-resource setting and using incremental PCA allows for the analysis of the extensive FBHM dataset without requiring significant computational resources. By processing the data in chunks, it can help reduce the time required for computation. Then, we applied D-SMOTE to the reduced features to generate synthetic samples for the minority class.

During the model training, we chose an Adam optimizer with a learning rate from $1e-5$ to $1e-2$, with a mini-batch size of 12 and a weight decay of 0.1. We conducted the

experiment using PyTorch 2.0, python 3.11, and Windows 10 on a machine with Nvidia A200 GPU, and 32GB RAM. For adversarial training of our model against non-technical attacks and technical attacks, we decreased the batch size to 8. The same non-technical and technical attacks described in Chapter 3 were used. The perturbation coefficient (ϵ) was set at 0.09 and our probability score was 0.9 for technical attacks.

4.3.3. Classification Results

In this section, we perform a quantitative and a qualitative analysis of the results obtained from the experiments we carried out. In Table 14, we display the performance of our model compared to several state-of-the-art models. We also show the results of the ablation studies we carried out and examples of memes detected by our model on the test set under various adversarial conditions.

A) Quantitative Performance Evaluation

Table 14 displays the performance of all the compared models on the FBHM dataset. It can be observed that our model outperforms the other multimodal-based networks. It should be noted that we considered adding adversarial examples for both modalities during training and testing. The study by Kiela et al., [65] illustrated the difficulty in multimodal classification and they observed that the more advanced fusion, the better the model performs. They included the ViLBERT-CC model and it achieved a score of 66.10%. Aggarwal et al., [34] carried out experiments on three multimodal models namely, RoBERTa-ResNet, Visual-BERT, and UNITER without considering adversarial training. They added adversarial examples for both modalities on the test set of the FBHM dataset to investigate the robustness of hate meme detection against adversarial examples. The study they performed revealed UNITER as the best model. Chhabra et al., [73] proposed a multimodal model that uses multi-scale visual kernels

and knowledge distillation architecture to achieve an impressive 76.07 % on the FBHM dataset.

Table 14. Results of different models on FBHM dataset

Author	Model	Macro-F1 Scores (%)	Accuracy (%)	Precision (%)
Aggarwal et al (2023), [34]	RoBERTa-ResNet (Baseline model)	63.10	N/A	N/A
Aggarwal et al (2023), [34]	Visual-BERT	60.88	N/A	N/A
Aggarwal et al (2023), [34]	UNITER	63.53	N/A	N/A
Kiela et al (2020), [65]	VilBERT-CC	N/A	66.10	N/A
Chhabra et al (2023), [73]	DistilBERT- SKResNet	N/A	76.07	N/A
HateXcel	RoBERTa-ResNet50	81.53	81.80	83.81

B) Ablation Study

In order to evaluate the effectiveness of adversarial training in making our model more robust against technical attacks described in Chapter 3, we carried out many other experiments to examine their corresponding performances. Also, for non-technical attacks described in chapter 3 we report the performance of our model in Table 14.

In Table 15, we report the accuracy of the model after each of the attacks has been applied to it. The accuracy before adversarial training is obtained by applying adversarial attacks on our model with accuracy 81.80%.

Table 15. Models' performance after applying technical adversarial attacks

Attacks	Accuracy after attacks on base model (%)	Accuracy after attacks on our model (%)	Accuracy after adversarial training on our model (%)
FGSM	40.61	60.2	89.8
PGD	19.7	39.8	70.2
AdvDrop	22.9	59.96	68.4

It should be noted that when performing adversarial training of our model on the attacks listed, the dataset used was not balanced. The choice of not balancing the dataset before adversarial training was due to the fact that we carried out these experiments in a low resource setting.

From Table 15, It can be seen that the PGD attacks is the attack that affects the most our model, dropping the performance by 42%, followed by AdvDrop that drops the model's accuracy by 21.84% and finally FGSM that drops the model's performance by 21.6%. This drastic drop in performance compared to the drop in performance in Chapter 3 Table 12 is due to the fact that the FBHM dataset is not balanced while the MAMI dataset is balanced. Also, even after adversarial training, for 20 epochs on each attack, the model least improves on the AdvDrop attack and highly improves on the FGSM attack. This is due to the fact that the FGSM attack is the least strong making the model to be easily robust. Meanwhile AdvDrop is the strongest.

In order to evaluate the effectiveness of combining D-SMOTE and Incremental PCA in an adversarial setting for non-technical attacks, we conducted a series of experiments to examine the corresponding performances. In Figure 40 we display the results obtained.

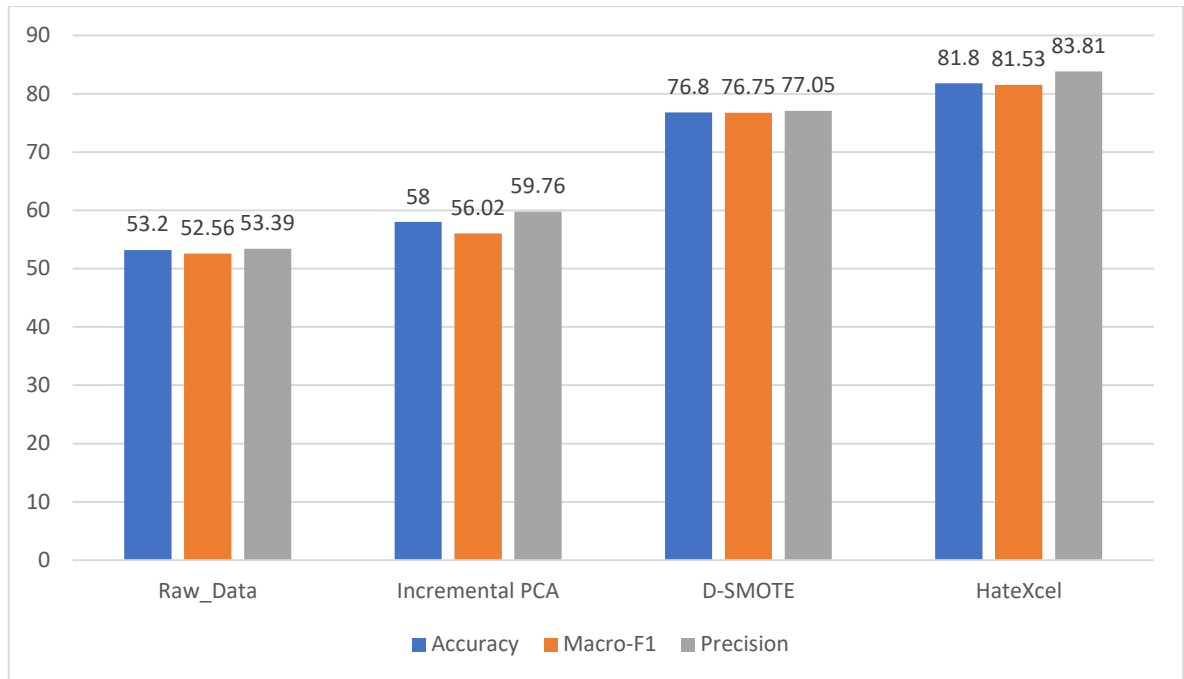


Figure 40. Evaluation of Model under Non-Technical Attacks

- **Raw_Data.** This was designed as our multimodal network without using SMOTE or Incremental PCA. We still added adversarial examples for non-technical attacks on both input modalities.
- **Incremental_PCA.** This was created as our multimodal network using incremental PCA to reduce the features and directly pass them on to our model. Still, this was carried out in an adversarial setting for non-technical attacks.
- **D-SMOTE.** This was created to balance the dataset and perform training in adversarial setting for non-technical attacks
- **HateXcel.** This was created as our multimodal network applying incremental PCA and D-SMOTE in an adversarial setting for non-technical attacks.

Based on the results in Figure 40, we can observe that applying D-SMOTE only has a greater effect on the performance than applying PCA only. It can also be observed that introducing either PCA or D-SMOTE enhances the performance of our model than

without. Overall, combining D-SMOTE and PCA achieves a better performance than the other methods.

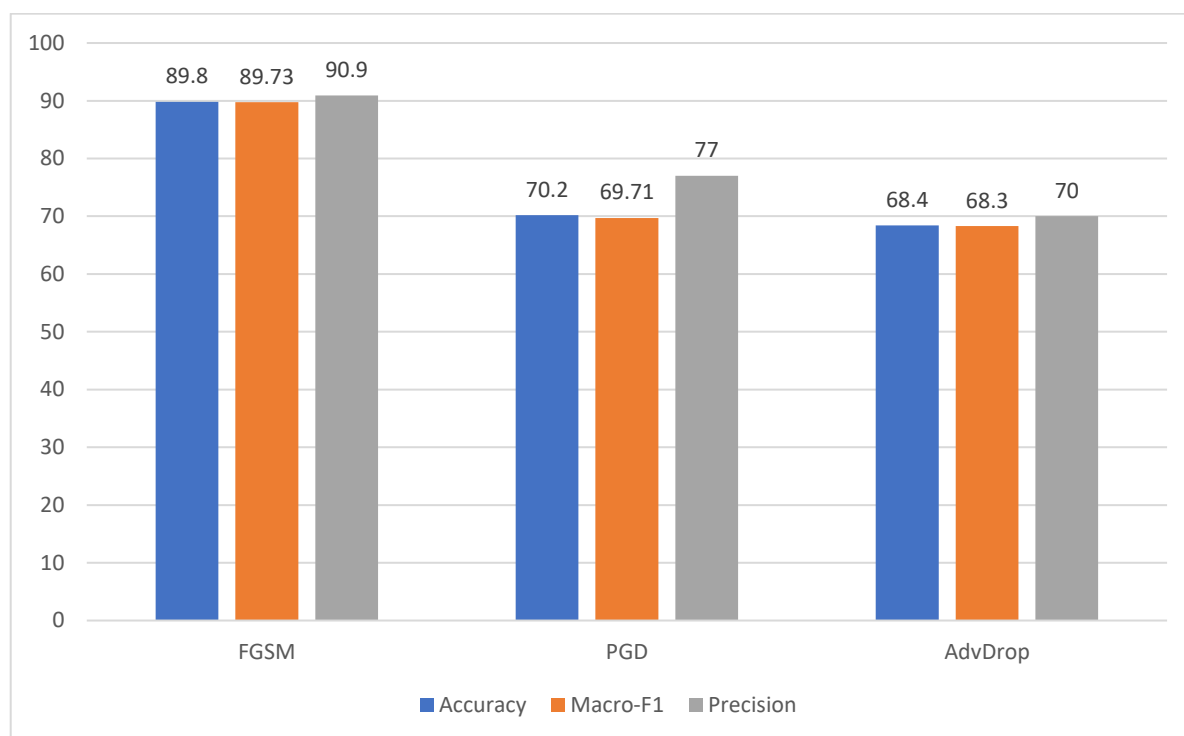


Figure 41. Model's performance on technical attacks

- **FGSM:** This is the model designed to undergo adversarial training to be robust against FGSM attacks on unbalanced dataset.
- **PGD:** This is the model designed to undergo adversarial training to be robust against PGD attacks on unbalanced dataset.
- **AdvDrop:** This is the model designed to undergo adversarial training to be robust against AdvDrop attacks on unbalanced dataset.

C) Qualitative examples

In Figure 42 we present examples of memes detected as not-hateful by our model and in Figure 43 we present examples of memes detected as hateful. For some post detected it was obvious that our model could obtain reliable features to make predictions but for others we leave it up for human level judgment. These detections were done under non-

technical attack. More examples are provided in Appendix B: Figure B1 for non-hateful posts and Figure B2 for hateful posts.



Figure 42. Some examples of posts detected as not-hateful



Figure 43. Examples of posts detected as hateful by our model

For technical attacks, Figure 44 shows some posts that had the same prediction label across the 3 attacks. Whether the model labelled it correctly or not is left for human judgment level.



Figure 44. Similar prediction across all 3 attacks

The post on the left in Figure 44 represents a post that was classified non-hateful and the post on the right side represents a post that was classified hateful by our model after undergoing all 3 attacks. More examples of hateful and non-hateful posts detected across all 3 attacks can be found in Appendix B: Figure B3 for non-hateful and Figure B4 for hateful posts.

In Figure 45, we show some examples of failure cases, the top two memes were detected as hateful and the bottom two memes were detected as not-hateful. The same observation made for failure cases in Chapter 3 on the MAMI dataset can still be observed here. Cultural context and nuanced understanding are still a challenge for our model. Stereotypes and cultural bias present in the FBHM dataset affect our model, for example, the post at row 1, column 1 of Figure 45 was detected as hateful because of our model's inability to eventually recognize visual stereotypes.



Figure 45. Failure cases on the FBHM dataset

4.4. Conclusion

In this Chapter, we highlighted the need to solve class imbalance in datasets to improve classification accuracy. We applied the SMOTE and Incremental PCA on the FBHM dataset to enhance the classification accuracy of our multimodal model. Our model achieves an accuracy of 81.80 % on the validation set when combining PCA with SMOTE, thereby outperforming existing models. Also, to improve the model's robustness we focused on adversarial training against attacks that don't require any technical background, but when used can greatly influence the classification accuracy of the model and notable attacks in the field like FGSM, PGD, and AdvDrop. Our model achieves a robustness accuracy of 89.8% against FGSM attack, 70.2% against PGD and 68.4% against AdvDrop. The increase in accuracy of our model after performing the FGSM attack can be attributed to the fact that our model was already pretrained on a balanced dataset that was attacked with FGSM. We stored the model's prediction on the test set for all the attacks in JSON files for human-level judgment.

CHAPTER FIVE

GENERAL CONCLUSION

5.1. Summary

Social media platforms have become an integral part of our daily lives. With communication on these platforms mostly occurring through images and text, it allows users to express and share their thoughts and opinions on all kinds of topics and events. Sometimes the thoughts expressed can cause more harm than good. Tools to detect harmful content online have been developed to moderate the spread of hateful content, but some users with malicious intentions continue to find ways to evade these detectors. This calls for a need to create robust online detectors that will still detect these disguised online posts.

In response to the limitations of existing tools, it is essential to explore the current state of research in multimodal content analysis and adversarial robustness. We conducted a comprehensive literature review focusing on the various methods that have been developed to address multimodal data in online environments. This review encompassed a wide range of approaches, highlighting their theoretical aspects, implementations, and comparative effectiveness. In addition, we carried out a detailed analysis of existing adversarial attacks on text and images, with particular attention to the techniques used to craft these attacks. Furthermore, we examined studies that used adversarial training as their defense methods. After which we discussed on several research frontiers in the field of multimodal adversarial learning, pointing to potential directions for future research and improvement.

Building upon the insights and gaps identified in the literature review, we develop AdverMIRA, a robust multimodal intelligence to recognize abuse, and specifically designed to handle both non-technical and technical attacks. AdverMIRA improves the accuracy of the baseline model by 13.5 % by utilizing a slower learning rate ($1e-5$) and a smaller batch size (6). This parameter configuration differing from traditional choices allows for more precise convergence and reduced overfitting. The result highlights the significance of parameter tuning in achieving state-of-the-art performance. Specifically, AdverMIRA achieves 98.98% macro-F1 score, accuracy of 99%, and 97.98% precision against non-technical attacks thereby outperforming existing models. For technical attacks it achieved a macro F1-score of 97.97%, accuracy of 98%, and precision of 97.98% for FGSM; 95.9%, 96%, and 91.66% for PGD; and 73.7%, 74%, and 69.56% for AdvDrop, respectively. This significant improvement is attributed to the new features, which capture previously unaccounted-for patterns in the data. This increased accuracy also enables more reliable predictions in real-world applications under adversaries. The model's prediction on the test set is stored in a JSON file for human-level judgment.

Recognizing the limitations imposed by class imbalance and high-dimensional multimodal data, we developed HateXcel, which is a combination of Incremental Principal Component Analysis (PCA) and Distance-based Synthetic Minority Over-sampling Technique (D-SMOTE) and adversarial learning. We used PCA and D-SMOTE not only to ensure scalability and memory efficiency when dealing with large datasets but also to improve feature representation and model generalization. Additionally, to improve the robustness of HateXcel and acknowledging that even non-technical attacks can significantly affects our model's performance, we decided to add

adversarial learning after balancing the dataset. We noticed that the combination of adversarial training on non-technical attacks with PCA, and SMOTE enhanced the accuracy of HateXcel to 81.80%. For technical attacks, HateXcel attained a macro F1-score of 89.73%, accuracy of 89.80%, and precision of 90.9% for FGSM; 69.71%, 70.2%, and 77% for PGD; and 68.3%, 68.4%, and 70% for AdvDrop, respectively.

5.2. Critical Analysis of Results

The results we obtained from carrying out this research demonstrate the effectiveness and the limitations of our proposed AdverMIRA model for the detection of misogynistic and HateXcel model for hate speech content on social media, particularly in an adversarial environment. AdverMIRA achieved an impressive performance metrics, with accuracy rates, macro-f1 and precision scores above 97% on the MAMI dataset under non-technical attacks, and it maintained a relatively high performance under technical attacks such as FGSM, PGD, and AdvDrop. This underscores the value of integrating advanced textual and visual features with adversarial training. The use of D-SMOTE and incremental PCA further contributed to an improved handling of class imbalance, as evidenced by the 18% gain in accuracy of HateXcel over baseline models.

However, a deeper analysis reveals several important nuances. While AdverMIRA demonstrated robustness against basic adversarial attacks, its performance notably declined under sophisticated attacks such as PGD and AdvDrop. This suggests that although adversarial training improves the resilience of the model, it does not fully immunize it against all forms of adversarial manipulation. The model's performance was also dataset-dependent, with the HateXcel model having a slighter lower robustness level observed on the Facebook Hate Meme (FBHM) dataset compared to AdverMIRA model on the MAMI dataset, this indicates that generalization across different data

distributions remains a challenge. Additionally, while the combination of D-SMOTE and incremental PCA improved class balance and overall accuracy, it introduced additional computational overhead, which could limit scalability for real-time deployment.

Another critical observation is that the model's effectiveness is closely tied to the quality and representativeness of the data it is trained on. Finally, while the model's predictions were stored in a JSON file for human-level judgment, we need to further our research to assess the interpretability and transparency of our model's decisions, which are crucial for deployment in sensitive applications such as online content moderation.

5.3. Difficulties Encountered

The research process for this thesis was marked by several significant challenges, both technical and practical, which influenced the direction and outcomes of the study. A primary difficulty was the collection and preparation of multimodal datasets. Publicly available datasets such as MAMI and FBHM are invaluable but often suffered from issues such as varying standards of data quality and class imbalance. Addressing these issues required extensive manual cleaning and the application of advanced techniques like D-SMOTE, which increased the time and complexity of the data preparation phase.

Also, implementing and evaluating adversarial attacks and defenses posed another major challenge. Adversarial machine learning is a rapidly evolving field and the computational demands of adversarial training are substantial, necessitating access to high-performance hardware and efficient resource management. Training deep multimodal models with adversarial robustness resulted in long training times and frequent hyperparameter tuning. From a practical standpoint, ensuring the

interpretability and transparency of our model decisions was challenging. While storing predictions in JSON files facilitated human-level judgment, explaining the model’s reasoning, especially in borderline or misclassified cases, remains an open problem for future work.

Throughout the course of this research, the frequent electricity outages and poor internet connectivity posed significant challenges that impacted the progress of this research. Unstable power supply often interrupted long-running experiments, particularly during training of our deep learning model, leading to unexpected shutdowns and loss of computational progress. This necessitated repeated restarts which extended the overall time required for our model development and evaluation. Similarly, inconsistent internet access hindered access to cloud-based resources essential for adversarial machine learning experiments. Despite these obstacles, careful planning, use of offline resources, and backup power solutions helped us to mitigate their impact, allowing us to complete the research successfully.

5.2. Further Works

In future, we intend to enhance the robustness and applicability of AdverMIRA and HateXcel. In order to do so, we divided our plans into short-term, mid-term and long-term.

On the short run, we plan to incorporate additional and more severe adversarial attacks, including but not limited to DeepFool, AutoAttack, and Carlini&Wagner (C&W) to allow for a more comprehensive evaluation of our models resilience under stronger attacks. More so, we plan to optimize the data preprocessing steps by experimenting

with other dimensionality reduction techniques and feature normalization strategies, especially for high-dimensional multimodal datasets.

On the mid run, we plan to incorporate multilingual user comments paired with memes to improve the model's ability to generalize across linguistic and cultural contexts. To achieve this, we will adapt our text processing pipeline to support multiple languages and evaluating multilingual language models such as mBERT or XLM-R. Also, we plan to test the model's adaptability across other domains (e.g., political, health-related, or entertainment content) to assess its transferability and robustness in real-world applications.

On the long run, we intend to develop a fully operational real-time multimodal adversarial content detection system, that will be capable of flagging harmful or manipulated content in dynamic online environments, such as social media platforms or news aggregators. More so, we plan on integrating human-level feedback mechanisms into our models retraining cycles to validate and improve our models predictions over time. Also, to support a safe and responsible deployment in real-world applications, we plan to align our system with ethical AI standards and content moderation policies.

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LIST OF PUBLICATIONS

The following articles have been published or submitted for publication in connection with the research presented in this thesis:

- Tchuindjang Tchokote, E.L. and Fute Tagne, E., "Effective multimodal hate speech detection on Facebook hate memes dataset using incremental PCA, SMOTE, and adversarial learning," *Machine Learning with Applications*, Elsevier, Vol.20, 2025.
- Tchuindjang Tchokote, E.L. and Fute Tagne, E., "Detecting online misogyny with robust multimodal fusion: The RobRes framework," *Array*, Elsevier, Under review, 2025.



Contents lists available at ScienceDirect

Machine Learning with Applications

journal homepage: www.elsevier.com/locate/mlwa

1. Introduction The rapid advancement of machine learning (ML) and artificial intelligence (AI) has led to significant breakthroughs in various fields, including healthcare, finance, and autonomous systems. These technologies have the potential to revolutionize industries by enabling more efficient and accurate data analysis and decision-making. However, the widespread adoption of ML and AI is still hindered by several challenges, such as the need for large amounts of high-quality data, the complexity of model development, and the lack of interpretability of many models. This paper addresses these challenges by proposing a novel framework for ML and AI that combines deep learning with explainable AI (XAI) techniques. The proposed framework aims to improve the performance and interpretability of ML models, making them more suitable for real-world applications. The paper is organized as follows: Section 2 discusses the related work in the field of ML and XAI. Section 3 describes the proposed framework in detail. Section 4 presents the experimental results and analysis. Finally, Section 5 concludes the paper and discusses future research directions.	
2. Related work This section reviews the existing literature on ML and XAI. In the field of ML, deep learning has emerged as a powerful paradigm for handling complex data. Various deep learning architectures, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have achieved state-of-the-art performance in many tasks. However, these models are often considered "black boxes" due to their lack of interpretability. XAI techniques aim to address this issue by providing explanations for the model's predictions. Popular XAI methods include LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Value SHaring). This paper compares the proposed framework with these existing methods to demonstrate its superior performance and interpretability.	3. Proposed framework The proposed framework consists of three main components: data preprocessing, model training, and model interpretation. In the data preprocessing stage, the input data is cleaned and normalized to ensure the quality of the training process. The model training stage involves training a deep learning model using a combination of supervised and unsupervised learning techniques. The model interpretation stage uses XAI techniques to generate explanations for the model's predictions. The proposed framework is designed to be flexible and scalable, allowing it to be applied to a wide range of ML tasks. The experimental results show that the proposed framework outperforms existing methods in terms of both performance and interpretability.
4. Experimental results and analysis This section presents the experimental results and analysis. The proposed framework is evaluated on a variety of datasets and tasks to assess its performance and interpretability. The results show that the proposed framework achieves higher accuracy and better interpretability compared to existing methods. The analysis also reveals that the proposed framework is robust to different data distributions and model architectures. These findings suggest that the proposed framework is a promising approach for ML and XAI.	5. Conclusion and future research This paper concludes by summarizing the main contributions of the proposed framework. The framework successfully addresses the challenges of ML and XAI by combining deep learning with XAI techniques. The experimental results demonstrate the framework's superior performance and interpretability. Future research directions include extending the framework to handle more complex data types and tasks, as well as exploring new XAI techniques to further improve the interpretability of the models.

managing abnormal instances within the minority class while applying re-balancing techniques. Many current approaches find it difficult to sustain prediction accuracy when creating synthetic data in the context of these abnormal instances.

Overall, since communication on social media primarily occurs through images and text, the innovative techniques developed for detecting online hate content remain vulnerable to multimodal adversarial attacks. These attacks aim to disrupt the classification abilities of detection systems, leading to the misclassification of hate content. Achieving high accuracy in predictions is important for effective hate content detection, as both false positives and false negatives can have serious repercussions. Frequent false positives may provoke backlash from users who believe their posts are being unjustly suppressed and censored, while false negatives can allow hate content to spread unchecked. The primary goal of this research is to develop a robust multimodal model for detecting hate online content, specifically utilizing the combination of Incremental PCA, SMOTE, and adversarial training on the Facebook Hate Meme Dataset to address challenges related to class imbalance and adversarial robustness. This combination enhances data representation, balances class distributions, and improves robustness against adversarial attacks.

The rest of this research is organized as follows: Section 2 delves deeper into previous works related to multimodal deep learning frameworks, adversarial machine learning techniques, dimensionality reduction, and class imbalance handling. Section 3 describes the proposed approach, and Section 4 presents the results from the experiments conducted and compares them with previous works. Finally, Section 5 concludes and outlines future open research directions.

2. State of the art

2.1. Multimodal deep learning frameworks

Chhabra et al. (Chhabra & Vishwakarma, 2023) introduced a novel architecture that employs multi-scale visual kernels combined with knowledge distillation techniques for detecting hateful memes. This framework extracts relevant features from visual content while ensuring the model learns effectively from visual and textual data sources. Their results demonstrate improved performance in identifying memetic hate speech compared to conventional methods.

A notable study by Singh et al. (Singh, Vajrobal & Aggarwal, 2023) investigates the effectiveness of using XLM-Roberta-base for detecting hate speech in multimodal data, particularly in text-embedded images. The research utilized a dataset comprising over 4700 images related to the Russia-Ukraine conflict and found that XLM-Roberta-base outperformed other classifiers, achieving an F1 score of 84.62 for identifying hate speech targets. This study emphasizes the importance of integrating advanced deep-learning models to improve detection accuracy across different content types.

Today, memes represent the most prevalent forms of multimodal communication on social media. Although often created with ironic intent, they can also convey hateful messages directed at various individuals or groups. Recent research by Hee et al. (Hee, Lee & Chong, 2022) suggests that the most effective multimodal approach combines textual and visual features utilizing transformer-based architectures, such as ViLBERT and Visual-BERT. These methods have been benchmarked against other unimodal and multimodal strategies, establishing them as a baseline for this challenge and underscoring the necessity for multimodal analysis in detecting hate content. Despite achieving approximately 75 % accuracy, these models still lag behind human performance, highlighting the need to develop new machine-learning models.

2.2. Adversarial machine learning techniques

Several innovative defense strategies have been proposed in the

literature to protect hate speech detection models from adversarial attacks. Most of these strategies utilize adversarial training to manage perturbations effectively. Tran et al., (Tran et al., 2020) implemented adversarial training to defend against Fast Gradient Method (FGM) attacks by augmenting FGM with a customizable and detailed noise magnitude, adding noise (perturbation) to misleading samples. The hate speech detection model they developed demonstrates a superior understanding of hate speech datasets, operates 4–5 times faster than the conventional BERT model, consumes less than one-third of the memory, and achieves enhanced classification performance.

Wang et al., (Wang, Wang, Ke & Wang, 2021) introduce a defensive tool called TextFirewall designed for sentiment analysis algorithms. This tool leverages the disparity between the predictions made by sentiment analysis models and the calculated impact value of words, which reflects their positive or negative influence on sentiment polarity. TextFirewall successfully detects adversarial texts produced by three leading attack methods and surpasses earlier defense strategies. Specifically, it achieves an average accuracy of 90.7 % on the IMDB dataset and 96.9 % on the Yelp dataset when countering these attacks. Notably, TextFirewall can function as a third-party tool without requiring modifications to the target model and is effective against various types of adversarial texts.

Aggarwal et al., (Aggarwal et al., 2023) conducted a case study to examine the weaknesses of current hateful meme detection systems in the face of external adversarial attacks. They identified nine distinct types of attacks to assess how susceptible these detection systems are to human-induced adversarial inputs, they discovered that all models tested exhibited significant vulnerabilities, with some experiencing up to a 10 % decrease in macro-F1 performance under certain conditions. To resolve these vulnerabilities, they proposed two countermeasures: one based on contrastive learning and the other on adversarial training. For the countermeasure involving adversarial training, the model was not only trained on existing data points but also on adversarial examples created by altering one modality of the input data while keeping another constant. Specifically, for image data, they focused on image-region features instead of modifying individual pixels.

2.3. Dimensionality reduction and class imbalance handling

Basgall et al., (Basgall, Hasperue, Naiouf & Fernandez, 2018) introduced a fully scalable preprocessing technique (SMOTE-BD) designed to solve imbalanced classification in Big Data. They used the SMOTE algorithm, to create new synthetic samples based on the proximity of minority class instances. SMOTE-BD is designed to operate independently of the number of partitions or processes, thereby enhancing its efficiency.

Mohammed et al., (Mohammed, Hassan & Kadir, 2020) used a novel dataset called ZADA, which includes medical information from over 7000 patients in Zakho, Iraq, to build a system to assess and detect patients with disproportionate diabetes. They diagnosed diabetes using six machine learning techniques and they corrected class imbalance by using a classification analysis that included the SMOTE resampling approach with three normalization strategies. Their study showed that the performance of the classification models based on minority class distribution was much enhanced by both the resampling and normalizing techniques.

Mulla et al., (Mulla, Demir & Hassan, 2021) performed a method that combined SMOTE oversampling with PCA to solve imbalance problems in high dimensional data. They used Logistic Regression, K-Nearest Neighbor, and Decision Tree methods on two separate datasets to evaluate their method efficiency and to determine their classifiers' performance. They used WEKA with the algorithms above to analyze raw datasets and converted datasets by PCA, SMOTE, and SMOTE+PCA. The results they obtained showed that almost all classification algorithms improved their classification accuracy using PCA, SMOTE, and SMOTE+PCA methods, with the SMOTE method giving the most efficient result for data rebalancing and K-Nearest Neighbor having the

highest classification performance among the other algorithms.

3. Methodology

3.1. Data preparation and processing

Principal Component Analysis (PCA). It is a statistical procedure (Gewers et al., 2021) concerned with elucidating the covariance structure of a set of variables. In particular, it permits us to identify the principal directions in which the data varies. The main goal of PCA is to reduce the dimensionality of a dataset while maintaining the most important patterns between the variables without any prior knowledge of the target variables. Before applying PCA, the features are standardized to have a mean of zero and a variance of one.

$$X' = \frac{X - \mu}{\sigma} \quad (1)$$

Where:

- X is the feature vector.
- X' is the standardized value.
- μ is the mean of the feature values.
- σ is the standard deviation.

Incremental PCA in our case allows for the processing of data in smaller batches, this enables us to update the principal components rather than loading the entire dataset into memory as shown in Eq. (3). This approach is very effective when dealing with large datasets or streaming data. Before applying Incremental PCA to the FBHM dataset, the features we extracted from both modalities (image and text) are standardized as shown in Eq. (1). This helps us to ensure that each feature contributes equally to the distance calculations during PCA. Given that our dataset X is represented as a matrix where each row is a sample and each column is a feature, we can find the lower-dimensional representation of Z such that:

$$Z = XW \quad (2)$$

Where:

- W is the matrix of principal components obtained from PCA
- The principal components are computed by maximizing the variance of Z .

The update rule for Incremental PCA can be expressed as:

$$W_{new} = W_{old} + \Delta w \quad (3)$$

Where Δw represents the change in principal components based on the new batch of data.

To efficiently control memory use, we processed the dataset in batches. For each batch of features, we updated the scaler with the new batch of features. After scaling, we updated the principal components based on the scaled features. After processing every batch, we looped through the dataset once more to transform all features, thereby reducing the dimensionality of the features while retaining the most essential information. We created a single feature set that captures information from both modalities by concatenating the reduced picture and text features after dimensionality reduction. After that, this merged feature set was ready for additional processing, including using SMOTE to address class imbalance.

SMOTE Implementation. After dimensionality reduction, we applied SMOTE to generate synthetic samples to balance the class distribution. For a minority class sample x_i , SMOTE selects one or more nearest neighbors x_j based on a distance metric (Euclidean distance). Eq. (4) shows how a synthetic sample x_{new} can be generated.

$$x_{new} = x_i + r(x_j - x_i) \quad (4)$$

Where:

- r is a random number in the range $[0,1]$
- the equation creates a point along the line segment between x_i and its neighbor x_j

To mitigate potential biases present in the FBHM dataset, we implemented the following strategies in conjunction with incremental PCA and SMOTE:

- **Data augmentation:** While SMOTE helps address the class imbalance thereby reducing bias towards the majority class, the use of pre-trained models (ResNet50 and RoBERTa) can be considered a form of transfer learning. These pre-trained models have learned features from features from larger datasets, which is beneficial for smaller and potentially biased datasets.
- **Class weighting:** To explicitly reduce bias, we calculated the class weights based on the inverse class frequencies in the training data and applied them to the loss function. By doing so, we address bias by penalizing misclassification more heavily on minority classes

3.2. Model architecture

Classification of data with class imbalance problems as well as building robust models against adversarial attacks is a great center of interest for researchers in different scientific domains. This study proposes a new method based on combining adversarial learning with SMOTE (for rebalancing) and PCA (for dimensionality reduction) to enhance the robustness and improve the accuracy of our classification model.

Multimodal Neural Network. We proposed an ensemble model that combines pre-trained transformer model (RoBERTa) for text processing and ResNet-50 for image processing. Fig. 1 shows an overview of the ensemble model used.

The pretrained models are combined using a simple feedforward neural network. The aim is to enable the model learn features from both forms of data and fuse them into a unified representation as shown in Fig. 1. Studies performed by Alec et al., (Alec et al., 2021) revealed deep CNN as a state-of-the-art method in image classification tasks. We choose ResNet50 because it uses pretrained deep CNN to extract visual features. We used google vision API to extract the text from the memes, the extracted text is then passed to RoBERTa to learn the textual features. RoBERTa's main innovation is that it uses a more robust and efficient pretraining approach, which allows it to obtain better results on most NLP tasks. We concatenated the combined features from both modalities and passed it through a fully connected layer to produce class scores.

3.3. Training procedure

As shown in Fig. 2, the method starts with using the original dataset, and then we started by checking if it is imbalanced (i.e. if the number of labels in the hateful class is not equals to the number of labels in the non-hateful class for the training set). If the training set is imbalanced, then we balanced it by applying incremental PCA (to first reduce the dimensionality of the data) and SMOTE method. Else, we proceeded to creating adversarial dataloaders to be able to add adversarial examples in the training set to be used during the training process. For the adversarial examples used during training, we mainly focused on the model-independent attacks which can be introduced by humans with little technical expertise. Some examples of these attacks are developed later in another section of this research. A random number between 1 and 10 was generated for each batch of data, if the number generated

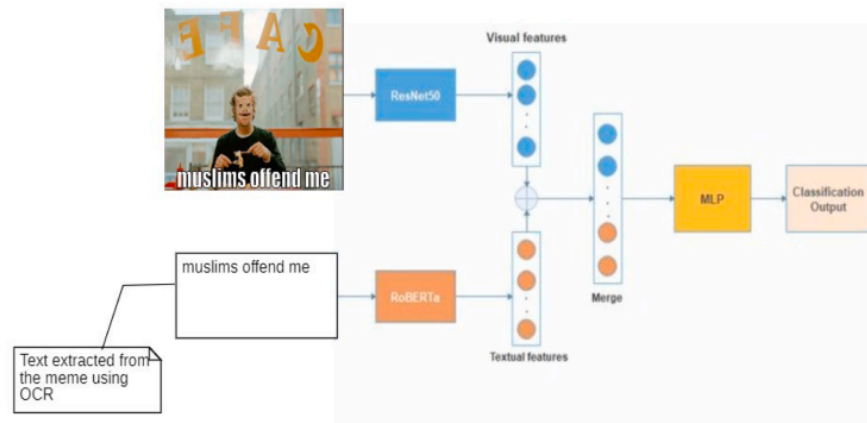


Fig. 1. Overview of multimodal model.

was greater than 1 then the text and image data were further augmented using one of the various augmentation operations offered by the Augly library we used. For each epoch, we computed the loss, validated the model on the validation set and stored the model with the highest validation accuracy.

Adversarial Examples. Adversarial attacks in machine learning are designed to manipulate or deceive models by creating inputs that are specifically crafted to cause misclassifications. These attacks can be categorized based on their nature, such as the level of access to the model (white-box, black-box, gray-box) and the type of manipulation (evasion, poisoning, exploratory). In this section, we focused on specific types of attacks related to text and image manipulation, such as those found in Augly. The attack nature of the attacks considered are black-box attacks and the types are evasion attacks. Let's consider the text extracted from the meme shown in Fig. 3, "Muslims offend me". We thought of the different attack forms for text, as shown in Table 1. In text manipulation, adversarial attacks often involve modifying the input text to deceive the NLP model. Some specific examples are shown in Table 1.

For scenarios where we cannot predetermine the internal behavior of the threat model, the attacker cannot appropriately generate attacks using gradient information. However, the attackers can introduce random modifications to the image while preserving its message. We considered the attacks as shown in Fig. 3. In image manipulation, adversarial attacks often involve modifying images to deceive computer vision models. The perturbation factor for the attack listed below was set to 0.2. Below are some specific examples:

- **Blur:** Applying a blur effect to an image reduces its clarity and potentially misleads the model.
- **Color Jitter:** Modifying the color balance or intensity of an image can affect how models perceive it
- **Stripes:** adding stripes or patterns to an image can disrupt the model's ability to recognize objects
- **Random Noise:** Adding random noise to an input image can disrupt clarity and mislead the model.
- **Shuffle pixels:** Randomly rearranging pixels in an image can significantly alter its appearance and challenge the model to correctly classify it.

4. Implementation and results

In this section, we described the environment in which the study was carried out and we presented the results obtained for the classification of memes into hate and non-hateful content.

4.1. Dataset

We used the Facebook Hate Meme (FBHM) dataset (Price, 2020) provided by Facebook AI. The dataset contains 10,000 memes collected from Getty images and has been annotated in a semi-artificially manner by benign confounders. The dataset is made up of five classes of memes which are multimodal hate, unimodal hate, benign image, benign text, and random not-hateful examples. We labeled the first four classes of memes as hateful and the rest as non-hateful. The dataset was split into 85 % training, 5 % development, and 10 % test sets.

The content types of the dataset include racist, and sexist memes and also contents that incites violence or targets protected categories like religion, gender, and sexual orientation. Table 2 describes the dataset size and label distribution

4.2. Experimental setup

Image Branch. We resized and normalized the input image before passing it to our pretrained ResNet50 model. The output of the last convolutional layer of the ResNet50 model was extracted as the image feature map with a tensor shape of (16, 2048).

Text Branch. We extracted the text from the meme, and divided it into subwords through a process called tokenization. We accomplished it using the WordPiece algorithm, which divides words into sub-words based on their frequency and patterns of co-occurrence. We used a learned embedding matrix to subsequently transform each token into a numerical representation known as embedding. These embeddings encapsulate the semantic meaning of each token. We passed the text tensor of shape (16, 768) through the RoBERTa encoder to enable the model to capture the contextual relationships between the tokens.

Classifier and Training. To obtain the multimodal comprehensive features, we concatenated the image and text tensors along the feature dimension to create a new tensor of shape (16, 2 × 768). We passed the concatenated tensor through two fully connected layers with ReLU activation functions, followed by a final fully connected layer with no

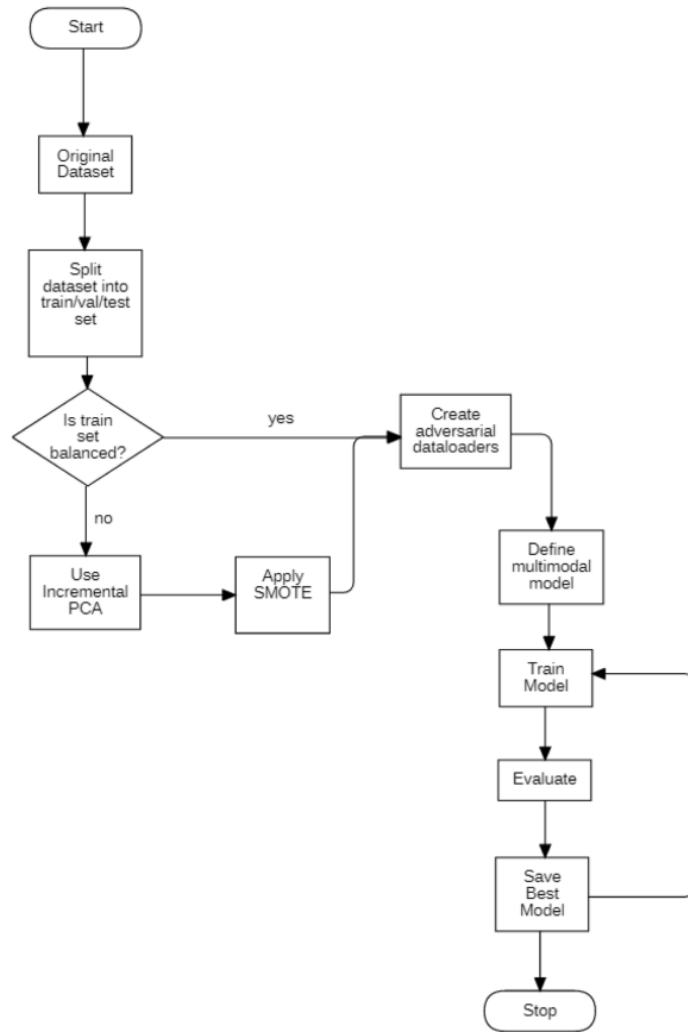


Fig. 2. Design of the proposed methodology.

activation function. And we finally passed the output tensor through another feedforward neural network defined by the MLP module as shown in Fig. 1. Our approach uses late fusion to effectively capture the unique contributions of each modality. This is crucial when the text and images convey conflicting information, as it will allow the model to weigh the importance of each modality's contribution to resolve the ambiguity.

We trained the model for 20 epochs on Adam optimizer with learning rate $1e-5$, weight decay 0.1. We chose a batch size of 12, and Adam optimizer with early stopping to avoid overfitting and implement a balanced approach for generating adversarial examples i.e., to have

equal number of adversarial examples and original samples in the batch during the training process whenever a batch is to contain adversarial examples. The number of components selected for incremental PCA was 512 and the batch size 200 because of the limited memory setting in which we carried out the experiment. This leads to more computational time to process the batches and transform the data before training. For SMOTE parameters, we reduced the number of k neighbors to 3, to make the process of selecting neighbors and interpolating between points faster, this also decreases the computational effort required for distance calculations and sorting. These experiments were run using PyTorch 2.0, python 3.11, and Windows 10 on a laptop with Nvidia A200 GPU, and

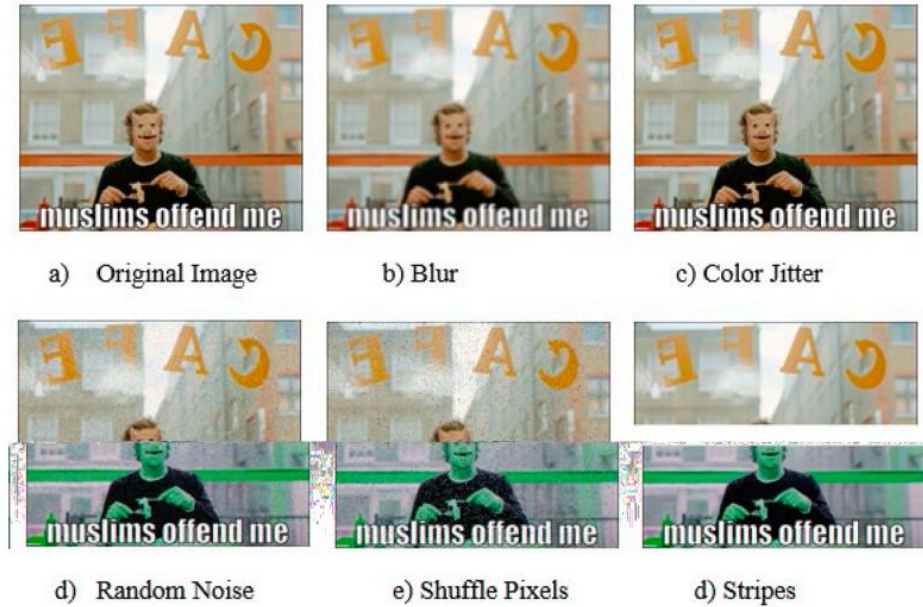


Fig. 3. Various forms of adversarial attacks on the image supported by our model.

Table 1
Various attack forms for text.

Attacks on Text	Description	Output
Replace Bidirectional	This involves replacing words/characters with their bidirectional equivalents to affect the model's sensitivity to directionality	EM DNEFFO SMILSUM
Replace Similar Chars	This involves substituting characters with visually similar ones to create adversarial examples that are hard to detect.	MUZZLIMS OFF3ND M3
Simulate Typos	This involves introducing common typos or keyboard errors into the text to test the model's robustness against human-like mistakes	MUSLIMS OFFEND LME
Change Case	This involves changing the case of letters in the text to affect the model	MuSLim OFFeNd mE
Insert Punctuation Chars	This involves adding or removing punctuation to alter the meaning/structure of sentences	M.U.S.L.I.M.S.O. F.F.E.N.D..M.E..
Insert White Space Chars	This involves adding extra spaces to affect how the models parse sentences	M U S L I M S O F F E N D M E

Table 2
FBHM dataset sizes and label distribution.

Dataset	Size	Hateful Memes	Non-Hateful memes
Train	8500	3049	5451
Test	1000	456	544
Validation	500	251	249

32GB RAM.

4.3. Classification results

In this section, we compare our model to several state-of-the-art models. Table 2 displays the performance of all compared models. We show the results of the ablation study we carried out and we also show some examples of memes detected by our model on the test set.

Quantitative Performance Comparison. Table 3 displays the performance of all the compared models on the FBHM dataset. It can be observed that our model outperforms the other multimodal-based

Table 3
Results of different models on the FBHM dataset.

Author	Model	Macro-F1 Scores (%)	Accuracy (%)
Aggarwal et al., (Aggarwal et al., 2023)	RoBERTa-ResNet (Baseline model)	63.10	N/A
Aggarwal et al., (Aggarwal et al., 2023)	Visual-BERT	60.88	N/A
Aggarwal et al., (Aggarwal et al., 2023)	UNITER	63.53	N/A
Kiela et al., (Kiela, Mohan & Firooz, 2020)	ViBERT-CC	N/A	66.10
Chhabra et al., (Chhabra & Vishwakarma, 2023)	DistilBERT-SKResNet	N/A	76.07
RobRes_adv	RoBERTa-ResNet50	52.56	53.2
RobRes_adv_iPCA	RoBERTa-ResNet50	56.02	58
RobRes_adv_SMOTE	RoBERTa-ResNet50	76.75	78
RobRes_adv_iPCA_SMOTE	RoBERTa-ResNet50	81.53	81.80

networks. It should be noted that we considered adding adversarial examples for both modalities during training and testing. The study by Kiela et al., (Kiela, Mohan & Firooz, 2020) illustrated the difficulty in multimodal classification and they observed that the more advanced fusion, the better the model performs. They included the ViBERT-CC model and it achieved a score of 66.10 %. Aggarwal et al., (Aggarwal et al., 2023) carried out experiments on three multimodal models namely, RoBERTa-ResNet, Visual-BERT, and UNITER without considering adversarial training. They added adversarial examples for both modalities on the test set of the FBHM dataset to investigate the robustness of hate meme detection against adversarial examples. The study they performed revealed UNITER as the best model. Chhabra et al., (Chhabra & Vishwakarma, 2023) proposed a multimodal model that uses multi-scale visual kernels and knowledge distillation architecture to achieve an impressive 76.07 % on the FBHM dataset.

Ablation Study. To evaluate the effectiveness of combining SMOTE and Incremental PCA in an adversarial setting, we conducted more experiments to examine their corresponding performances:

Raw_Data. This was designed as the proposed multimodal network without using SMOTE or Incremental PCA on the original dataset. We still added adversarial examples on both input modalities.

Incremental_PCA. This was created as the proposed multimodal network using incremental PCA to reduce the features and directly pass them on to our model. Still, this was carried out in an adversarial setting.

SMOTE_PCA. This was created as the proposed multimodal network applying incremental PCA and SMOTE in an adversarial setting.

Based on the results in Fig. 4, we can observe that applying SMOTE only has a greater effect on the performance than applying PCA only. It can also be observed that introducing either PCA or SMOTE enhances the performance of our model than without. Overall, combining SMOTE and PCA achieves a better performance than the other methods.

Qualitative Examples. In Fig. 5, we present examples of post detected by our multimodal network, we can observe that the model can obtain reliable cues to make accurate predictions based on the contents of the image and the text. As demonstrated in Table 2, our model achieved 81.53 % macro-F1 score on the FBHM dataset. This is still a long way to a perfect performance, so it is worth analyzing the failure case for

the future work. For failure cases some posts don't contain explicit hate content, if our model can infer the context relation among posts and the knowledge from human experts, it will improve the model performance.

Fig. 6 shows some examples of failure cases, the top three memes were detected as hateful and the bottom three memes were detected as not-hateful. It can be observed that cases where the textual and visual elements require nuanced understanding and cultural context are particularly challenging. For instance, our model frequently mislabeled memes that reported sarcasm or irony due to the model's lack of pragmatic reasoning. Also, the model is affected by cultural bias and stereotypes present in the dataset, for example, the meme at row 1, column 1 of Fig. 6 was detected as hateful because of our model inability to recognize visual stereotypes.

5. Conclusions and future works

In this paper, we highlighted the need to build robust systems to be able to effectively detect hate content online as it has become common to spread hate through images and text on social media and masking the posts to evade online detectors. We tackle this challenge by addressing both data imbalances and model vulnerabilities. We uniquely combine SMOTE, to mitigate class imbalance, with incremental PCA, which offers memory-efficient dimensionality reduction which is critical for handling large, multimodal datasets like the Facebook Hateful Memes (FBHM) dataset. This combination enabled our multimodal model to achieve an accuracy of 81.80 % on the validation set, thereby outperforming existing benchmarks. Beyond accuracy, we acknowledge the ethical imperative of building models that are resilient to adversarial manipulation. Recognizing that even non-technical attacks can significantly affect the models' performance, this work provides a practical approach to fortifying hate post detection in real-world scenarios. We stored the model's prediction on the test set in a JSON file for human-level judgment and we discussed some predictions it made.

We noticed that the adoption of incremental PCA proved to be particularly beneficial, offering the scalability and adaptability essential for modern machine learning workflows dealing with ever-growing data streams. To enhance model training by ensuring that both images and text features are appropriately scaled and reduced before further processing, we used incremental PCA as a crucial preprocessing step to enable efficient handling of multimodal data while addressing class

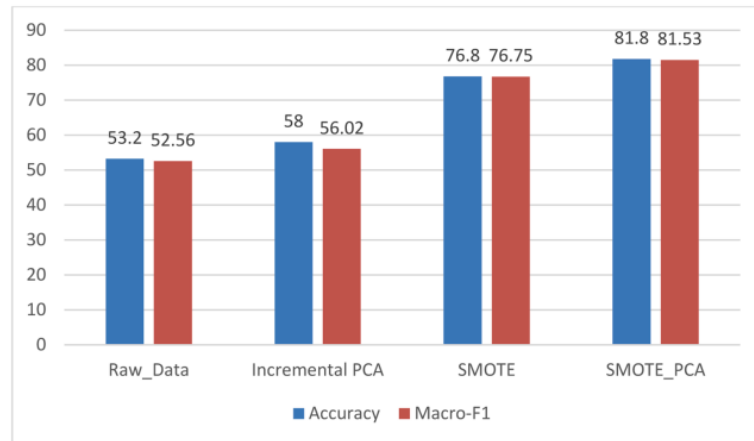


Fig. 4. Ablation Study of the Proposed Model.



Fig. 5. Examples of posts that were detected successfully by our model.

imbalance through synthetic sample generation (SMOTE).

In future, we intend to train our model against more severe adversarial attacks with perturbation norms from l_0 , l_2 and l_∞ to make it even more robust, we intend to try more variants of attacks. We equally intend to perform experiments under varying degrees of dataset imbalance to provide a deeper insight into the robustness of our approach. Also, we intend to use multilingual comments with memes to extract better features.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly in order to improve readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Author statement

***Conceptualization:** E. L. Tchuindjang Tchokote: Developed the main idea and research questions.

***Methodology:** E. L. Tchuindjang Tchokote: Designed the study and selected appropriate methodologies.

***Data Collection:** E. L. Tchuindjang Tchokote: Conducted experiments and collected data.

***Data Analysis:** E. L. Tchuindjang Tchokote: Analyzed the data and interpreted results.

- Elie Fute Tagne: Assisted in statistical analysis and validation of findings.

***Writing – Original Draft:** E. L. Tchuindjang Tchokote: Wrote the initial draft of the manuscript.

- Elie Fute Tagne: Contributed sections related to literature review and discussion.



Fig. 6. Failure cases examples.

***Writing – Review & Editing:** Elie Fute Tagne: Reviewed and edited the manuscript for clarity and coherence.

***Supervision:** Elie Fute Tagne: Oversaw the project, provided guidance, and ensured compliance with ethical standards.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendices

Appendix A: Model Prediction Outputs on MAMI dataset

Figure A1: Some misogynous posts detected under non-technical attacks



Figure A2: Some non-misogynous posts detected under non-technical attacks

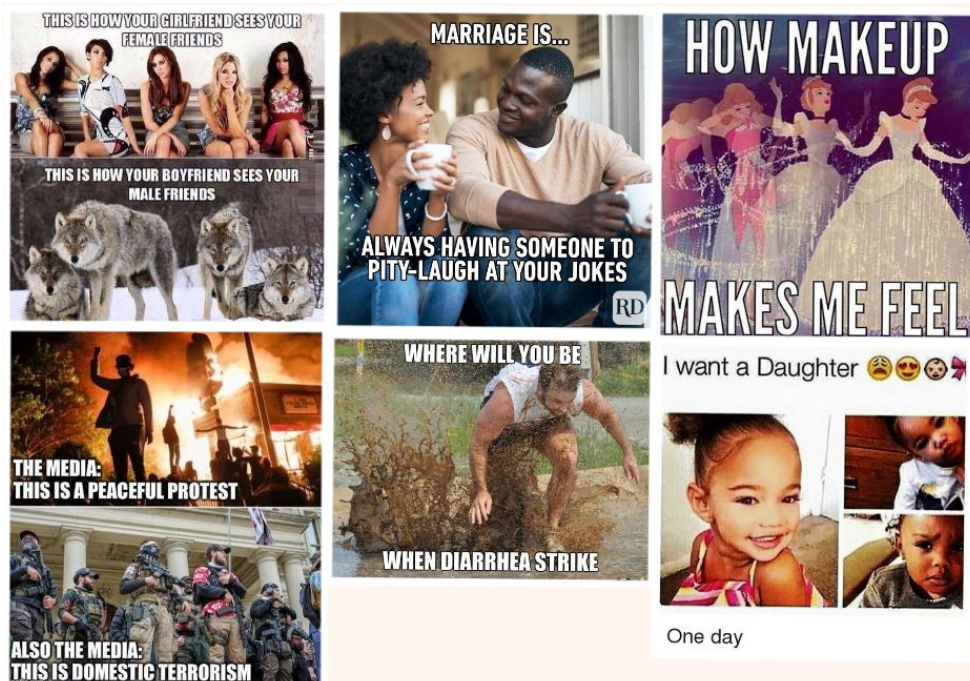


Figure A3: Some non-misogynous posts detected under FGSM, PGD, and AdvDrop attacks

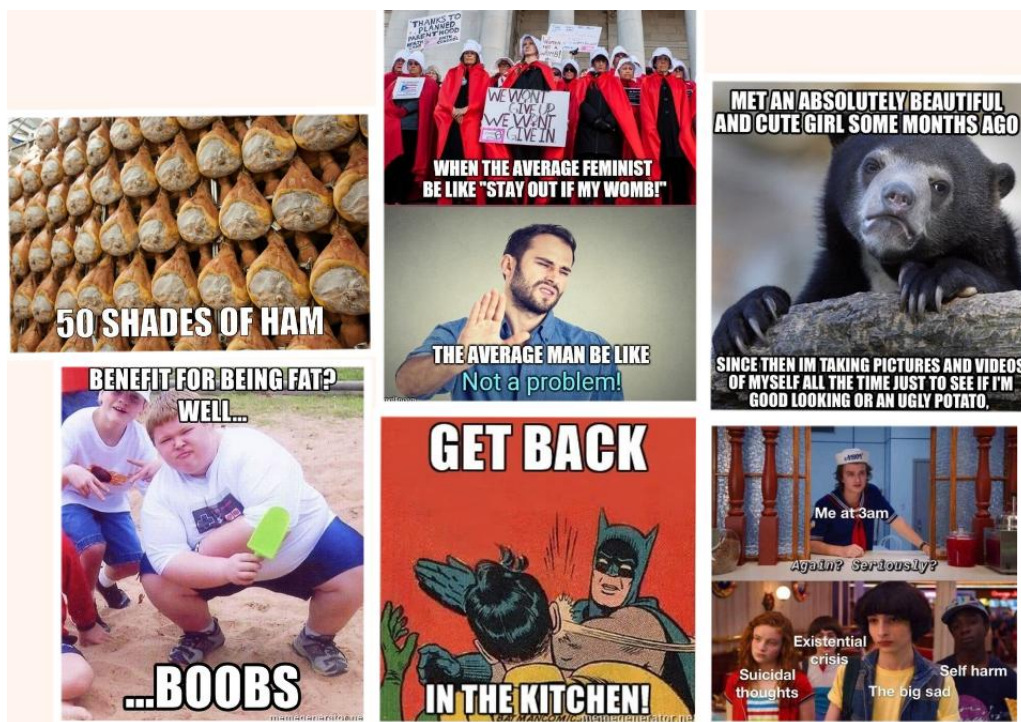
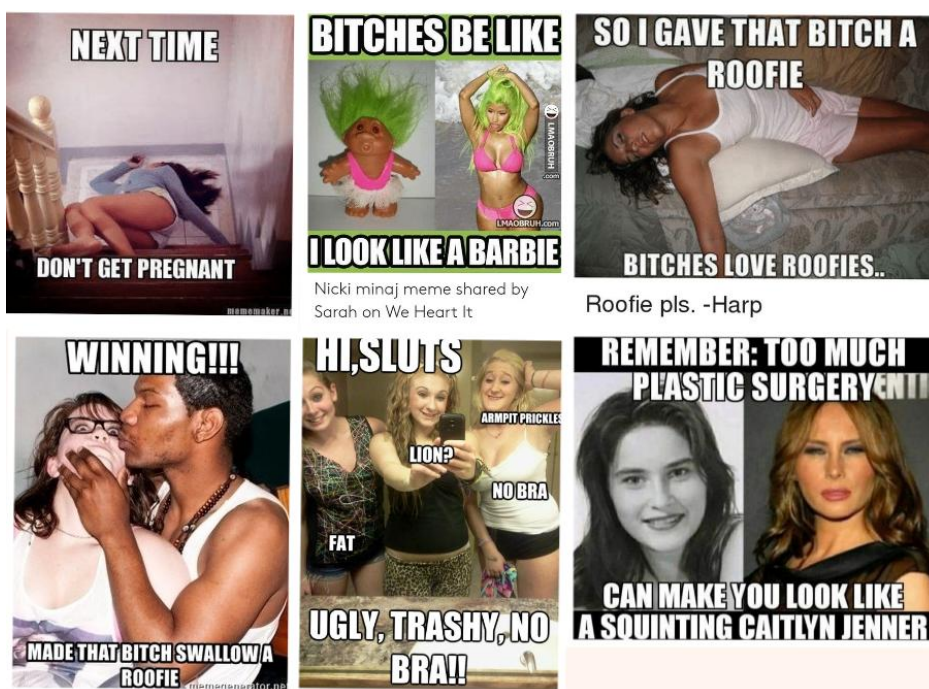


Figure A4: Some misogynous posts detected across FGSM, PGD, and AdvDrop attacks



Appendix B: Model Prediction Outputs on FBHM dataset

Figure B1: Some hate posts detected under non-technical attacks



Figure B2: Some non-hate posts detected under non-technical attacks

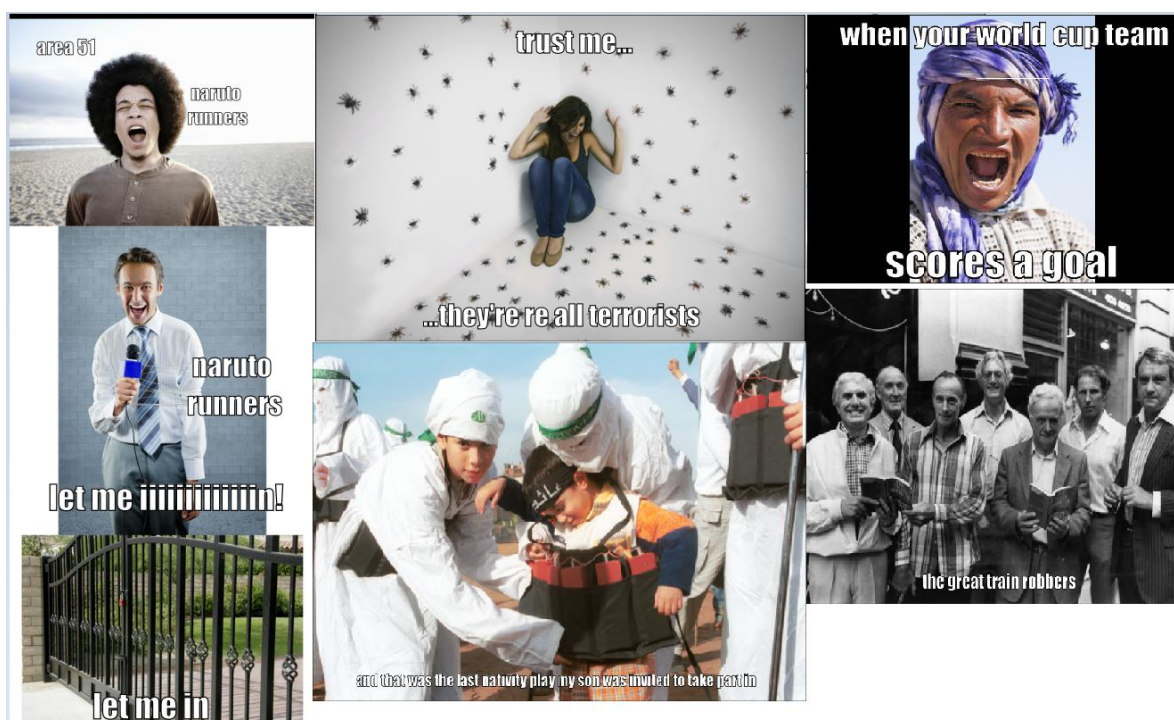


Figure B3: Some non-hate posts detected under FGSM, PGD, AdvDrop attacks

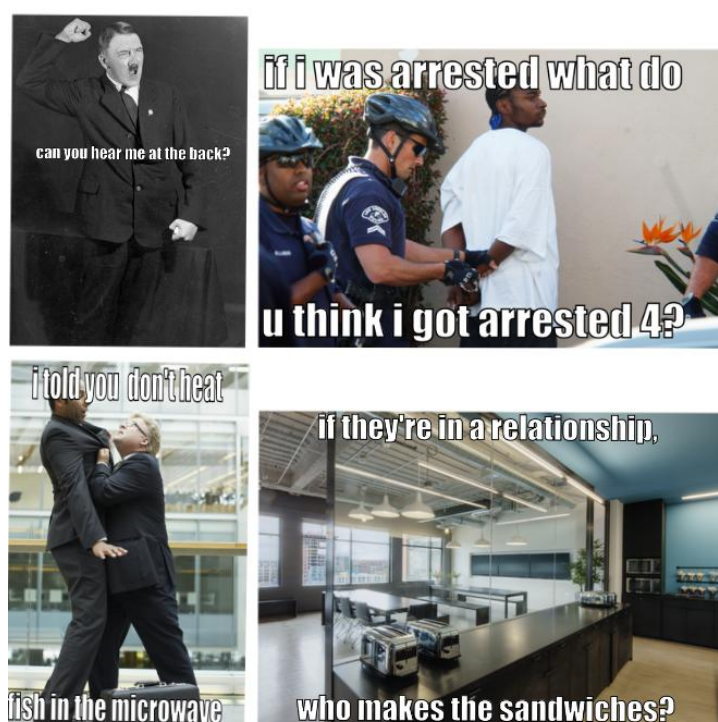


Figure B4: Some hate posts detected under FGSM, PGD, AdvDrop attacks

